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820-0228 – Install Instructions for Cryo Comp. Flow Meter (P/N: 205-0121) GE P/N: 5156316

Tools Required:

- 7/8" wrench or slip wrench
- Adjustable wrench or slip wrench
- Tie Wraps (Lytron P/N 101601-01)
- 2' foam insulation tape

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Drain system. Drain system in accordance with Lytron manual #820-0190. Alternate draining through the pump housing is also permissible.
4. Move Electrical Power Box out of the way by following Lytron procedure #820-0233.
5. Remove cabling to flow meter by disconnecting from Signal Box and cutting retaining tie wraps along wire route.
6. Disconnect incoming fluid from Cryo meter (top) by using the adjustable wrench and 7/8" wrench. (Refer to Photo 1)

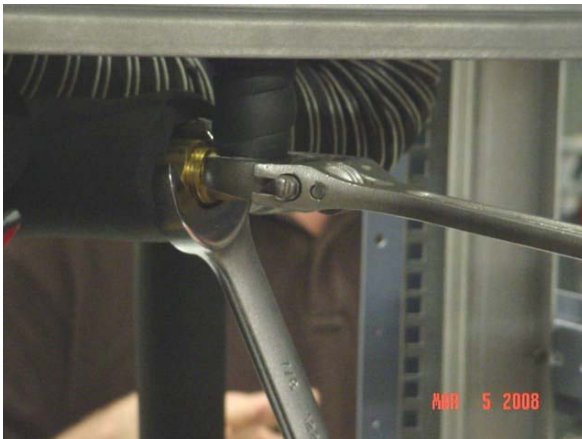
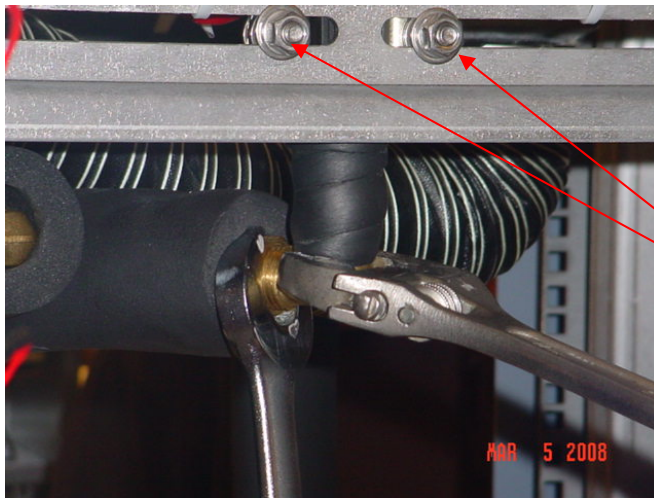


Photo 1

7. Disconnect outgoing fluid from Cryo meter (bottom).
8. Loosen 3/8" nuts securing Cryo meter (2 places).

NOTE: There are 4 flat washers and 1 lock washer per bolt. Care should be taken so that they do not drop into machinery.



3/8" Nuts & supporting washers

Photo 2

9. After removing 3/8" nuts and supporting washers & bracket, remove the Cryo meter assembly from the top of the machine. (Refer to Photo 2)

End of Procedure

Re-install of Cryo Comp. Flow Meter (P/N: 205-0121)

1. From the top of the machine, place the Cryo meter so that it will easily mate with supply and return lines (top and bottom).
2. Use mounting bracket and 3/8" nuts and supporting washers that were saved during the removal of the old meter. Use 4 washers and a lock washer per bolt.

CAUTION: Do not over tighten the 3/8" nuts as this might impede flow through the meter.

3. Connect the upper and lower fluid connections using an adjustable wrench and a 7/8" wrench.
4. Route cable for Cryo meter along support beam toward the left edge of the machine. Secure with suitable tie wraps.
5. Continue to route cable until it terminates at the Signal box jack XX to jack 26. Secure with tie wraps.
6. Fill system in accordance with Lytron manual #820-0190.
7. Check for leaks around unit just installed.
8. Wrap exposed joints with foam insulation tape to prevent condensation.
9. Return machine to running condition per Lytron manual #820-0190.

End of Procedure



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820-0229 - REPLACEMENT INSTRUCTIONS FOR PN 205-0121-01 PUMP AND 205-0121-25 (GE PN 5156321)

PPE REQUIRED: Safety glasses, Safety shoes, Gloves

TOOLS REQUIRED:

- Tools included with FRU
 - Strap Wrench
 - Pliers for VFD Chip
- Hoist Service Kit 5196226
- Non-Magnetic Tool Kit 5113258
- Bucket
- DVMR Coolant 5174313-4 (1 kit = 4 US Gal)

REMOVAL PROCEDURE:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Disable MagMon from sending alarms during procedure.
3. Lockout/Tagout system in accordance with GE procedures.
4. Drain the respective pump by:
 - a. Closing the gate valve beneath the reservoir.
 - b. Direct the hose at the bottom of the pump into bucket.
 - c. Open the drain valve and remove the bolt on the face of the pump. Refer to picture below for pump locations.

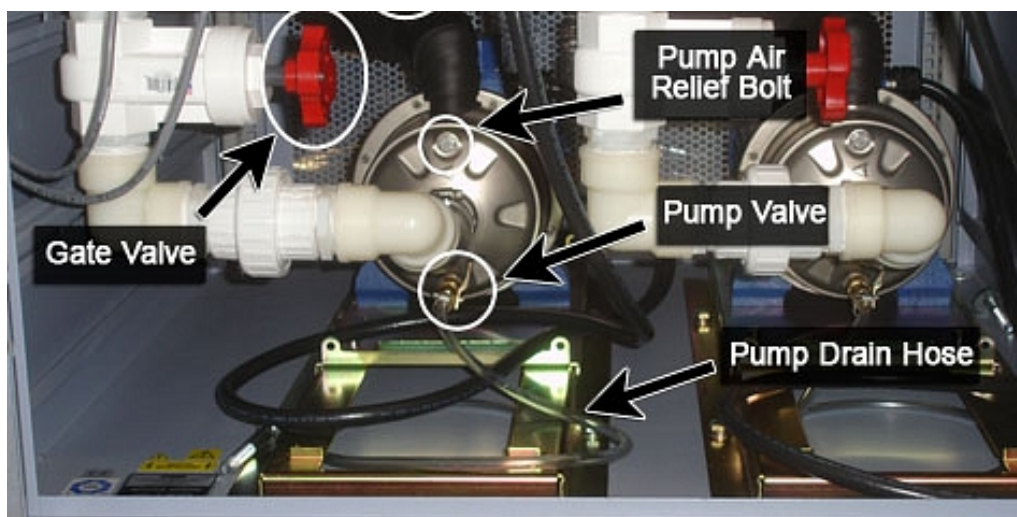


Photo 1 Component Locations for Draining (Ebara Pump)

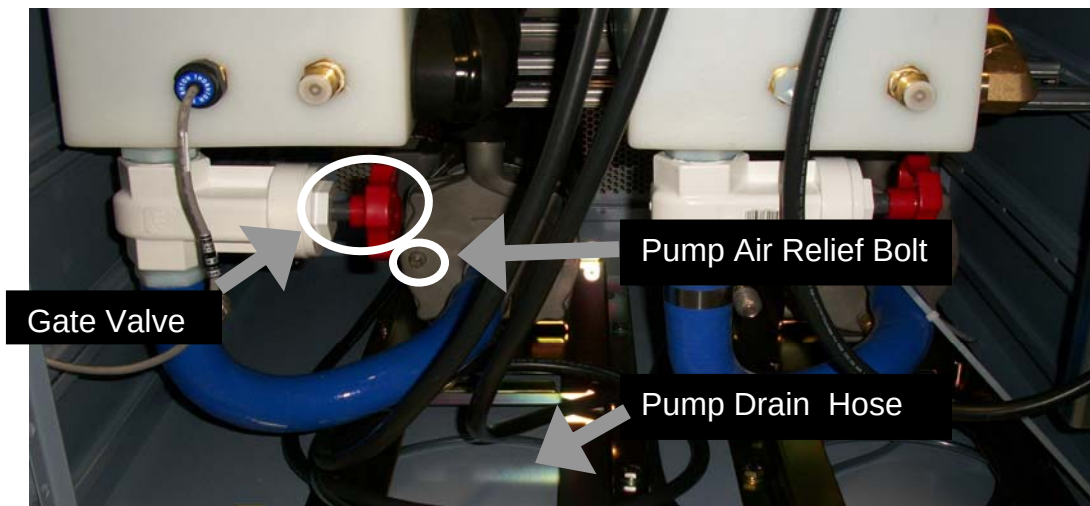


Photo 1A Component Locations for Draining (Price Pump)

5. Install Hoist Assembly (reference GE service procedure) and Chain lift assembly included in Pump FRU kit.
6. Open Front cover of Power box inside HEC.
7. Locate Variable Frequency Drive (VFD) for pump in question.
8. Remove pump supply wiring from VFD (record wire locations for re-installation).
9. Remove power cord from bottom of Power Box.
10. Remove discharge hose from top of pump using Slip-joint pliers on SAE flare fitting.
11. Break suction hose from tank at union (see Photo #2) using strap wrench (applies to Ebara pump only).



Photo 2 - Breaking the Suction Hose Connection to the Tank (Ebara Pump Only)

12. Loosen hose clamp to remove blue suction hose (applies to updated Ebara configuration and Price pump).
13. Loosen retaining screws for pump tray (2 places – Refer to photo)

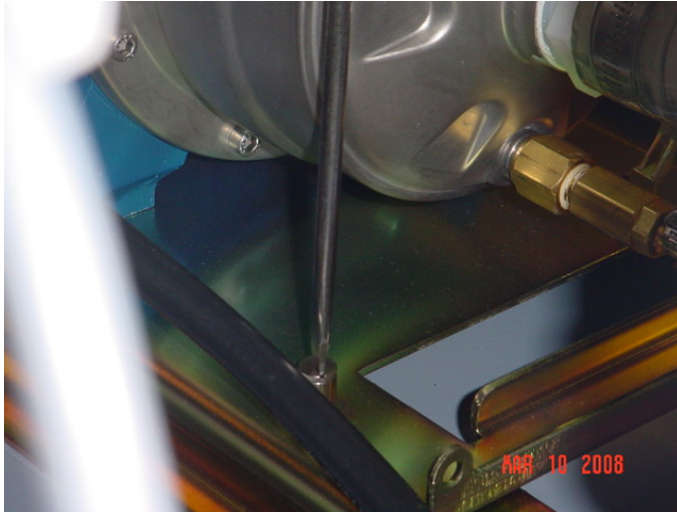


Photo 3 Loosening the Pump Tray Retain Screws

14. Grab tray by built in handle and pull forward until it stops. Lift up over lip of cabinet to facilitate lifting with hoist.
15. Assemble Hoist Kit to HEC.
16. Attach Chain lift assembly to pump lifting brackets on 4 corners of pump plate. Ensure that both chains going to back brackets are mated to the same link. Ensure the front bracket chains are likewise mated. This will provide a balanced lift.
17. Attach Chain lift assembly to Hoist assembly.



Photo 4 Hoist Assembly Hook and Chain

18. Place Hoist Assembly handle in “Up” position.



Photo 5 Hoist Handle in “UP” Position

19. Ratchet Hoist Assembly handle to begin lifting the pump.

20. When pump is high enough to clear cabinet, carefully swing Chain lift assembly toward the packing crate

21. When pump is in proper position, place Hoist assembly ratchet in “Down” position.

22. Lower pump until it is settled in crate.

End of Procedure

INSTALLATION PROCEDURE

NOTE: Depending on the revision of the HEC, it is possible that there is a mismatch between pump suction hoses. The molded blue hose should be used whenever possible.

1. If the system has a rigid PVC hose present, it must first be removed. Refer to the boxed outlined section below titled "INSTALLING THE MOLDED HOSE" on how to remove the rigid PVC suction line.
2. Machine should be in a shutdown condition as required by above procedure.
3. Position Pump crate as close to machine as possible to facilitate lifting of the pump.
4. Attach Hoist Assembly chain hook to the pump through 4 eyelets located on the pump tray.

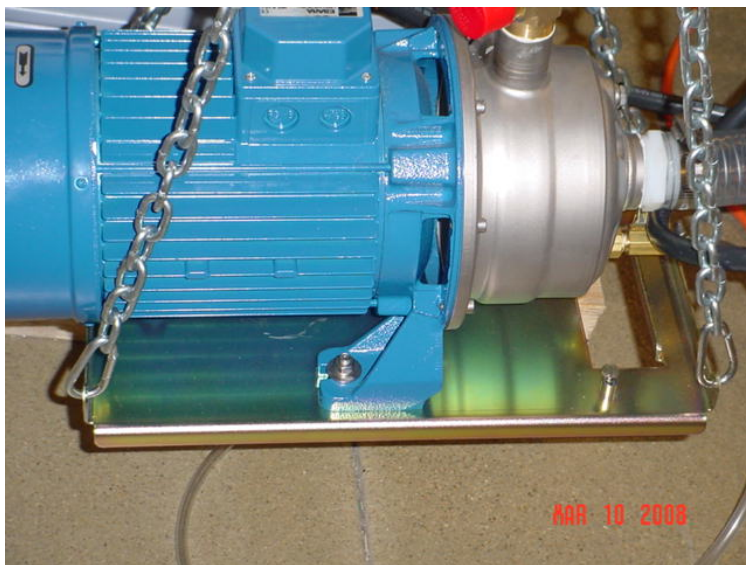


Photo 6 Lifting the Pump Assembly (EBARA pump pictured)

5. Place Hoist Assembly handle in the "UP" position.
6. Take slack out of chain length by ratcheting the handle.
7. While guiding the tightened chain with your free hand, slowly lift the pump until it clears the lip of the machine.
8. Slide the pump into the machine along the Hoist Assembly track.

NOTE: The pump will be unable to go the complete distance due to the angle with the machine.

9. Lower the pump until the pump front end rests on the lip of the machine.
10. Remove the Hoist Assembly chain hooks from the pump.
11. Move the pump onto the pump positioning tracks.
12. Slide the pump back into the machine to align and tighten the two thumbscrews that secure the pump.
13. Remove the Hoist Assembly from the machine.
14. Reattach discharge hose at top of pump.

INSTALLING THE MOLDED HOSE:

1. Remove existing pump suction hose assembly from HEC (leave gate valve in place: ensure it does not rotate when removing fittings with slip-joint pliers)



2. Wrap threads of supplied plastic barbed fittings with the supplied PTFE tape. Wrap entire area of threads with two full wraps (in a clockwise direction when looking at the threaded end).
3. Thread new hose barb fitting into bottom of gate valve to make a leak-tight connection.



4. Place the supplied Breeze-brand hose clamps over pump suction hose and install over barb on front of the pump and beneath the gate valve (hose is orientation specific and can only be installed one way). Refer to photo below



5. Tighten Breeze hose clamp until the stacked washers flatten.

15. Fill the system in accordance with Lytron manual #820-0190. Ensure gate valve beneath tank is fully open.
16. Route the power cable for the pump through the machine to the Power Box.
17. Wire the pump to the appropriate Variable Frequency Drive (VFD). Wires are phase sensitive and need to be installed in correct location (see labels on wires)

IMPORTANT: The HEC may have pumps from different manufacturers. (Price or Ebara). It is imperative that the pump type be exactly matched with the correct VFD chip. The Pump FRU kit will contain the correct VFD chip for the pump, as well as a pliers to facilitate removal and replacement of the VFD chip.

An incorrectly-matched pump and VFD chip will result in either high pressure or low flows, risking equipment damage and/or systems errors.

Price pumps/chips and the Ebara pump/chips can be identified according to the following photos and descriptions:



Price



Price Pump:

Irregular pump face shape.
Bolt at 9 o'clock position on pump face.

Price VFD Chip:

Identified with 'P'



Ebara



Ebara Pump:

Circular pump face shape.
Bolt at 12 o'clock on pump face.

Ebara VFD Chip:

Plain, colored chip.

NOTE: Suction hose is independent of pump type (Ebara cabinets may have white or blue hose type)

The gradient coil pump in on the left-hand side of the HEC and corresponds to the VFD in the left most location in the Power Box. The power electronics pump is on the right-hand side of the HEC and corresponds to the VFD in the middle location in the Power Box. The right most VFD is for the chilled air blower and is not impacted by this replacement procedure.

If the style of chip matches the style of pump according to the above descriptions, the next section can be skipped and you may proceed with the next step in the installation procedure.

The chip is replaced using the pliers shipped with the Pump kit. Remove the tape holding the chip in the front of the respective VFD in the Power Box. Grab with the pliers and pull straight out. To install the new chip, align it according to its keyed shape and press in fully with your finger. Place a piece of tape (in the FRU kit) over the newly-installed chip.

(Note: If the VFD chip is missing, lost or broken from the Pump FRU, see the GE FRU Manual and order the correct VFD chip.)



Figure 6 Remove the existing VFD chip with replacing the Ebara pump with a Price Pump.



Figure 7 Complete installation. (This HEC has an Ebara gradient coil pump and a Price power electronics pump, as noted by the 'P' on the VFD chip in the power electronics VFD)

18. Replace front cover of Power Box.

FINALIAZTION

1. Clear the Lockout/Tagout in accordance with GE procedures.
2. Re-start HEC and turn pump ON. Add new DVMR coolant as necessary.
3. Check for any leakage. Repair if necessary.
4. Perform a TPS reset.
5. Ensure HEC is in a scan-ready state and not in the low power mode; this is done by prescribing a scan at least to the point where the scan protocol.
6. Return to the HEC and view the displayed speeds on the two pump VFDs.

Note if the VFD is displaying 'F1' the VFD chip is either not installed or not seated fully.

Review the charts below and circle 'YES' or 'NO' below, indicating you have confirmed the correct speed based on pump and system configuration:

GRADIENT COIL PUMP:

<u>Pump Type</u> (identify as described above):	EBARA	PRICE
<u>Displayed Value</u> <u>On left-most VFD</u> (VFD GPMP)	50 +/- 1	61 +/- 1

CORRECT GC Pump Speed?

YES / NO

POWER ELECTRONICS PUMP:

Pump Type: (identify as described above):	EBARA		PRICE	
RF-amplifier type in PGR cabinet (identify by presence of red and blue cooling hoses hooked up to front of RF-amplifier. Hoses will not be present on air cooled RF amplifiers.	Liquid	Air	Liquid	Air
Displayed value on middle VFD ('VFD PPMP')	50 +/- 1	43 +/- 1	61 +/- 1	53 +/- 1

Correct PE pump speed

YES / NO

7. If either of the pump speeds do not match the charts above, turn OFF the pumps and:

A.Re-inspect that the VFD chips match the pump type.

B.Ensure the HEC is in a scan-ready state and not in low-power mode.

If you do not have the proper VFD chips to match up with the pumps present, consult the FRU manual and order the required VFD chips.

Further Troubleshooting information can be found in the GE Service Methods document "HEC Troubleshooting".

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	APPRV'D
A	INITIAL RELEASE	6/20/08	HB
B	REV PER ECN 17636 CHANGES TO PUMP PLUMBING PER CUSTOMER REQUEST.	3/17/11	MG
C	CHANGES REQUESTED BY CUSTOMER TO INCLUDE VFD CHIP AND OTHER EDITORIAL CHANGES	01/02/12	HB



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820-0311 – Install instructions for PN 205-0121-23 Pump Hose (GE P/N 5402967)

Tools Included in Kit:

- Strap wrench (Lytron PN 530-0790)
- Teflon (PTFE) tape

Tools Required:

- 3/8" wrench or adjustable wrench
- Large slip-joint pliers

Optional Tool:

Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190. (GE P/N 5161931 Heat Exchanger Cabinet Installation) found in the cabinet or on the Service DVD.
2. Lockout/Tagout system in accordance with GE procedures.
3. Drain system. Drain system in accordance with Lytron manual #820-0190. *It is also acceptable to drain the system through the pump drain.*

To install the molded hose:

4. Remove existing pump suction hose assembly from HEC. (Leave gate valve in place; ensure gate valve does not rotate when removing fittings with slip-joint pliers.)



Figure 1 Removing existing hose assembly detail.

5. Wrap threads of supplied plastic barbed fitting with the supplied PTFE tape. Wrap entire area of threads with two full wraps (in a clockwise direction when looking at the threaded end).
6. Thread new hose barb fitting into bottom of gate valve to make a leak-tight connection.



Figure 2 Installing new hose bard fitting detail.

7. Place the supplied Breeze-brand hose clamps over pump suction hose and install over barb on front of the pump and beneath the gate valve. (Hose is orientation specific and can only be installed one way) Refer to photo below:



Figure 3 Detail of Hose orientation

8. Tighten Breeze hose clamps with 3/8" wrench until the stacked washers flatten.
9. Fill the system in accordance with Lytron manual #820-0190 (GE P/N 5161931).
10. Clear the Lockout/Tagout in accordance with GE procedures.
11. Perform a normal startup according to Lytron manual #820-0190.
12. Check for any leakage. Repair if necessary.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPV'D	SERV APPRV'D
A	Initial Release	5/5/2011	NG	HB



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820-0339 – Replacement Instructions GC and PE Mixing Valves PN 205-0121-30 (GE P/N 5156318)

Purpose: The purpose of this document is to provide instruction on how to remove and replace the GC and PE mixing valves. Note that there are 2 different types of valves (refer to illustrations within this document to determine mixing valve type). The mixing valve will be supplied with a mounting bracket and hoses making installation and removal in all cabinets possible. The GC and PE valve assemblies in a new cabinet are identical in form, only the hose lengths are different. This kit has been designed with hose lengths to replace either a GC or PE valve.

Scope: This instruction applies to all DVMR Heat Exchanger Cabinets (HEC).

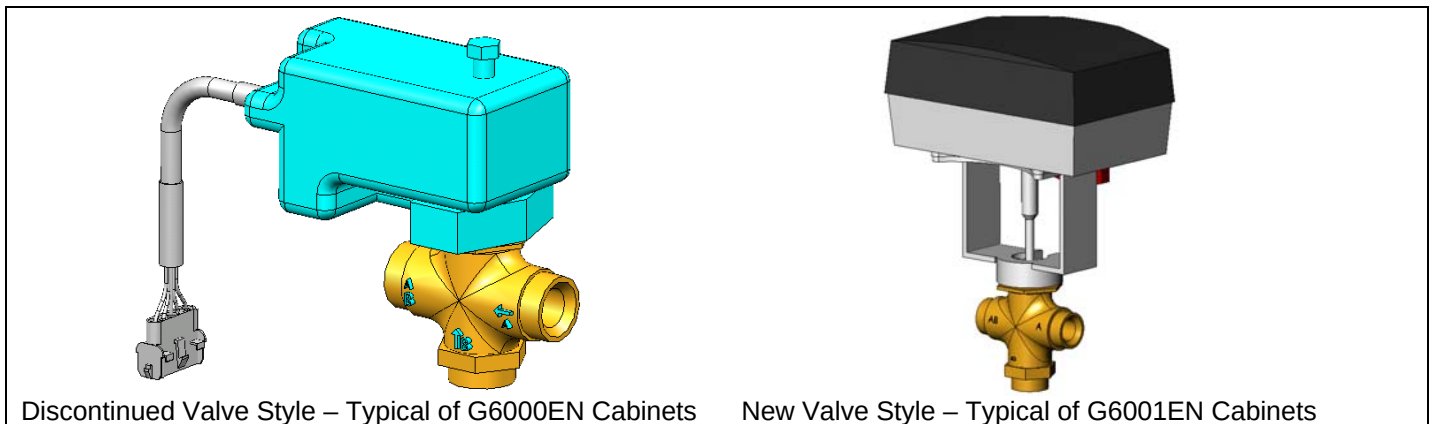
Tools Required:

- 1-1/4" open end or adjustable wrench
- Adjustable wrench (opening up to 1-1/8" for backing stationary fittings)
- 7/16" deep welled socket or wrench
- Absorbent towels
- Wet/Dry Vacuum or Bucket for draining

People Required: 2

Time Required: 2 hours per valve

Refer to the illustrations below to determine mixing valve type.



Discontinued Valve Style – Typical of G6000EN Cabinets New Valve Style – Typical of G6001EN Cabinets

Illustration 1 – Mixing Valve Styles

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE "LOTO for Heat Exchanger Cabinet (HEC)" procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by disconnecting the ground strap (The ground strap nut can either be removed using a 10mm or adjustable wrench) and lifting up on the door.

3. Close Facility Supply and Return Ball valves located at the top of the HEC to isolate HEC from facility water.
4. Locate either GC or PE mixing valve (Refer to Illustration 2).

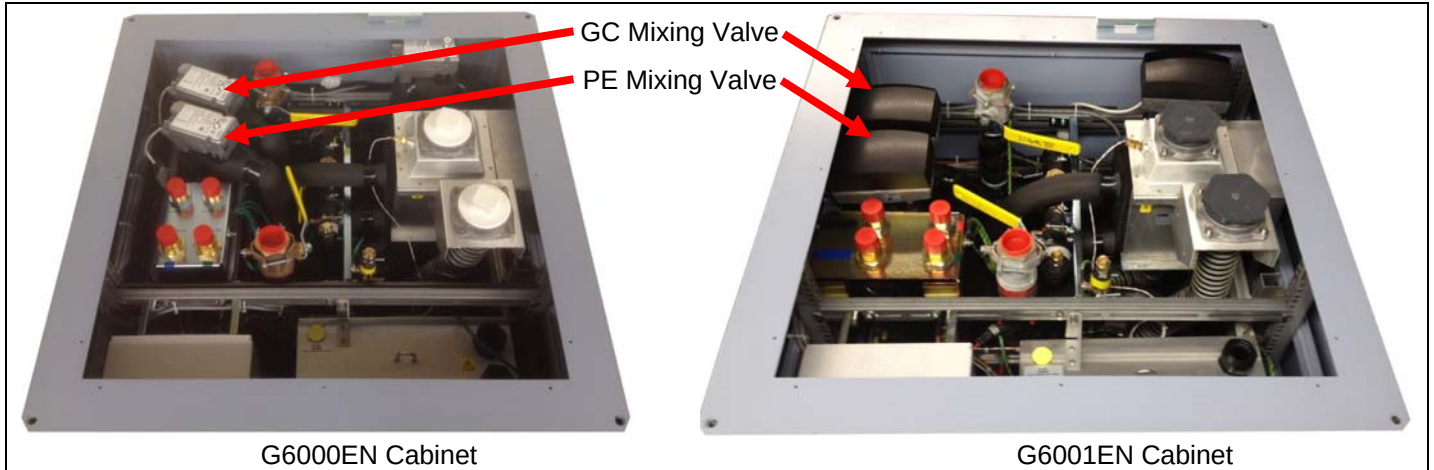


Illustration 2 – GC and PE Mixing Valve Locations in G6000EN and G6001EN Cabinets

5. Drain Facility side plumbing per GE draining procedures.

Mixing Valve Assembly Kit

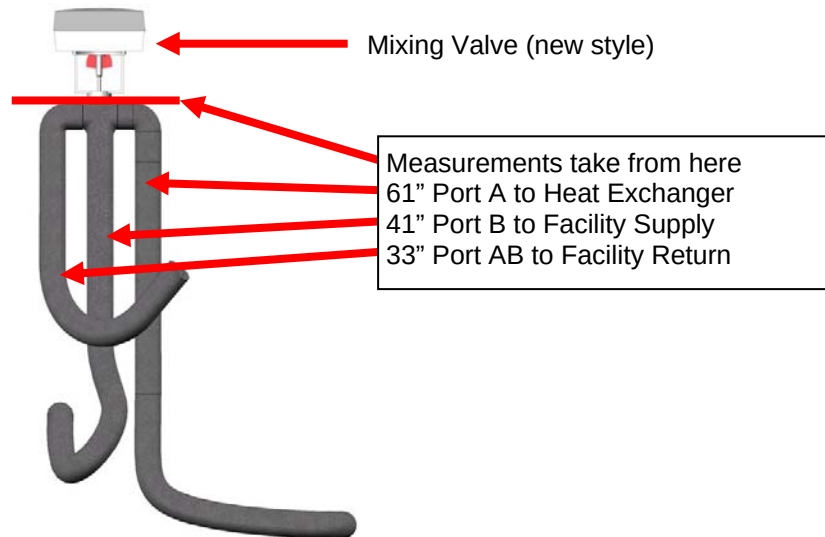


Illustration 3 - Connections for Either the GC or PE Mixing Valve Kit



CAUTION When removing items weighing greater than 35lbs, the hoist system must be utilized. Failure to use the hoist system may result in damage to the equipment or personal injury.

6. It is recommended to move/remove either the Signal Box or the Electrical Box to gain access to connection points on Facility Return header. In order to slide the Electrical Box to the side, disconnect the cables between the Electrical Box and the Signal Box. Refer to Lytron Procedures #820-0233 (Electrical Box) and #820-0234 (Signal Box) for details on removing these parts. It is also recommended to remove the blower to get better access to the hoses per GE main Blower replacement procedure under Heat Exchanger Cabinet Setup.

7. The Photos below detail the locations for connecting ports A, AB, and B to the system. Refer to Illustration 3 above for locations of ports detailed in following photos. NOTE: Both Signal Box and Electrical Box have been removed in the photos below for ease of locating connections.

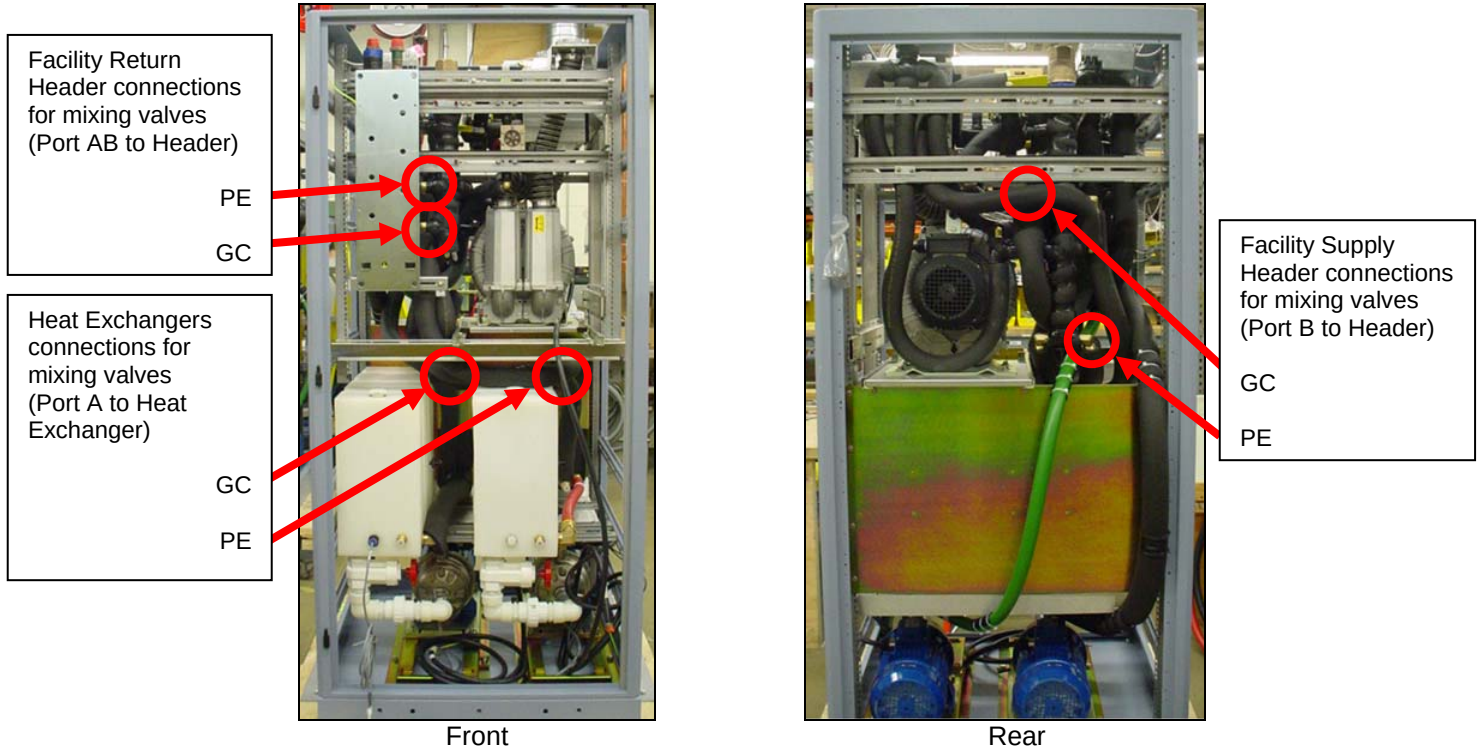


Illustration 4 - Detail of Cabinet with Signal Box and Electrical Box removed (Front and Rear Views)

8. Remove 5 pin connector for appropriate mixing valve (GC or PE).

NOTICE

Fittings within the HEC should be backed with a second wrench in order to prevent them from turning. Failure to do so could result in fluid leaks and damage to the equipment.

9. Using a 1-1/4" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting, remove the Port A hose from the Heat Exchanger (top, left fitting of heat exchanger when looking from the front). It may be necessary to cut the insulation around the fitting in order to expose the wrench flats. There may be residual coolant in the hose that will need to be drained.
10. Again using the 1-1/4" wrench and the adjustable wrench, remove the Port AB hose from Facility Return connection at the header. As above, insulation may need to be removed at the header and the hose drained.
11. Again using the 1-1/4" wrench and the adjustable wrench, remove the Port B hose from the Facility Supply connection at the header. As above, insulation may need to be removed at the header and the hose drained. To locate the correct hose, follow the hose connection from the valve. Refer to Illustration 5 for locations of Port B on the Facility Supply Header.

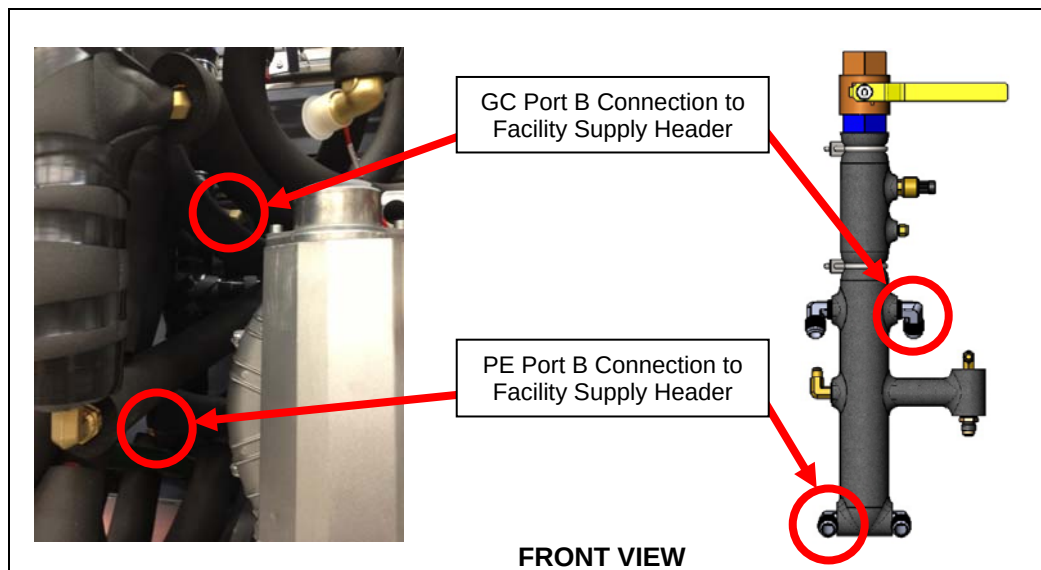


Illustration 5 – Front View Location of Port B Connections to Facility Supply Header for GC and PE Valves

12. **a. (G6000EN Cabinet Valve Replacement)** - While supporting the valve assembly with one hand, remove the 1/4-20 nuts that secure the valve to the system using a U bracket assembly using the 7/16" deep well socket or wrench. There are 2 U brackets, on each U bracket there are 2 nuts (Total of 4 nuts per valve). Refer to Illustration 6 for location of the 4 nuts for each of the valves.
- b. (G6001EN Cabinet Valve Replacement)** - Loosen hex screw on side of mounting bracket using the 7/16" deep well socket or wrench.

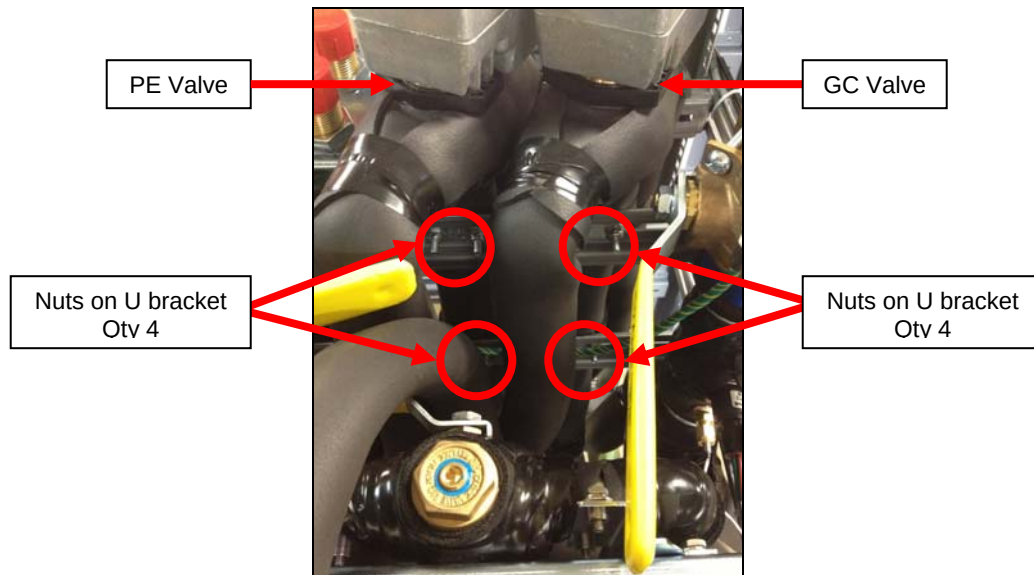


Illustration 6 – Location of 4 Nuts for Each of the GC and PE Valves

13. Remove the valve assembly (valve, 3 attached hoses, and bracket hardware) by pulling it up through the top of the cabinet. It may be necessary to alternate feeding the hoses up through the cabinet and pulling the valve assembly out the top of the cabinet.

Installation Procedure:

NOTICE

The mixing valve has a RED MANUAL/AUTO SWITCH. The valve must be placed in AUTO prior to energizing the system. Identify the correct position of the switch prior to

energizing system (Refer to Illustration 7). Verify prior to operating unit. Failure to place the switch in AUTO could result in failure of the unit to operate correctly.



Illustration 7 – Manual/Auto Switch Position on Mixing Valve

1. Insert the new valve assembly into the cabinet from the top by feeding the hoses down into the cabinet. Both style cabinets will have one hose (Port AB) inserted between the frame strut and cabinet side panel and the other two hoses (Ports A and B) will be inserted between the two frame struts on the left of the cabinet (Reference Illustration 8).



Illustration 8 – New Style Valve Installed in G6000EN Cabinet (System without Insulation)

2. Once in the cabinet, the four tabs of the mounting bracket should go behind the two frame struts. Secure the assembly to frame with hex screw on the left side of the bracket using the 7/16" deep well socket or wrench. (Reference Illustration 9)

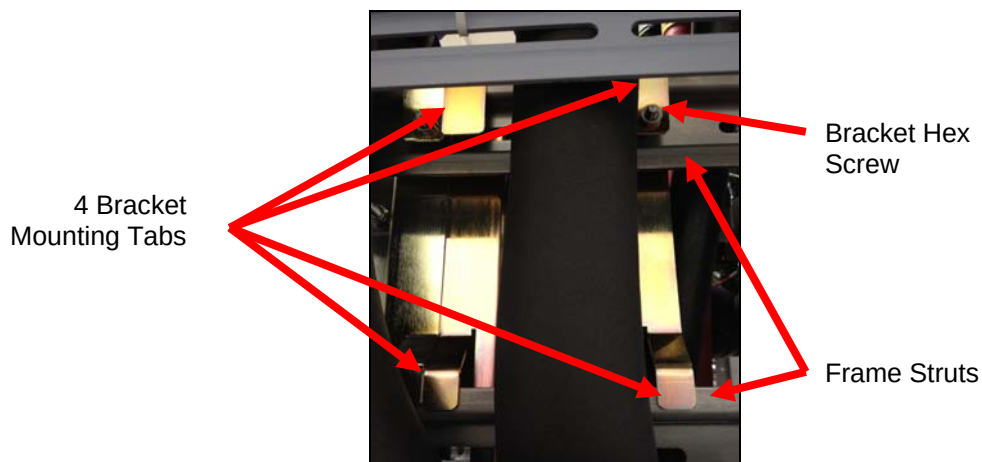


Illustration 9 – Rear View of PE Mixing Valve Installed in Cabinet

3. Using a 1-1/4" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting install the Port B hose to the Facility Supply connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.

4. Again using the 1-1/4" wrench and the adjustable wrench, install the Port A hose to the Heat Exchanger (top, left fitting of heat exchanger when looking from the front). Tighten hand tight and then tighten an additional 1/4 turn.
5. Again using the 1-1/4" wrench and the adjustable wrench, install the Port AB hose to Facility Return connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
6. **(G6000EN Systems Only)** - After installation, the new mixing valves will require the fuse in the Signal Box to be changed to a higher value (T2AL250V). The fuse is part of the New Mixing Valve Kit and needs to be installed on the inside left hand side wall of the Signal Box. (Reference Illustration 10)

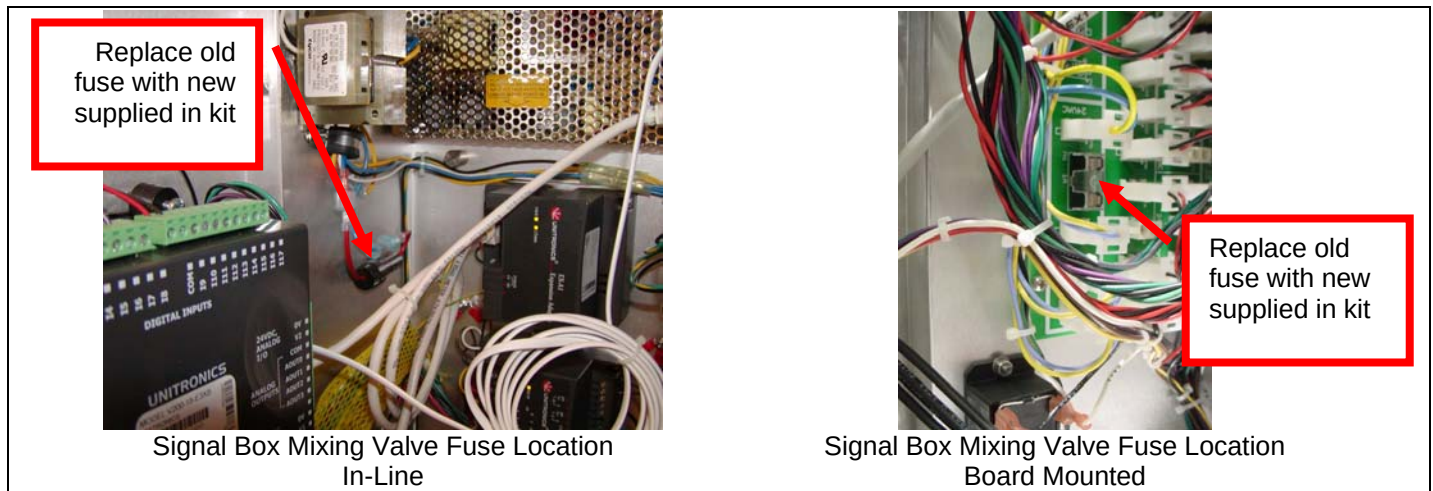


Illustration 10 – G6000EN Signal Box Fuse Locations (Two Design Styles)

7. Label the new fuse with the supplied label. For the inline fuse, place the label in the vicinity of the fuse. For the circuit board mounted fuse, place the label over the existing labeling on the circuit board.
8. Remove any tubing utilized for draining in previous steps.
9. Ensure all connections are tight and properly insulated.
10. Reconnect 5 pin connector to mixing valve assembly.
11. Open Facility Supply Ball valve.
12. Open Facility Return Ball valve.
13. Check for leaks around the manifolds of mixing valve(s) replaced.
14. Use the 2" insulation supplied to rewrap the terminations of the hose insulation within the system as necessary.
15. Reinstall any assemblies (Electrical Boxes, Doors, etc) that were removed during the process to gain access to mixing valves within the cabinet.
16. Clear Lockout/Tag out for the system in accordance with GE procedures.
17. Restart system.
18. Place system back into operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
1	Engineering Release	01/08/2013	GDD	
2	Changed HEC1 to G6000EN and HEC1.5 to G6001EN.	01/08/2013	GDD	
3	Updated with requested changes from GEHC. Changed required people to 2, made removal of the Signal Box or Electrical Box recommended in removal step 6, corrected valve callout in removal step 7, added illustrations to removal steps 11 and 12, and inserted installation step 7 to label the new fuse.	01/15/2013	GDD	
4	Revised and clarified dimensions in illustration 3.	01/18/2013	GDD	



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820-0340 – Replacement Instructions Chilled Air Blower Mixing Valves PN 205-0121-31 (GE P\N 5156320)

Purpose: The purpose of this document is to provide instruction on how to remove and replace the Chilled Air Blower mixing valve. Note that there are 2 different types of valves (refer to illustrations within this document to determine mixing valve type). The mixing valve will be supplied with a mounting bracket and hoses making installation and removal in all cabinets possible.

Scope: This instruction applies to all DVMR Heat Exchanger Cabinets (HEC).

Tools Required:

- 7/8" open end or adjustable wrench.
- Adjustable wrench (opening up to 3/4" for backing stationary fittings)
- 7/16" deep welled socket or wrench.
- Flat-head screwdriver (for removing cover of valve actuator for installation into G6000EN cabinets)
- Absorbent towels.
- Wet/Dry Vacuum or Bucket for draining.

People Required: 2

Time Required: 2 hours per valve.

Refer to the illustrations below to determine mixing valve type.

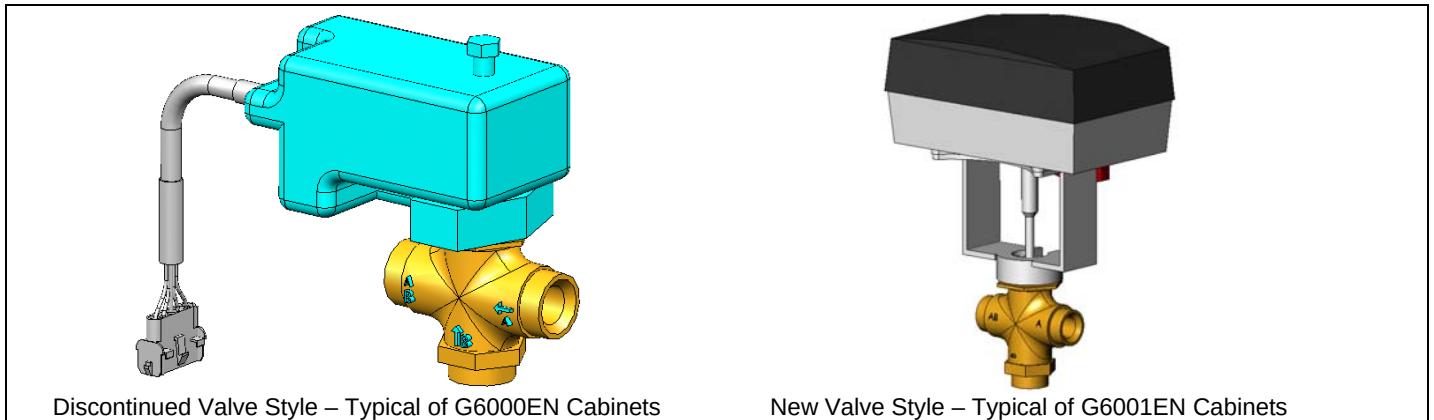


Illustration 1 – Mixing Valve Styles

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE "LOTO for Heat Exchanger Cabinet (HEC)" procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by disconnecting the ground strap (The ground strap nut can either be removed using a 10mm or adjustable wrench) and lifting up on the door.

3. Close Facility Supply and Return Ball valves located at the top of the HEC to isolate HEC from facility water.
4. Locate the chilled air mixing valve (Refer to Illustration 2).

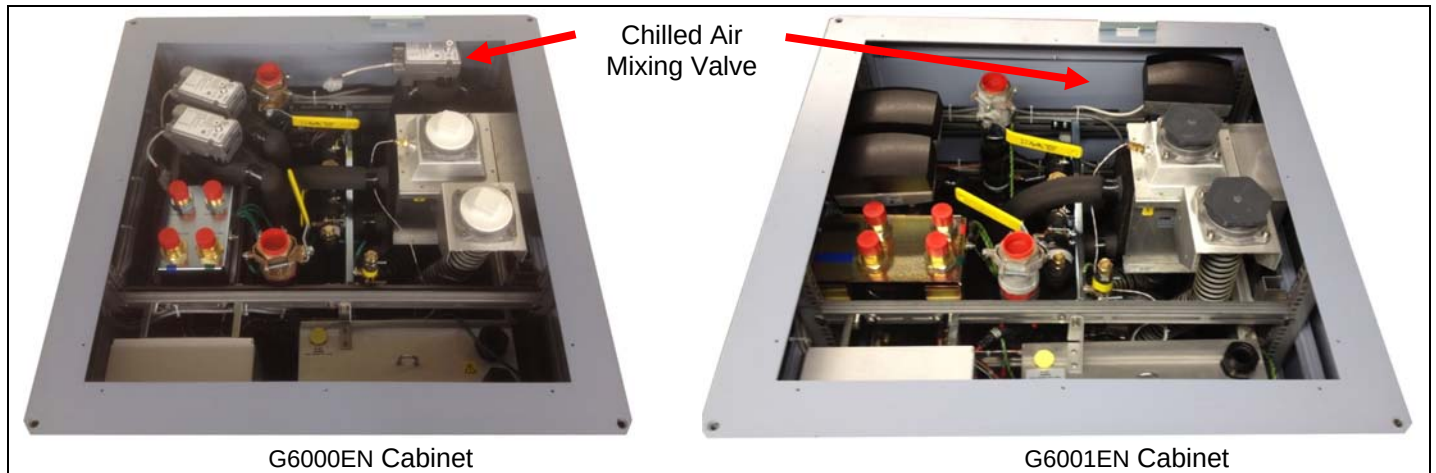


Illustration 2 – Chilled Air Mixing Valve Locations in G6000EN and G6001EN Cabinets

5. Drain Facility side plumbing per GE draining procedures.

Mixing Valve Assembly Kit

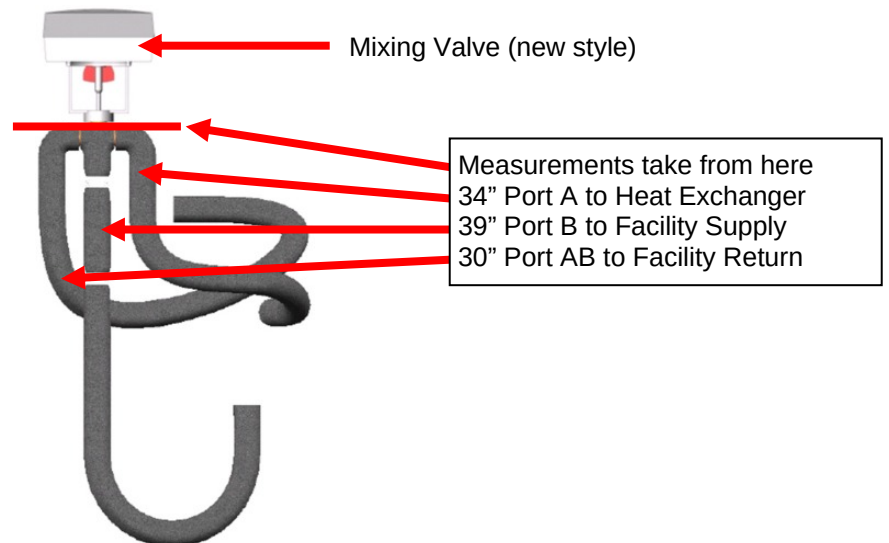


Illustration 3 - Connections for the Chilled Air Mixing Valve Kit



When removing items weighing greater than 35lbs, the hoist system must be utilized. Failure to use the hoist system may result in damage to the equipment or personal injury.

6. It is recommended to move/remove either the Signal Box or the Electrical Box to gain access to connection points on Facility Return header. In order to slide the Electrical Box to the side, disconnect the cables between the Electrical Box and the Signal Box. Refer to Lytron Procedures #820-0233 (Electrical Box) and #820-0234 (Signal Box) for details on removing these parts. It is also recommended to remove the blower to get better access to the hoses per GE main Blower replacement procedure under Heat Exchanger Cabinet Setup.

7. The Photos below detail the locations for connecting ports A, AB, and B to the system. Refer to Illustration 3 above for locations of ports detailed in following photos. NOTE: Both Signal Box and Electrical Box have been removed in the photos below for ease of locating connections.

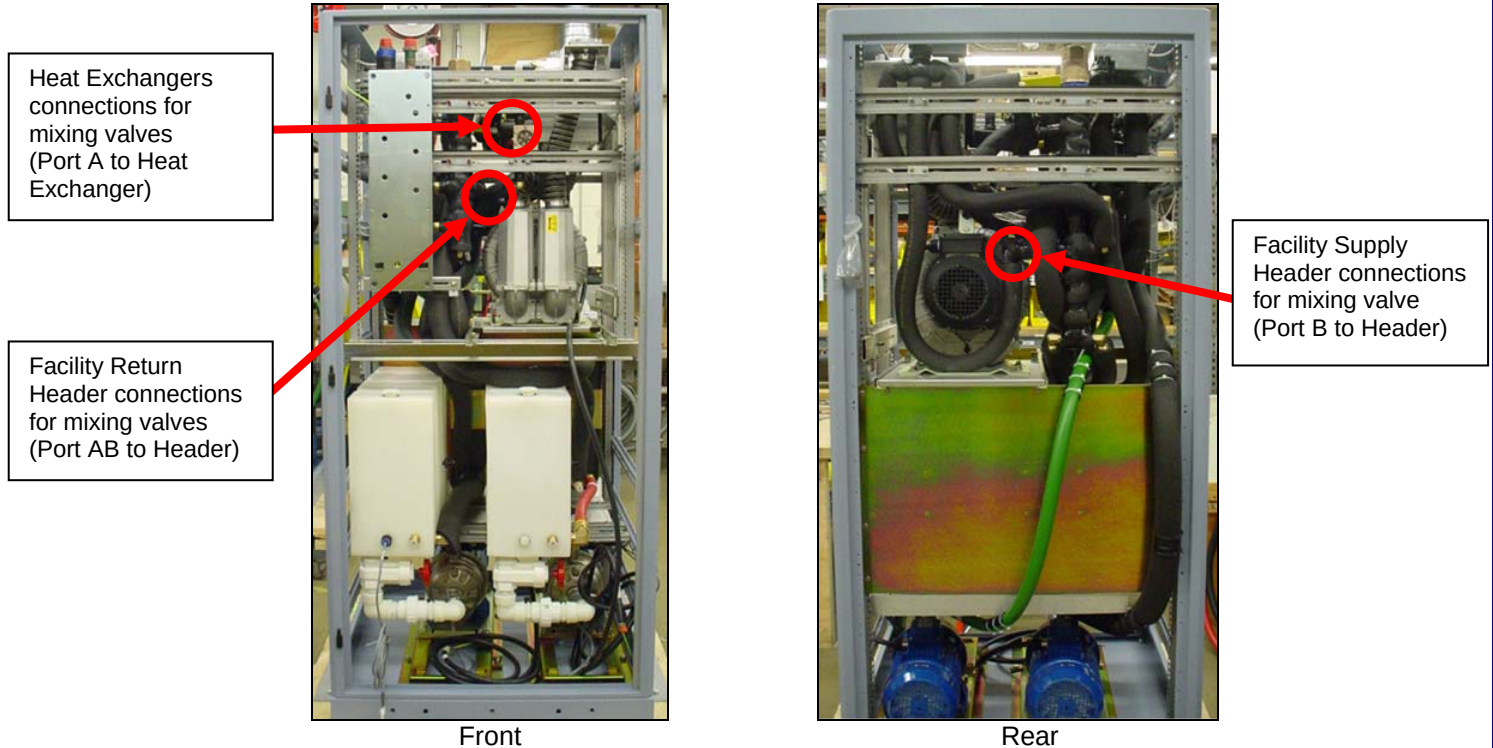


Illustration 4 - Detail of Cabinet with Signal Box and Electrical Box removed (Front and Rear Views)

8. Remove 5 pin connector for the mixing valve.

NOTICE

Fittings within the HEC should be backed with a second wrench in order to prevent them from turning. Failure to do so could result in fluid leaks and damage to the equipment.

9. Using a 7/8" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting, remove the Port A hose from the Heat Exchanger (top fitting of Liquid-to-Air heat exchanger). It may be necessary to cut the insulation around the fitting in order to expose the wrench flats. There may be residual coolant in the hose that will need to be drained (Reference Illustration 5)

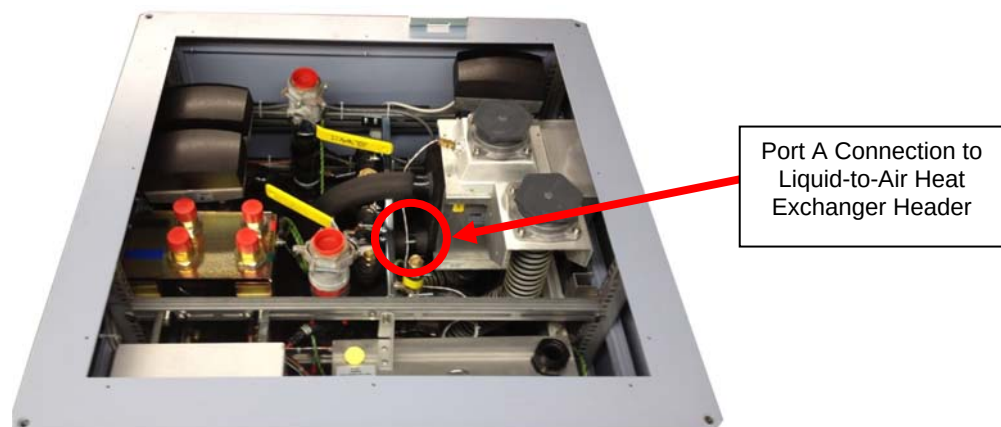


Illustration 5 - Port A Location of the Chilled Air Mixing Valve

10. Again using the 7/8" wrench and the adjustable wrench, remove the Port AB hose from Facility Return connection at the header. As above, insulation may need to be removed at the header and the hose drained.
11. Again using the 7/8" wrench and the adjustable wrench, remove the Port B hose from the Facility Supply connection at the header. As above, insulation may need to be removed at the header and the hose drained. To locate the correct hose, follow the hose connection from the valve. Refer to Illustration 6 for locations of Port B on the Facility Supply Header.

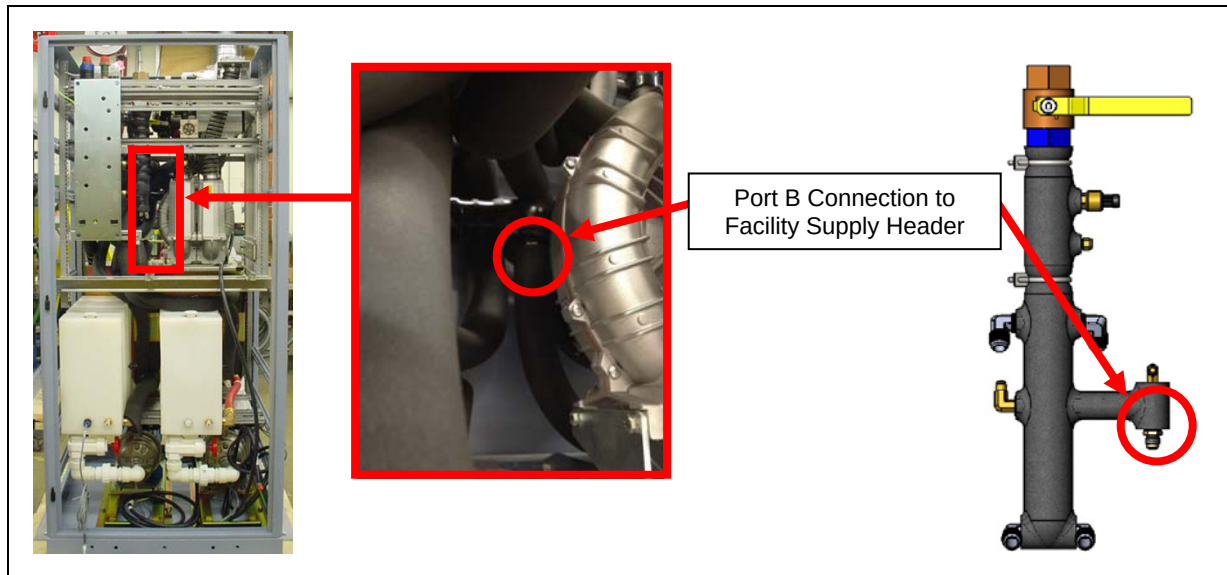


Illustration 6 – Front View Location of Port B Connection to Facility Supply Header

12. **a. (G6000EN Cabinet Valve Replacement)** - While supporting the valve assembly with one hand, remove the 1/4-20 nuts that secure the valve to the system using a U bracket assembly using the 7/16" deep well socket or wrench. There are 2 U brackets, on each U bracket there are 2 nuts (Total of 4 nuts per valve). Refer to Illustration 7 for location of the 4 nuts for each of the valves.
- b. (G6001EN Cabinet Valve Replacement)** - Loosen hex screw on side of mounting bracket using the 7/16" deep well socket or wrench.

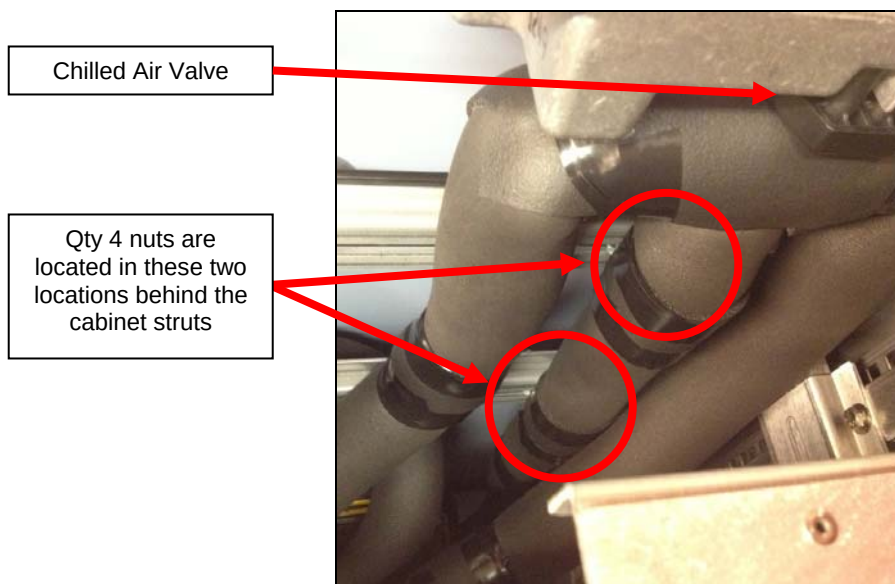


Illustration 7 – Location of 4 Nuts for Each of the GC and PE Valves

13. Remove the valve assembly (valve, 3 attached hoses, and bracket hardware) by pulling it up through the top of the cabinet. It may be necessary to alternate feeding the hoses up through the cabinet and pulling the valve assembly out the top of the cabinet.

Installation Procedure:

NOTICE

The mixing valve has a RED MANUAL/AUTO SWITCH. The valve must be placed in AUTO prior to energizing the system. Identify the correct position of the switch prior to energizing system (Refer to Illustration 8). Verify prior to operating unit. Failure to place the switch in AUTO could result in failure of the unit to operate correctly.



Illustration 8 – Manual/Auto Switch Position on Mixing Valve

1. **(G6000EN Cabinet Valve Installation Only)** – It may be necessary to remove the top cover of the actuator that is part of the mixing valve assembly in order to fit the Chilled Air Valve into a G6000EN cabinet. In order to do this, insert a flat-head screwdriver under the tab of black actuator cover and lift up (Reference Illustration 9).

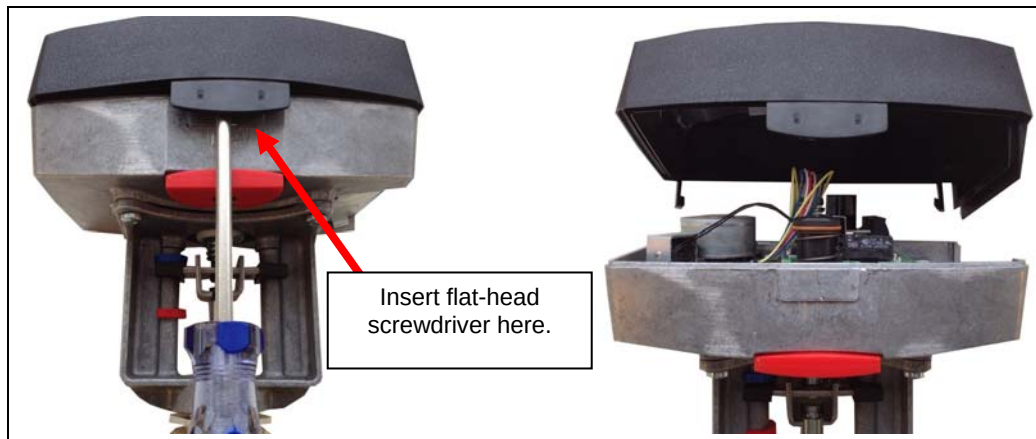


Illustration 9 – Removal of the Actuator Cover

2. Insert the new valve assembly into the cabinet from the top by feeding the hoses down into the cabinet. Both style cabinets will have the new valve installed such that all three hoses will be in front of the two cabinet struts that run horizontally across the rear of the cabinet.

3. Once in the cabinet, the four tabs of the mounting bracket should go behind the two frame struts. Secure the assembly to frame with hex screw on the left side of the bracket using the 7/16" deep well socket or wrench. (Reference Illustration 10)

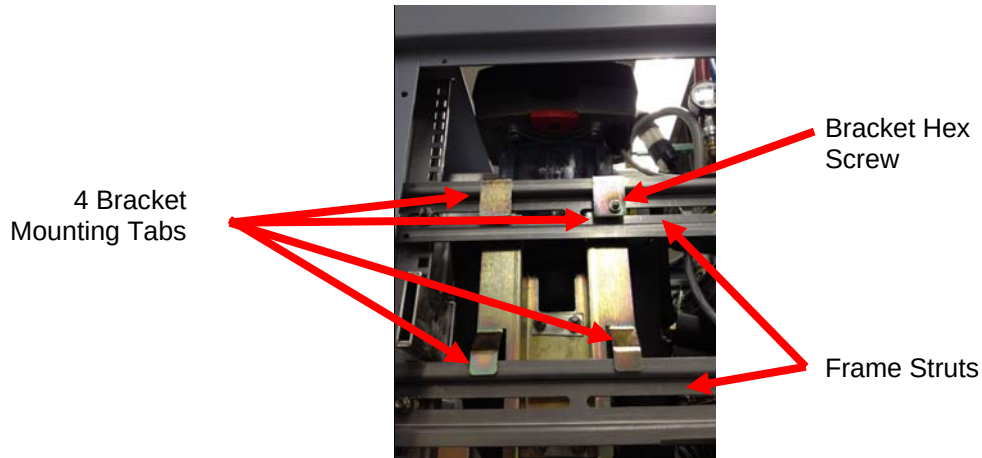


Illustration 10 – Rear View of Chilled Air Mixing Valve Installed in Cabinet

4. Once the valve assembly has been secured in place, re-install the actuator cover if removed during step 1.
5. Using a 7/8" wrench on the swivel fitting and an adjustable wrench to back the stationary fitting install the Port B hose to the Facility Supply connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
6. Again using the 7/8" wrench and the adjustable wrench, install the Port A hose to the Heat Exchanger (top fitting of Liquid-to-Air heat exchanger). Tighten hand tight and then tighten an additional 1/4 turn.
7. Again using the 7/8" wrench and the adjustable wrench, install the Port AB hose to Facility Return connection at the header. Tighten hand tight and then tighten an additional 1/4 turn.
8. **(G6000EN Systems Only)** - After installation, the new mixing valves will require the fuse in the Signal Box to be changed to a higher value (T2AL250V). The fuse is part of the New Mixing Valve Kit and needs to be installed on the inside left hand side wall of the Signal Box. (Reference Illustration 11)

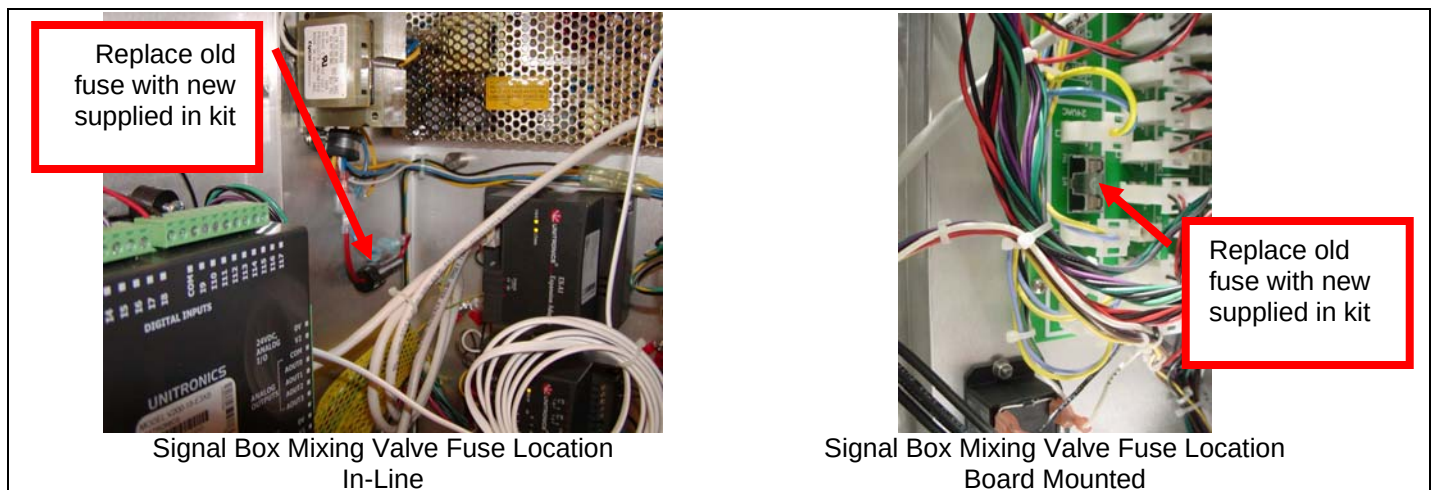


Illustration 11 – G6000EN Signal Box Fuse Locations (Two Design Styles)

9. Label the new fuse with the supplied label. For the inline fuse, place the label in the vicinity of the fuse. For the circuit board mounted fuse, place the label over the existing labeling on the circuit board.

10. Remove any tubing utilized for draining in previous steps.
11. Ensure all connections are tight and properly insulated.
12. Reconnect 5 pin connector to mixing valve assembly.
13. Open Facility Supply Ball valve.
14. Open Facility Return Ball valve.
15. Check for leaks around the manifolds of mixing valve(s) replaced.
16. Use the 2" insulation supplied to rewrap the terminations of the hose insulation within the system as necessary.
17. Reinstall any assemblies (Electrical Boxes, Doors, etc) that were removed during the process to gain access to mixing valves within the cabinet.
18. Clear Lockout/Tag out for the system in accordance with GE procedures.
19. Restart system.
20. Place system back into operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
1	Engineering Release	01/16/2013	GDD	
2	Changed Illustration 10 to say Chilled Air and updated arrows in illustration 3.	01/17/2013	GDD	
3	Revised and clarified dimensions in illustration 3.	01/18/2013	GDD	
4	Corrected location of Port A in illustrations 4 and 5.	01/30/2013	GDD	



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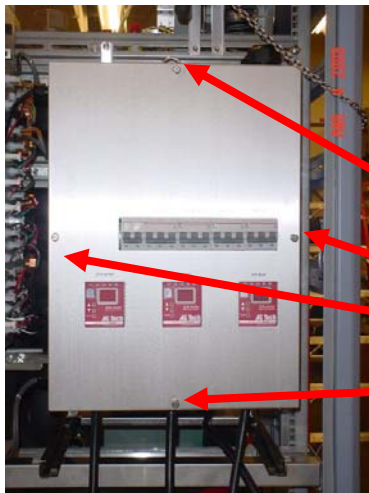
820-0232 – Install Instructions Main Blower (P/N: 205-0121-04) GE P/N: 5156315

Tools Required:

- Hoist Assembly (P/N: 205-0121-17)
- Flat bladed screwdriver
- 1/2" wrench or slip wrench

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Install Hoist Assembly (P/N: 205-0121-17)
4. Remove Electrical Box cover by loosening 4 thumb screws (Refer to photo 1).



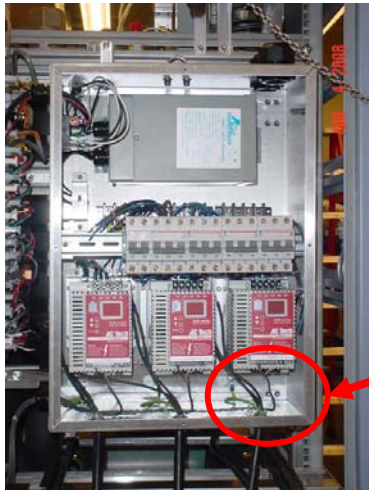
Loosen these 4 thumbscrews

Photo 1 Electrical Box exterior

5. Remove leads U, V, and W as well as GND connection from Blower Variable Frequency Drive (VFD) (Refer to Photo 2).

CAUTION: OBSERVE PROPER SAFETY PROCEDURES WHEN INSIDE THE ELECTRICAL CABINET. HIGH VOLTAGE IS PRESENT WHEN COVER IS REMOVED.

6. Remove Blower power cable from bottom of Electrical Box.
7. Remove the following cable from the connectors at the Signal Box:
 - a. P1 115v Power to the PLC
 - b. P2 DI Kit
 - c. P19 GPMP (GC Pump) VFD
 - d. P20 PPMP (PE Pump) VFD
 - e. P7 Blower VFD



Leads U, V, and W as well as GND

Photo 2

8. Pull Electrical Box forward by loosening 2 thumb screws to the left of the cabinet (top and bottom) securing it to the frame.
9. Pull forward and swing out of the way to the right.
10. Remove supply and return air hoses by loosening worm gear clamps with 1/4 " nut driver or regular flat blade screwdriver. Carefully move hoses out of the way in order to facilitate blower removal.

CAUTION: Care should be taken when removing hoses. The hoses are easily torn.

11. Coil power cord into 6" coils and attach to blower using tie wraps.
12. Loosen 2 thumbscrews securing blower plate to frame.
13. Slide out blower until it stops on the rails. Be careful that the blower hoist lift bracket does not damage the removed air hoses.
14. Attach hoist chain eyelet to Blower lifting bracket.

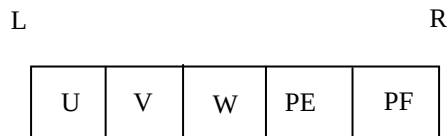
CAUTION: Ensure hoist ratchet handle is selected to "UP" position.

15. Take out slack from lift chain.
16. Remove the 1/2" bolts, lock washers, and flat washers that secure Blower to frame. Use a 1/2" wrench or socket.
17. Raise Blower using hoist. Ratchet in the "UP" direction using handle until the Blower clears its' mounting tray.
18. Push mounting tray in toward the rear of the chiller.
19. Put Hoist ratchet in the "DOWN" position and lower the Blower to a cart or pallet.
20. Disconnect hoist chain eyelet from Blower.

End of Procedure

Installation of Blower:

1. Place Blower close to front of cabinet.
2. Attach Hoist chain eyelet to Blower lift bracket.
3. Place Hoist ratchet in the "UP" position.
4. Lift Blower steadily to a height above the mounting tray.
5. Pull mounting tray forward.
6. Lower Blower onto tray and align with bolt holes.
7. Use a 1/2" wrench or socket to tighten 4 bolts, lock washers, and flat washers that secure Blower to mounting tray.
8. Push Blower and tray toward back of machine and lock into position with 2 thumbscrews in the front.
9. Ensure power leads are uncoiled from the Blower and ready to attach to the Variable Frequency Drive (VFD).
10. Connect supply and return air hoses to the Blower. Be careful not to tear the hoses as they are delicate. Tighten with flat bladed screwdriver.
11. Swing Electrical Box to the left and secure with 2 thumbscrews on the top and bottom of the Electrical Box.
12. Route the power cable for the Blower to the appropriate VFD and wire as follows:



13. Reattach the following cables at the connectors on the Signal Box:

- a. P1 115v Power to the PLC
- b. P2 DI Kit
- c. P19 GPMP (GC Pump) VFD
- d. P20 PPMP (PE Pump) VFD
- e. P7 Blower VFD

End of Procedure

Released



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820-0233 – Replacement Instructions for Electrical Power Box (P/N: 205-0121-05) GE P/N: 5156319

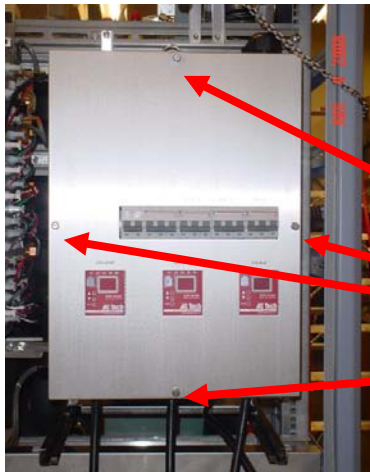
WARNING: Lytron Part # 205-0121-05 – GEHC #5156319 is equipped with Lenze AC Tech VFD's and has been produced since 2007. This part is **NON-RoHS Compliant**. **DO NOT USE** in **RoHS Compliant** HEC cabinets.

Tools Required:

1. #2 Philips screwdriver
2. Flat bladed screwdriver (large and small)
3. Hoist Service Kit 5196226

Removal Procedure:

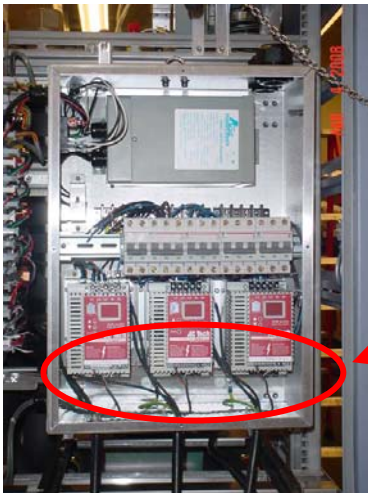
1. Lockout/Tag out HEC electrical power in accordance with GE procedures.
2. Attach Hoist Assembly GE 'Hoist Service Kit' instructions found on the service methods CD.
3. Remove Power Box cover by loosening 4 thumb screws (Refer to photo 1).



Loosen these 4 thumbscrews

Photo 1 Power Box exterior

4. Remove leads U, V, and W as well as GND connection from Blower Variable Frequency Drive (VFD), Gradient Coil (GC) VFD, and Power Electronics (PE) VFD.
5. Remove GC power cable, PE power cable, and the Blower power cable from the bottom of the Power Box.



Leads U, V, and W as well as GND from all three VFDs.

Photo 2 Interior of the Power Box

6. From the left side of the Power Box, remove the following cables going to the Signal Box (SB):
 - a. P1 115V power to the Signal Box.
 - b. P2 DI Kit
 - c. P19 GPMP VFD (GC Pump)
 - d. P20 PPMP-VFD (PE Pump)
 - e. P7 Blower VFD
7. Disconnect the incoming facility and outgoing cryo compressor power cored.
IMPORTANT: label or mark cables to ensure phases can be hooked up the same later.
8. Attach Hoist Assembly chain hook to the lifting hook of the EB.

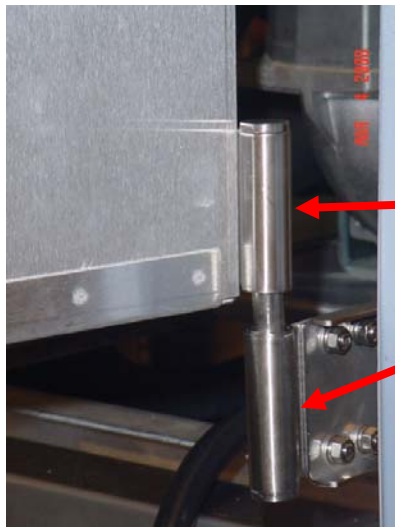


Picture 3 Hoist attachment to Power Box

9. Loosen upper & lower thumbscrews on the left side of the Power Box that secures it to the frame of the cabinet.
10. Pullout Power Box out until it meets the stops on the rail.

CAUTION: Ensure hoist ratchet handle is selected to "UP" position.

11. Raise the Power Box up off of its hinges in order to facilitate removal.



Lifting Power Box off
of the hinges

Photo 4 EB Hinge detail

12. While raising the Power Box, move it forward so that it clears the cabinet interior.

CAUTION: Ensure hoist ratchet handle is selected to "DOWN" position.

13. Lower Power Box until it rests on pallet or shipping container.



Photo 5 Lowering Power Box to floor

14. Note: Do not get rid of existing Power Box at this time; parts are required from it during the installation procedure. Specifically, remove the VFD EPM chip in each VFD. Since there are two different types (for different pumps), be mindful of which EPM chip goes with each VFD. These EPM chips will need to be re-installed in the new Power Box. Refer to next section for removal and installation of EPM chip.

End of Procedure

Installing Electrical Power Box (P/N: 205-0121-05)

WARNING: Lytron Part # 205-0121-05 - 5156319 is equipped with Lenze AC Tech VFD's and has been produced since 2007. This part is **NON-RoHS Compliant**. DO NOT USE in **RoHS Compliant** HEC cabinets.

Tools Required:

- #2 Philips screwdriver
- Flat bladed screwdriver (Large and small)
- Hoist Service Kit 5196226

Procedure:

1. Ensure LOTO is still applied as before.
2. Place Hoist ratchet handle in "DOWN" position. Let out enough chain to reach the floor.
3. Attach Hoist chain eyelet to the hoist eyelet located on the top of the Power Box.
4. Place Hoist ratchet handle into the "UP" position.
5. Slowly raise the Power Box up by taking out the slack in the chain.
6. Pull out the Power Box mounting rail with exposed hinge sleeves forward until it stops.
7. Continue raising up the Power Box until it can be just above the exposed hinge sleeves on the rail.
8. Place the hoist ratchet handle in the "DOWN" position.
9. Lower the Power Box until the hinge pins on the Power Box are fully seated in the hinge sleeves.
10. Lower chain slightly to induce slack in the chain line.
11. Remove the Hoist chain eyelet from the Power Box eyelet located on the top of the Power Box.
12. Push the Power Box back into the cabinet on the mounting rails.
13. Secure Power Box to HEC in two places by tightening the upper and lower thumbscrews on the left side of the Power Box.

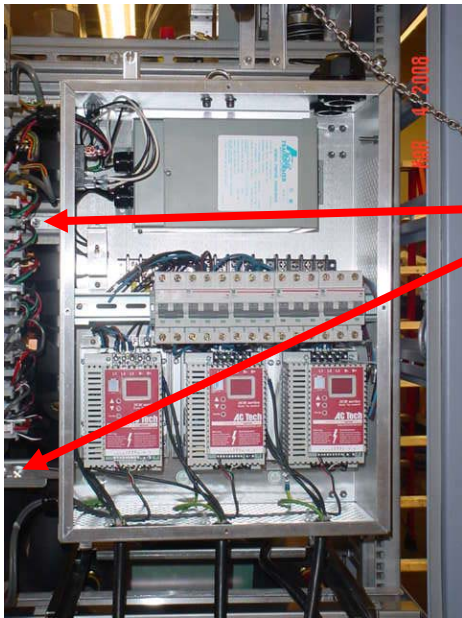


Photo 6

14. Install GC power cable, PE power cable, and the Blower power cable from the bottom of the Power Box.
15. Install leads U, V, and W as well as GND connection from Blower Variable Frequency Drive (VFD), Gradient Coil (GC) VFD, and Power Electronics (PE) VFD.

IMPORTANT: Ensure leads are installed in correct locations

16. From the left side of the Power Box, install the following cables going to the Signal Box:
 - a. P1 115V power to the Signal Box.
 - b. P2 DI Kit
 - c. P19 GPMP -VFD(GC Pump)
 - d. P20 PPMP-VFD (PE Pump)
 - e. P7 Blower VFD

IMPORTANT: The HEC may have pumps from different manufacturers. (Price or Ebara). It is imperative that the pump type be exactly matched with the correct VFD chip. The Pump FRU kit will contain the correct VFD chip for the pump, as well as pliers to facilitate removal and replacement of the VFD chip.

An incorrectly-matched pump and VFD chip will result in either high pressure or low flows, risking equipment damage and/or systems errors.

Price pumps/chips and the Ebara pump/chips can be identified according to the following photos and descriptions:



Price



Price Pump:

Irregular pump face shape.

Bolt at 9 o'clock position on pump face.

Price VFD Chip:

Identified with 'P'



Ebara



Ebara Pump:

Circular pump face shape.

Bolt at 12 o'clock on pump face.

Ebara VFD Chip:

Plain, colored chip.

NOTE: Suction hose is independent of pump type (Ebara cabinets may have white or blue hose type)

The gradient coil pump is on the left-hand side of the HEC and corresponds to the VFD in the left most location in the Power Box. The power electronics pump is on the right-hand side of the HEC and corresponds to the VFD in the middle location in the Power Box. The right most VFD is for the chilled air blower and is not impacted by this replacement procedure.

If the style of chip matches the style of pump according to the above descriptions, the next section can be skipped and you may proceed with the next step in the installation procedure.

The chip is replaced using the pliers shipped with the Pump kit. Remove the tape holding the chip in the front of the respective VFD in the Power Box. Grab with the pliers and pull straight out. To install the new chip, align it according to its keyed shape and press in fully with your finger. Place a piece of tape (in the FRU kit) over the newly-installed chip.

(Note: If the VFD chip is missing, lost or broken from the Pump FRU, see the GE FRU Manual and order the correct VFD chip.)



Photo 7 Remove the existing VFD chip with replacing the Ebara pump with a Price Pump.



Photo 8 Complete installation. (This HEC has an Ebara gradient coil pump and a Price power electronics pump, as noted by the 'P' on the VFD chip in the power electronics VFD)

17. Re-install VFD EPM Chips that were removed from previous Power Box.
18. Clear the Lockout/Tagout in accordance with GE procedures.
19. Re-start HEC and turn pump ON. Add new DVMR coolant as necessary.
20. Check for any leakage. Repair if necessary.
21. Perform a TPS reset.
22. Ensure HEC is in a scan-ready state and not in the low power mode; this is done by prescribing a scan at least to the point where the scan protocol.
23. Return to the HEC and view the displayed speeds on the two pump VFDs.
24. Note if the VFD is displaying 'F1' the VFD chip is either not installed or not seated fully.

Review the charts below and circle 'YES' or 'NO' below, indicating you have confirmed the correct speed based on pump and system configuration:

GRADIENT COIL PUMP:

<u>Pump Type</u> (identify as described above):	EBARA	PRICE
<u>Displayed Value</u> <u>On left-most VFD</u> (<u>'VFD GPMP'</u>)	50 +/- 1	61 +/- 1



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CORRECT GC Pump Speed? YES / NO

POWER ELECTRONICS PUMP:

Pump Type: (identify as described above):	EBARA		PRICE	
RF-amplifier type in PGR cabinet (Liquid cooled is identified by presence of red and blue cooling hoses hooked up to the front of the RF amplifier. Hoses will not be present on air cooled RF Amplifiers)	Liquid	Air	Liquid	Air
Displayed value on middle VFD ('VFD PPMP')	50 +/- 1	43 +/- 1	61 +/- 1	53 +/- 1

Correct PE pump speed YES / NO

If either of the pump speeds do not match the charts above, turn OFF the pumps and:

- A. Re-inspect that the VFD chips match the pump type.
- B. Ensure the HEC is in a scan-ready state and not in low-power mode.

If you do not have the proper VFD chips to match up with the pumps present, consult the FRU manual and order the required VFD chips.

- 25. Install main power leads through the top of the POWER BOX. Wire according to phasing as before. Reconnect cryo compressor power cord to respective circuit breaker.

IMPORTANT: Note incoming facility voltage and place voltage selector switch in the correct position (if questions, see GE installation procedure for 'System Power On'.

- 26. Replace cover on front of Power Box. Clear LOTO to HEC. Restart HEC according to GE Installation procedures for HEC power-on and startup.

- 27. If it is suspected that the phases were not hooked up correctly, the pumps and blower can be inspected for proper rotation – the rotation direction is marked on each unit with an arrow.



Photo 9 Complete Installation

End of Procedure



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820-0348 – Replacement Instructions for Electrical Power Box (PN 205-0121-33) GEHC # 5156319-100

Tools Required:

1. #2 Philips screwdriver
2. 3/32" Flat blade screwdriver
3. Wire strippers
4. 1/4" Flat blade screwdriver
5. Magnetic Pick-up tool (adjustable, telescopic)
6. Hoist Assembly (Lytron PN 205-0121-17, GE PN 5262029, 5196226)

WARNING: HIGH VOLTAGE PRESENT, LOTO FROM MAIN DISCONNECTS.

Identification of Power Box types

CAUTION: BEFORE YOU BEGIN, VERIFY THE POWER BOX TYPE YOU ARE REPLACING, AND THAT THE FRU HAS BEEN ORDERED AND RECEIVED

Power Box Comparison

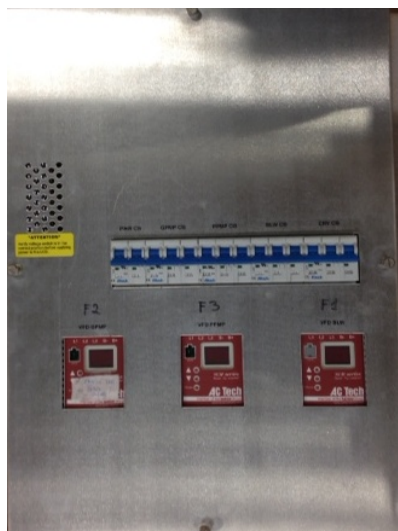
There are two types of variable frequency drives offered in HEC power boxes:

- Lytron Part # 205-0121-05 - 5156319 is equipped with Lenze AC Tech VFD's and has been produced since 2007. This part is **NON-RoHS Compliant**.
- Lytron Part # 205-0121-33 – 5156319-100 is equipped with Eaton VFD's and is released as a RoHS compliant design in 2014. This part **MUST** be used for **RoHS Compliancy**

While both boxes are functionally equivalent, speed scaling is accomplished differently in each box.

A summary of key differences are listed below:

- Lytron 205-0121-33 – 5156319-100 uses the same external dimension and cabinet interfaces as the original Lytron 205-0121-05 / 5156319 power box.
- To accommodate the larger Eaton VFD's, the internal box has been modified in the following way:
 - a. Circuit breakers moved up approximately 3.5"
 - b. 50/60 Hz switch moved up approximately 2"
 - c. Drives are now mounted on a plate that runs across the box. This is to maintain same distance between front of drive and inside of cover.
 - d. Ground studs 3X are now located on the floor of the box perpendicular to the floor.
 - e. Speed scaling is accomplished by wire jumper. (Speed scaling in AC Tech drives includes a switchable EEPROM chip.



Lenze / AC Tech Box Covered



Lenze / AC Tech Box Uncovered



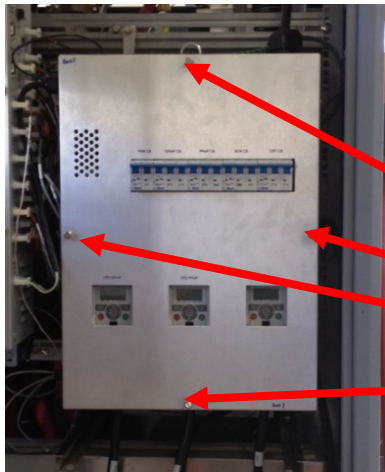
New Eaton Box Covered



New Eaton Box Uncovered

Removal Procedure:

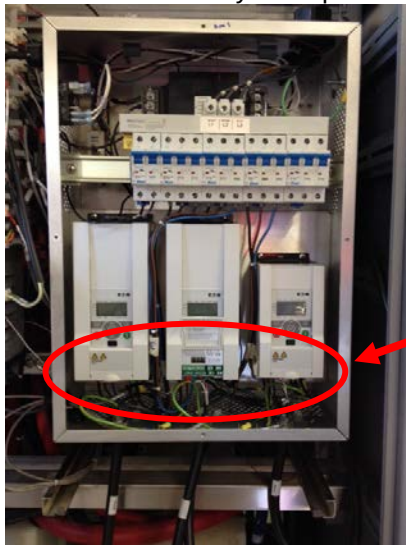
1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system from main disconnect in accordance with GE procedures.
3. Remove LCS7397GX door by opening door and removing door from its hinges.
4. Attach Hoist Assembly to cabinet per GE Procedures.
5. Remove Electrical Box cover by loosening 4 thumb screws (refer to photo below).



Loosen these 4 thumbscrews

Photo 1 Electrical Box exterior

6. Remove power leads U/T1, V/T2, and W/T3 as well as GND connection from Blower Variable Frequency Drive (VFD), Gradient Coil (GC) VFD, and Power Electronics (PE) VFD. Note each location for re-installation in the new electrical power box.
7. Remove GC power cable, PE power cable, and the Blower power cable from the bottom of the Electrical Box (Power Box).
8. Remove Cryo compressor power cable from circuit breaker. Note wire position and color.



Leads u/T1, V/T2, and W/T3 as well as GND from all three VFDs.

Figure 2 Interior of the Electrical Box

9. From the left side of the Power Box, remove the following cables going to the Signal Box (SB):
 - a. P1 115V power to the Programmable Logic Controller (PLC)
 - b. P2 DI Kit
 - c. P19 GPMP-VFD (GC Pump control)
 - d. P20 PPMP-VFD (PE Pump control)
 - e. P7 Blower VFD (Blower control)
10. Remove the main power cables that feed the Power Box. Note locations and wire colors of L1, L2, L3.
11. Attach Hoist Assembly chain hook to the lifting eyelet of the Power Box.



Figure 3 Hoist attachment to Electrical Box

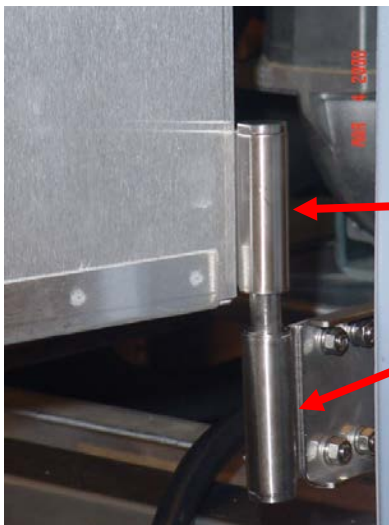


Figure 3a. Thumbscrew locations

12. Loosen upper & lower thumbscrews on the left side of the Power Box that secures it to the frame of the cabinet Refer to Figure 3a.
13. Pullout Power Box until it meets the stops on the rail.

CAUTION: Ensure hoist ratchet handle is selected to “UP” position.

14. Raise the Power Box up off of its hinges in order to facilitate removal.



Lifting Electrical Box off
of the hinges

Figure 4 Power Box Hinge detail

15. While raising the Power Box, move it forward so that it clears the cabinet interior.

CAUTION: Ensure hoist ratchet handle is selected to “DOWN” position.

16. Lower Power Box until it rests on pallet or shipping container.



Figure 5 Lowering Electrical Box to floor.

Installing Electrical Power Box (PN 205-0121-33) GEHC # 5156319-100

Tools Required:

1. #2 Philips screwdriver
2. 3/32" Flat blade screwdriver
3. Wire strippers
4. 1/4" Flat blade screwdriver
5. Magnetic Pick-up tool (adjustable, telescopic)
6. Hoist Assembly (Lytron PN 205-0121-17, GE PIN 5262029)

Procedure:

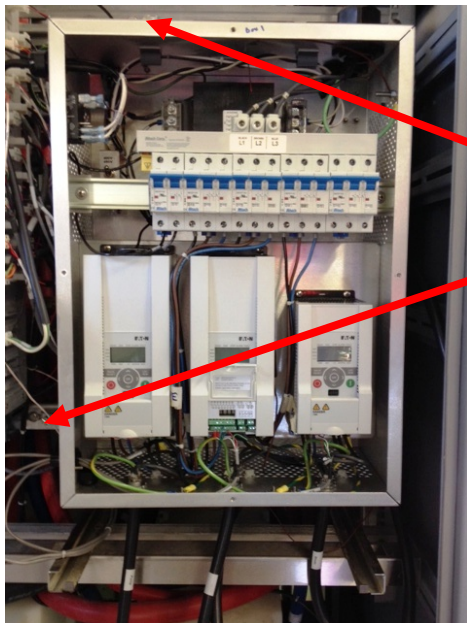
1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Remove LCS7397GX door by opening door and removing door from its hinges.
4. Attach Hoist Assembly to cabinet per GE procedures as in similar fashion described starting at Step 4 in **Removal Procedure** that started this document.
5. Place Hoist ratchet handle in "DOWN" position. Let out enough chain to reach the floor.
6. Attach Hoist chain hook to the lifting eyelet of the Power Box.
7. Place Hoist ratchet handle into the "UP" position.
8. Slowly raise the Power Box up by carefully pulling the chain and begin to raise the Power Box.

9. Pull out the Power Box mounting rail with exposed hinge sleeves forward until it stops.
10. Continue raising the Power Box until it can be just above the exposed hinge sleeves on the rail.
11. Place the hoist ratchet handle in the "**DOWN**" position.
12. Lower the Power Box until the hinge pins on the Power Box are fully seated in the hinge sleeves.



Place Electrical Box on the hinges

13. Lower chain slightly to induce slack in the chain line.
14. Remove the Hoist chain hook from the Power Box lifting eyelet located on the top of the Power Box
15. Push the Power Box back into the cabinet on the mounting rails.
16. Secure Power Box to machine frame using upper and lower thumbscrews locations. Refer to photo below.



Thumbscrew locations

17. Install GC power cable, PE power cable, and the Blower power cable from the bottom of the Electrical Box (Power Box). Ensure braided shield is folded over outer insulation and clamped in strain relief. Refer to photos below.



Pump/Blower Power Wires			
	VFD Terminal		Function
Lead Color	AC Tech	Eaton	Line Voltage
Black	L1	L1	Line 1 Power
Brown	L2	L2	Line 2 Power
Blue	L3	L3	Line 3 Power

18. Install leads U/T1, V/T2, and W/T3 as well as GND connection from Blower Variable Frequency Drive (VFD), Gradient Coil (GC) VFD, and Power Electronics (PE) VFD. Use table below to determine correct terminal location. Refer to table below.

Power Wires From Circuit Breakers to VFD's			
	VFD Terminal		Function
Lead Labeling	AC Tech	Eaton	Line Voltage
One	Pin U	Pin U/T1	Line 1
Two	Pin V	Pin V/T2	Line 2
Three	Pin W	Pin W/T3	Line 3

Table 1. Power Wires from CB to VFDs

Control Wires			
	VFD Terminal		Function
Lead Color	AC Tech	Eaton	
Red	1	8	Motor Start
Black	2	3	Ground
White	25	2	4-20 mA analog input

Table 1a – Control wire terminal locations

CAUTION: BEFORE YOU BEGIN, VERIFY THE PUMP TYPE AS EITHER A PRICE PUMP OR AND EBARA PUMP. IDENTIFICATION IS NECESSARY FOR COMPLETION OF NEXT SECTION.

Price pumps/chips and the Ebara pump/chips can be identified according to the following photos and descriptions



Price

Price Pump:

Irregular pump face shape.

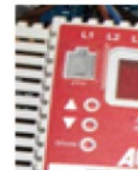
Bolt at 9 o'clock position on pump face.

Price VFD Chip:

Identified with 'P'



Ebara

Ebara Pump:

Circular pump face shape.

Bolt at 12 o'clock on

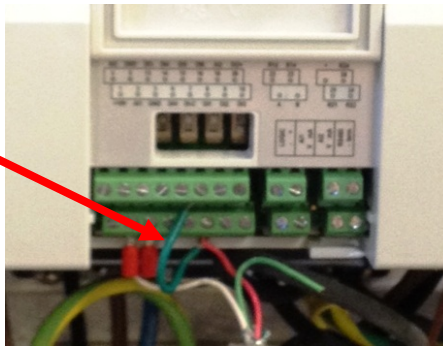
pump face.

Ebara VFD Chip:

Plain, colored chip.

19. If HEC cabinet has an EBARA pump, a jumper (qty 2 supplied in kit) will be needed on the EATON VFD to which the pump is attached. Add jumper wire from terminal 8 to terminal 16 if EBARA pump is present during installation of Power Box.

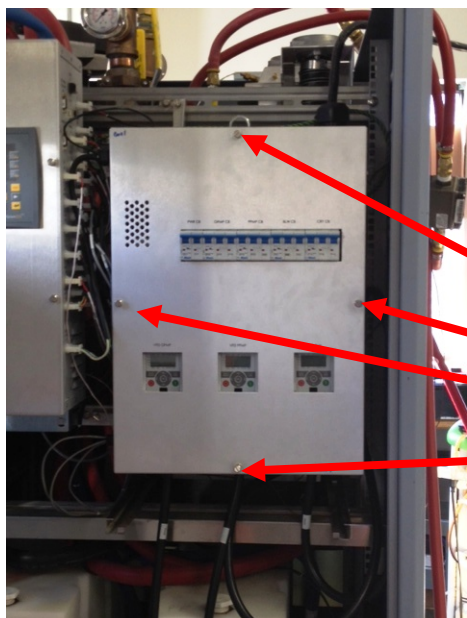
Jumper wire for
EBARA pump



20. From the left side of the Power Box, install the following cables going to the Signal Box (SB):
 - a. P1 115V power to the Programmable Logic Controller (PLC)
 - b. P2 DI Kit
 - c. P19 GPMP (GC Pump control)
 - d. P20 PPMP-VFD (PE Pump control)
 - e. P7 Blower VFD (Blower control)
21. Install main power leads through the top of the Power Box. Wire each breaker as required by electrical schematic Lytron PN 830-0239 and wiring diagram PN 830-0239-01.

CAUTION: OBSERVE PROPER SAFETY PROCEDURES WHEN INSIDE THE ELECTRICAL CABINET. HIGH VOLTAGE IS PRESENT WHEN COVER IS REMOVED.

22. Remove LOTO from HEC cabinet using standard GE procedures.
23. Turn on power in accordance with Lytron manual #820-0190.
24. To check proper rotation of pumps & blowers (including cryo compressor), turn on each breaker momentarily to observe the direction of the individual motor or blower. If rotation is reversed, verify wiring in Step 18. To correct a reverse rotation, switch two of the three phases for the appropriate breaker. If all motors and/or blowers rotate opposite, switch two of the three incoming power feeds.
25. Replace the cover to the Power Box by securing the 4 thumbscrews.



Secure these 4 places
with Thumbscrews

Figure 6 Front Cover in place (Thumbscrew locations).

26. Start system in accordance with startup procedures in Lytron manual #820-0190.

FINALIAZTION

1. Clear the Lockout/Tagout in accordance with GE procedures.
2. Re-start HEC and turn pump ON. Add new DVMR coolant as necessary.
3. Check for any leakage. Repair if necessary.
4. Perform a TPS reset.
5. Ensure HEC is in a scan-ready state and not in the low power mode; this is done by prescribing a scan at least to the point where the scan protocol has been downloaded.
6. Return to the HEC and view the displayed speeds on the two pump VFDs.

GRADIENT COIL PUMP:

<u>Pump Type</u> (Identify as described above):	EBARA	PRICE
<u>Displayed Value</u> <u>On left-most VFD</u> (‘VFD GPMP’)	50 +/- 1	61 +/- 1

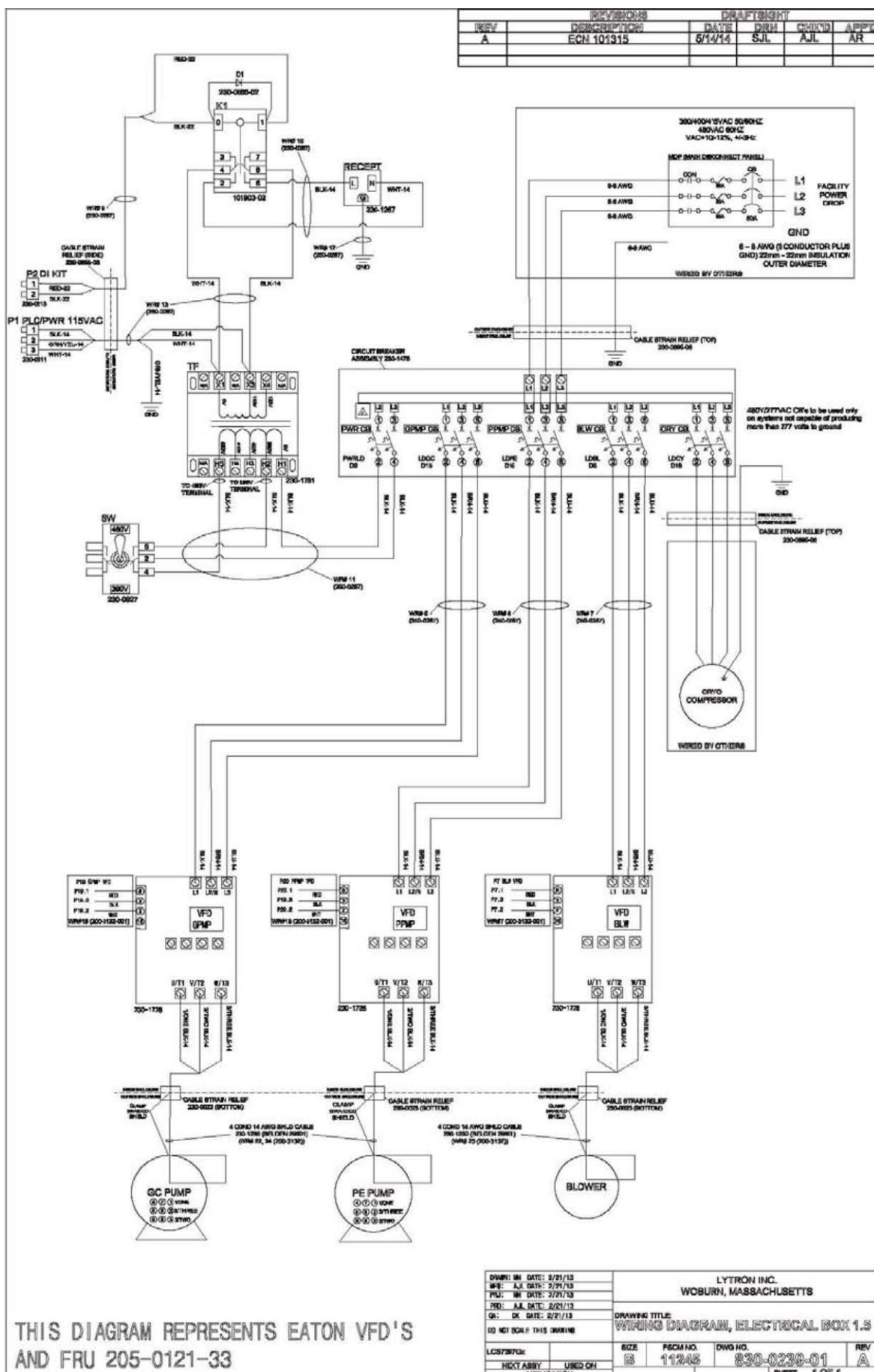
POWER ELECTRONICS PUMP:

Pump Type: (Identify as described above):	EBARA		PRICE	
RF-amplifier type in PGR cabinet (identify by presence of red and blue cooling hoses hooked up to front of RF-amplifier. Hoses will not be present on air cooled RF amplifiers.	Liquid	Air	Liquid	Air
Displayed value on middle VFD (‘VFD PPMP’)	50 +/- 1	43 +/- 1	61 +/- 1	53 +/- 1

7. If either of the pump speeds does not match the charts above, turn OFF the pumps and:
 - A. If the pump is an Ebara pump, check corresponding VFD for that jumper wire is installed and wired to the correct terminals.
 - B. Ensure the HEC is in a scan-ready state and not in low-power mode.

Further Troubleshooting information can be found in the GE Service Methods document “HEC Troubleshooting”.

END OF PROCEDURE



Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial Release	10/09/14	MN	HB



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820-0234 Replacement Instructions - Signal Box (PN 205-0121-06) GE PN 5269691

Tools Required:

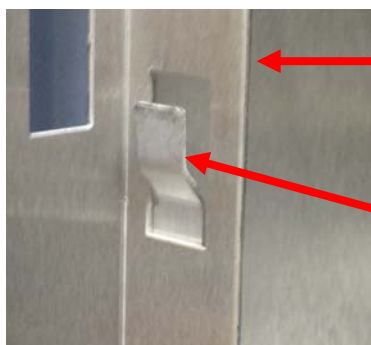
- Flat bladed screwdriver
- Wire Cutters(as needed to cut zip ties)

NOTICE

The signal box weighs approximately 14 lbs.

Removal Procedure:

1. Perform shutdown and lockout of system in accordance with the GE 'LOTO for Heat Exchanger Cabinet (HEC) procedure.
2. Open door to heat exchanger cabinet. If necessary, the door can be removed from its hinges by lifting up on the door.
3. Open front cover of signal box by loosening 6 thumbscrews.
4. Secure signal box cover to Signal box frame by hanging cover hinge holes over the Signal frame built-in tabs. Refer to Photo 1.



Signal box cover with hinge holes.

Signal box frame with built-in tab.

Photo 1 Detail showing mounting Signal Box cover onto built-in tabs of frame.

5. Remove all cables from their respective J## sockets -on the right side of the signal box.

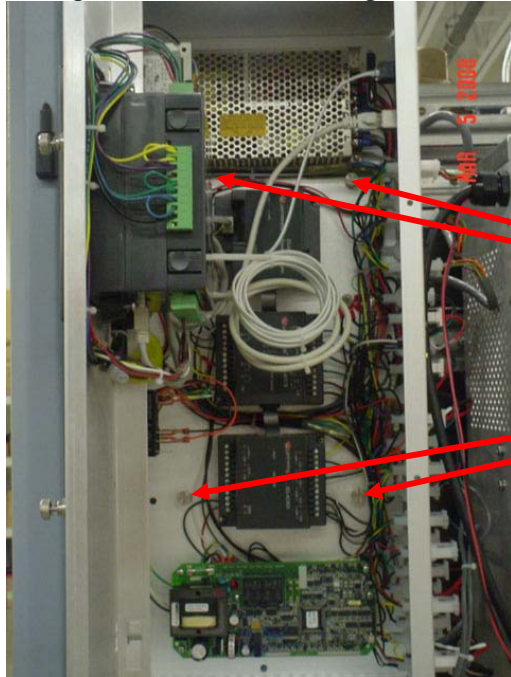


Photo 2 Detail of different connection locations



Photo 3 Detail showing HEC ID connector

6. While supporting the signal box with one hand, loosen 4 thumbscrews located inside the signal box that secure the signal box to the heat exchanger cabinet frame.



Thumbscrews – 1 in
each corner

Photo 4 signal box interior- thumbscrew location

7. Close the main cover to the signal box.
8. Crate signal box for shipment.

End of Procedure

Installation Procedure:

1. Remove new signal box front cover by loosening 6 screws.
2. Position the signal box so that the mounting holes in the main cabinet frame align with thumbscrews on the signal box.
3. Tighten one thumbscrew at a time to just finger tight. Tighten thumbscrew and then its opposite counterpart until all 4 thumbscrews are secure. Refer to attached photo 4.
4. Plug connectors J1 through J28 into the appropriate jack on the right side of the signal box. Refer to attached photo 1.
 - o Depending on revision of signal box, the location of the connectors J1 through J28 may be different (see attached photo 2). Later revision signal box will have a J29 'HEC ID' plug which will only have a connector plug on G6001EN cabinets (see attached photo 3). J10 'DI-2' and J27 'BLW-VEL' may also remain blank.
 - i. NOTE: When replacing the new signal box in a cabinet that had an earlier revision signal box some cable ties may have to be cut. The power box door will have to be opened to cut the J1 zip ties to gain enough length on the cable to make the connection (see attached photo 5).



Photo 5 Close-Up of J1 zip ties in power box

5. Replace front cover on the signal box.
6. Remove Lockout/Tagout in accordance with GE procedures.
7. Start system per Lytron manual #820-0190.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
C		1/24/2013		



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820-0235 – Install Instructions Liquid Pressure Transducer (XDCR) (P/N: 205-0121-07) GE P/N: 5267643

Tools Required:

- Adjustable or open end Wrench
- Loctite™ 565
- Loctite™ 7649 Primer

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Drain system in accordance with Lytron manual #820-0190.
4. Locate Pressure Transducer (xdcr) to remove.
5. Remove electrical connection by pressing down on xdcr retention tab and carefully extracting the connection. (Refer to Photo 1)



Photo 1

6. Using an adjustable or open end wrench, remove the pressure xdcr from the system. Leave the cable in place.

Install Instructions for Liquid Pressure Transducer (XDCR)

1. Prepare threaded end of transducer (xdcr) with Light coating of Loctite 7649 primer and allow to dry (approx 2 min) then apply a bead of Loctite 565 around the threads starting 1 thread from end of xdcr.
2. Carefully screw xdcr into manifold.
3. Using an adjustable or open end wrench, tighten to hand tight.
4. Connect signal harness to xdcr.
5. Fill system in accordance with Lytron manual #820-0190.
6. Check for leaks around unit just installed.
7. Return machine to running condition per Lytron manual #820-0190.
8. Check pressure reading associated with pressure transducer just replaced.

820-0235 Rev C

Released



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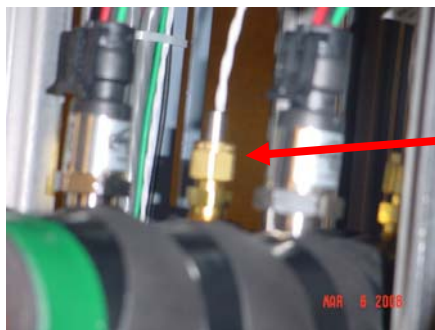
820-0236 – Install Instructions for Liquid Temperature Sensor (RTD) (P/N: 205-0121-08) GE P/N: 5268665

Tools Required:

- (2) 9/16" wrenches or slip wrench
- Loctite™ 565
- Tie Wraps (Lytron P/N 101601-01)

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Drain system. Drain system in accordance with Lytron manual #820-0190.
4. Locate Liquid Temperature Sensor (RTD) to remove. There are temperature detectors located in the GC Supply, PE Supply, and Facility Water. (Refer to Photo 1)



Liquid RTD
(Typical)

Photo 1

5. Disconnect the associated RTD from the Signal Box.
6. Cut tie wraps that secure the RTD from the Signal Box to the RTD position.
7. While holding the inner nut with a 9/16" wrench, remove RTD by loosening RTD outer nut with a second 9/16" wrench.
8. Remove nut and ferrule from temperature sensor by sliding off. Leave inner nut attached to manifold.

End of Procedure

Install Instructions for Liquid Temperature Sensor (RTD)

1. Install new ferrule and barrel connector by sliding them on over the temperature sensor.
2. Discard new bottom nut.
3. Prepare screw end of Liquid Temperature Sensor (RTD) with a moderate amount of Loctite 565 or equivalent.
4. Carefully screw RTD into manifold.
5. Using a 9/16" wrench, tighten to hand tight.
6. Route RTD cable along original path to the Signal Box. Tie wrap in place.
7. Connect signal harness to Signal Box.
8. Fill system in accordance with Lytron manual #820-0190.
9. Check for leaks around unit just installed.
10. Return machine to running condition per Lytron manual #820-0190.
11. Check reading associated with Liquid Temperature Sensor just replaced.

End of Procedure



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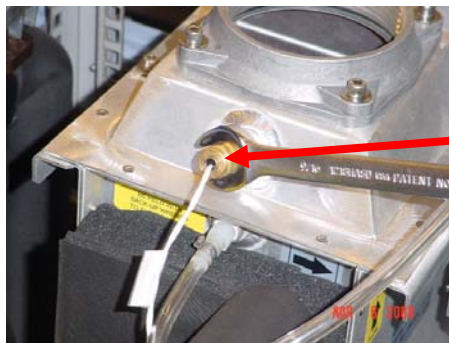
820-0238 – Install Instructions Air Temperature Sensor (RTD) (P/N 205-0121-10) GE P/N: 5269689

Tools Required:

- (2) 9/16" wrenches or slip wrenches
- Loctite™ 565
- Tie Wraps (Lytron PN 101601-01)

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Locate Air Temperature Sensor (RTD) to remove. Sensor is located on exhaust of heat exchanger.



Air RTD

Photo 1

4. Disconnect the associated RTD from the Signal Box.
5. Cut tie wraps that secure the RTD routed from the Signal Box to the RTD position.
6. While holding the inner nut with a 9/16" wrench, remove RTD by loosening RTD outer nut with a second 9/16" wrench.
7. Remove nut and ferrule from temperature sensor by sliding off. Leave inner nut attached to exhaust plenum.

End of Procedure

Install Instructions for Air Temperature Sensor (RTD)

1. Install new ferrule and barrel connector by sliding them on over the temperature sensor.
2. Discard new inner nut.
3. Prepare screw end of Air Temperature Sensor (RTD) with a moderate amount of Loctite 565 or equivalent.
4. Carefully screw RTD into manifold.
5. Using a 9/16" wrench, tighten to hand tight.
6. Route the RTD cable along original path to the Signal Box. Tie wrap in place.
7. Connect signal harness to Signal Box.

8. Check for air leaks around unit just installed.
9. Return machine to running condition per Lytron manual #820-0190.
10. Check reading associated with Air Temperature Sensor just replaced.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	APPRV'D
A	Initial Release	6/11/2008	H. Bufe
B	Updated	7/1/08	M. Golini
C	ECN 17682 changed GE P/N to 5269689	3/3/11	H. Bufe



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820-0239 – Install Instructions for Tank Level Switch (P/N: 205-0121-11) GE P/N: 5267644

Tools Required:

- 1" wrench or slip wrench
- 1 ½" wrench or slip wrench
- Loctite™ 739 or equivalent

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Drain system. Drain system in accordance with Lytron manual #820-0190. Alternate draining through the pump housing is also acceptable.

NOTE: Removal of lower Tank Level Switch requires removal of the pump discharge hose to the tank.

4. Remove pump discharge hose from left side of tank (looking from the front of the system).



Remove

Photo 1 Tank inlet location

5. Disconnect the switch leads from Tank Level switch.
6. Remove the Tank Level switch using a 1" wrench or slip wrench.

End of Procedure

Install Instructions:

1. Apply a generous amount of Loctite 739 or equivalent around the threads of the tank level switch.
2. Screw the tank level switch into the tank.
3. Allow some time for the Loctite to setup (approx. 10min).
4. Attach leads to tank switch.
5. Re-attach pump discharge to tank.
6. Fill system in accordance with Lytron manual #820-0190.
7. Start system normally.
8. Refer to PLC screen to verify proper level switch operation.

End of Procedure



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820-0240 Install Instructions Resistivity Sensor Kit (PN 205-0121-12) GE PN 5267649

Tools Required:

- Adjustable wrench or Slip Wrench
- Nylon Plug Fitting (Lytron PN 430-0277)
- Teflon™ Tape
- Loctite™ 565
- Tie Wraps

NOTE: For systems with 2 Resistivity sensors, please refer to Procedure below as well as the procedure starting on page 2 of this instruction for removal of duplicate sensor.

Procedure:

1. Shutdown and Lockout/Tag out system in accordance with GE procedures.
NOTE: The Gradient Pump should be off before proceeding - verify by looking at variable frequency drive for GPMP, the display should be blank.
2. Drain the gradient coil reservoir by following GE service procedures.
3. Unplug the failed DI sensor from the appropriate jack on the side of the Signal Box. Refer to the table and photo below:

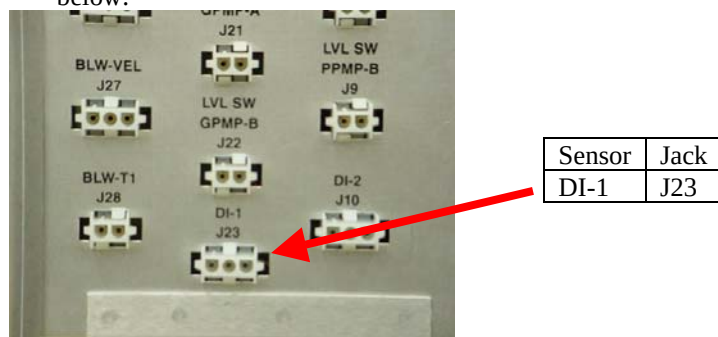


Illustration 1 Detail of DI Sensor Jack on side of Signal Box

4. Remove the DI sensor cable from its routing from the tank to the right side of the Signal Box.
5. Using an adjustable wrench, unscrew the DI Sensor from the tank.



DI Sensor

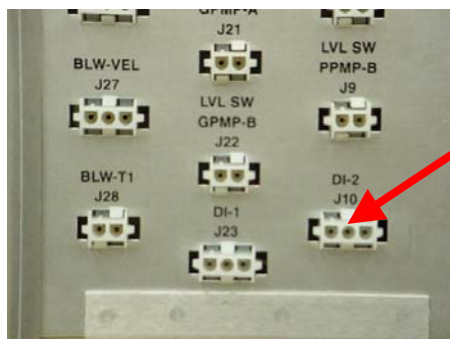
Illustration 2 DI Sensor Location

Installation of Resistivity Sensor Kit.

1. Apply adequate amount of Loctite™ 565 around the threads of the new sensor.
2. Using the adjustable wrench, screw new sensor into the tank. Allow at least 30 minutes for Loctite™ to set up.
3. After making a service loop by the tank, tie wrap cable in place along route to Signal Box.
4. Plug new sensor into DI-1 (J23) on the side of the Signal Box.
5. Fill the system with coolant previously removed using the funnel provided in the HEC and remove LOTO.
6. Restore power to the HEC and restart pumps and blower if they don't come on automatically (hold up and push F1, F2, F3).
7. Check for leaks around sensor just replaced.
8. Monitor system for proper operation.

Procedure (For systems having 2 Resistivity Sensors in Tank):

1. Shutdown and Lockout/Tag out system in accordance with GE procedures.
NOTE: The Gradient Pump should be off before proceeding - verify by looking at variable frequency drive for GPMP, the display should be blank.
2. Drain the gradient coil reservoir by following GE service procedures.
3. Unplug the failed DI sensor from the appropriate jack on the side of the Signal Box. Refer to the table and photo below:



Sensor	Jack
DI-2	J10

Illustration 3 Detail of DI Sensor Jack on side of Signal Box

4. Remove the DI sensor cable from its routing from the tank to the right side of the Signal Box.
5. Using an adjustable wrench, unscrew the DI Sensor from the tank.



DI Sensor
DI-2

6. Refer to ***Installation of Resistivity Sensor Kit*** for installation steps.

End of Procedure



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820-0242 – Install Instructions Fuse Kit (PN 205-0121-16) GE PN 5266827

Tools Required:

- No special tools required

Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Locate Fuse on DI board in Signal Box (Refer to picture).

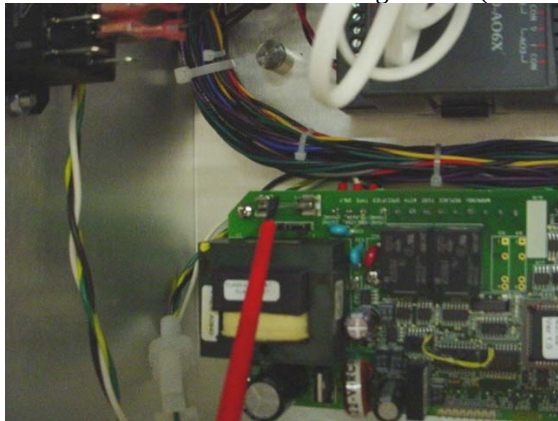


Photo 1 DI Board Fuse location

4. Replace fuse with same value from kit.
5. Locate fuse for mixing valve next to P5 on the top left inside the Signal Box. (Refer to picture).
6. Replace fuse with same value from kit.

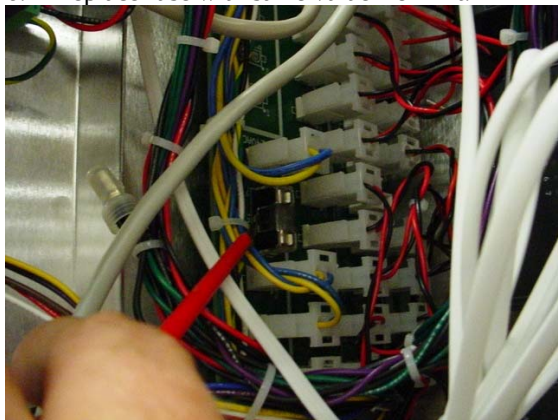
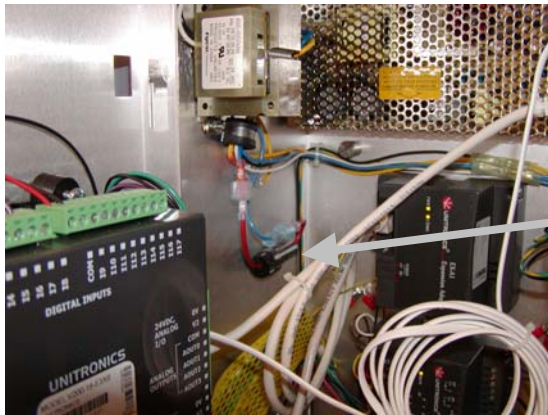


Photo 2 Fuse location on inside of Signal Box

NOTE: On older models of machine there is a fuse for the mixing valve located in the upper left inside of Signal Box. Refer to photo below for location.



Older style in-line
fuse holder

Photo 3 Older Style In-line Fuse holder.

7. Clear Lockout/Tag out of system per GE procedures.
8. Start system in accordance with Lytron manual #820-0190.
9. Observe system for proper operation - specifically in relation to fuse replaced.

End of Procedure



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820-0244 – Install Instructions Pressure Differential Sensor Board Kit (P/N: 205-0121-14) GE P/N: 5267648

Tools Required:

- Philips #2 screwdriver or equivalent
- 4mm Allen wrench or equivalent
- Small flat blade screwdriver
- Tie Wraps (Lytron P/N 101601-01)

Removal Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Remove air line to Pressure Differential Sensor Box at fitting marked "High" on the top of the box. Mark High line to differentiate from Low line.
4. Remove air line to Pressure Differential Sensor Box at fitting marked "Low" on the top of the box. Mark Low line to differentiate from High line.
5. Remove cover to the Pressure Differential Sensor Box to reveal internal components using the Philips screwdriver.
6. Use a 4mm Allen wrench to remove the Pressure Differential Sensor Box from the frame. Retain screws for re-installation.
7. Remove the following wires from the marked 3 position terminal board using a flat bladed screwdriver (Refer to photo 1 below).
 - a. "+" – Red Wire
 - b. "-" – White
 - c. "O" – Black

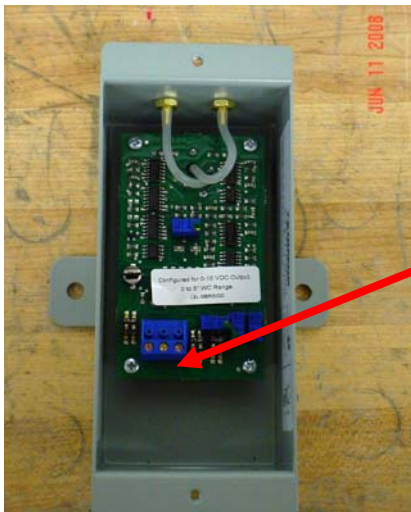


Photo 1 Pressure Differential Box

8. Remove wires through strain relief.

End of Procedure

Installation of Pressure Differential Sensor Board Kit

1. Remove cover from the new Pressure Differential Sensor Box.
2. Install new strain relief on new box.
3. Thread 3 wire cables (Red, White, and Black) through strain relief from the bottom.
4. Attach each wire to screw terminals in the following order:
 - a. "+" – Red Wire
 - b. "-" – White
 - c. "O" – Black
5. Visually check dip switches S1, S2, and S3 against the following table:

	DIP 1	DIP 2
S1	ON	OFF
S2	ON	OFF
S3	ON	OFF

6. Mount box using 4mm screws saved from the removal of the box.
7. Attach air line marked "High" to Pressure Differential Sensor Box air connection on the top marked "High".
8. Attach air line marked "Low" to Pressure Differential Sensor Box air connection on the top marked "Low".
9. Replace cover to Pressure Differential Sensor Box.
10. Return system to normal operation using GE guidelines.
11. Monitor system for proper operation.

End of Procedure



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820-0241 – Filter Replacement for Cryocooler Compressor Valves P/N: 205-0121-13 (GE P/N: 5269685 or 5269685-2)

NOTE: The Kit contains 3 styles of filter elements, depending on the configuration of the strainer valve. The filter elements are all 150 micron (100 mesh) screens. Configuration #1 is used in the plastic strainer; configurations #2 and #3 are used in the Jomar valves while configuration 3 uses the smaller of the two included screens.

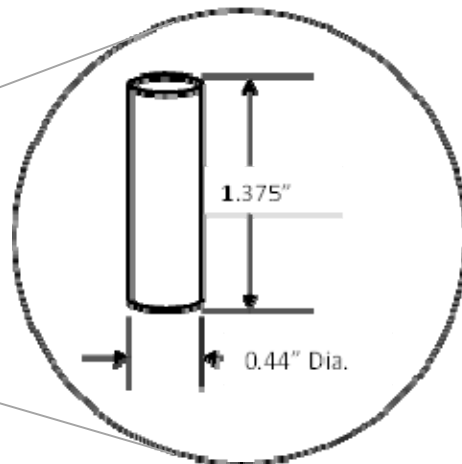
Procedure Timing:

20 min for all Configurations

Tools Required:

- Adjustable wrench
- Snap Ring Pliers utilized in configurations #2 and #3 (included in HEC or Cryogen Supply Header FRU Kit)

Configuration #1 (Old Style Filter Replacement):



1. Refer to the illustration below, and actuate the three yellow-handled valves in the order described:
 - a. Open lower right valve: Callout "A"
 - b. Close middle valve: Callout "B"
 - c. Close left valve: Callout "C"

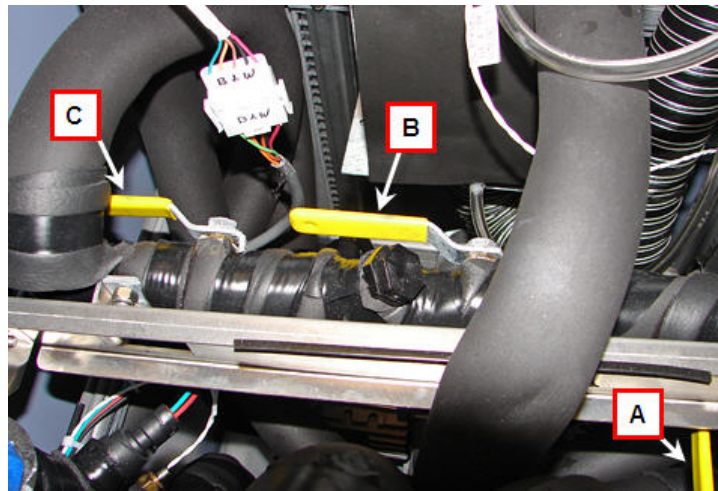


Illustration 1: Configuration #1 Location of Cryocooler Compressor Valves

2. Remove the black plastic hex cap shown below.



Illustration 2: Configuration #1 - Location of Hex Cap over Filter

3. Pull out the wire mesh cylindrical screen from the exposed valve and replace it.
4. Proceed as follows to complete the process:
 - a. Match the filter type to what was removed and install a clean filter, and tighten down securely.
 - b. Replace the hex cap.
 - c. Open the left valve (C).
 - d. Open the middle valve (B).
 - e. Close the right valve (A).

Configuration #2 and #3(New Style Filter Replacement):



Configuration #2
.985" Long x .70" Dia.



Configuration #3
.78" Long x .685" Dia.

NOTE: Configuration #2 and configuration #3 are identical except for length of assembly and filter size. Procedure for changing the filter is identical for both configurations. For illustration purposes, only configuration#2 is shown below.

1. Close the Jomar Filter Ball Valve in the header by turning the handle perpendicular to the pipe of the supply header. (Open and closed positions are shown below.)

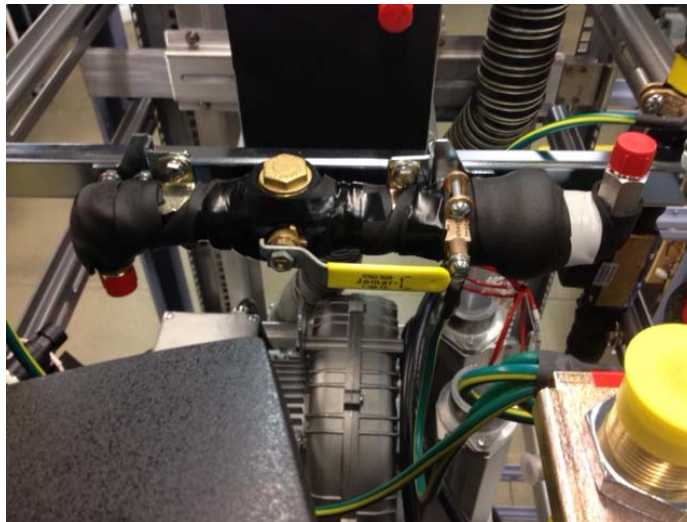


Illustration 3: Configuration #2 Filter Ball Valve Open



Illustration 4: Configuration #2 Filter Ball Valve Closed

2. Unscrew the top of the ball valve using an adjustable wrench.
3. Remove the snap ring from the valve with the snap ring pliers to expose the strainer basket (The snap ring pliers are included with the HEC).



Illustration 5: Configuration #2 - Using Snap Ring Pliers to remove spring clip.

4. Remove the strainer basket. Select the matching strainer from the FRU kit and replace (reference photos at beginning of section for strainer differences).



Illustration 6: Configuration #2 Removing Strainer Basket from Valve

5. Select the appropriate matching snap ring from what was removed and replace..
6. Replace the top of the ball valve, and tighten slightly past hand tight to ensure the O-ring is engaged.

END OF PROCEDURE

REVISION HISTORY		
REV	ECN / DATE	REASON FOR CHANGE
A	6/11/2008	Initial Release
B	ECN# 17216 / 6/04/2010	Documenting changes requested by GE
C	ECN# 100952 11/08/2013	Change Jomar valve to new style.



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820-0295 – Install Instructions for Cryo Fitting Kit GE PN 5357155

Tools Required:

- Cryo Fitting Kit (5357155)
 - o Kit Includes
 - Fitting
 - Teflon™ tape
 - Wrench
 - Insulating tape
 - Install Instructions

Removal of Installed Fittings

Procedure:

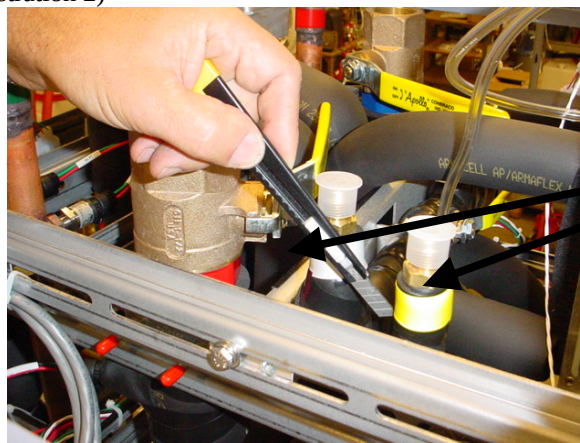
1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Shut off Facility water Supply at the top of the unit. Drain Facility water per GE procedures.

Note: Before shutting off the facility water supply, check with the facility maintenance to determine what measures are needed. The goal is to prevent dead head of pumps or other possible issues.

4. Disconnect cryo hoses from the HEC by loosening and disconnecting them at the swivel flare fitting.

CAUTION: There may be residual coolant in the lines, which may drip out during removal. Any coolant that escapes should be cleaned up with an absorbent towel.

5. Using a utility knife, cut away the black insulating covering around the fitting (see Illustration 1).
6. Using the pipe wrench supplied in the FRU kit to hold the pipe, loosen and remove the installed cryo fittings with an adjustable wrench. (See Illustration 2)



Cut away
insulation here

Illustration 1 Detail of Removing Insulation around fitting.

Place pipe wrench
here.



Illustration 2

Detail of Removing Fittings

Installation of Cryo Fitting Kit

1. Clear threads of tubes prior to cryo fitting installation.
2. Wrap threads with 2 turns of Teflon™ tape prior to installation of fittings.
3. Screw fitting onto pipe until finger tight.
4. Using the pipe wrench supplied in the FRU kit to keep the base from moving, tighten the fitting over the Teflon™ tape until finger tight and then 2-3 turns past finger tight.
5. Tape around the black insulating tape to secure the insulation.
6. Reconnect the cryo compressor hoses at top of HEC by tightening the female flare fitting onto the newly installed male flare cryo fitting. Hold the male cryo fitting with a wrench while tightening. Tighten female flare to finger tight and then ¼ turn past finger tight.
7. Open the Facility water Supply by opening valves on top of the system. Check for leaks before proceeding.
8. Check for any leakage around fittings just installed and correct before proceeding.
9. Replace insulating tape around joint.
10. Return system to operation.

Note: Pipe wrench is magnetic and a consumable. For safety reasons, it should be discarded once procedure is complete.



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820-0327 – HEC Variable Frequency Drive EPM Chip Replacement (PN 205-0121-27) GE PN 5443037

Purpose: This procedure is only required when a pump has been changed. The EPM chip modifies the frequency range of the VFD to match corresponding pump in HEC. The Price pump must have EPM chip with “P” in the corresponding VFD. Ebara pumps must be controlled by a VFD that does not have a letter/label on the EPM chip.

NOTE: The Ebara pump EPM chip replacement procedure is the same as for the Price pump with the exception of not having a label on the chip.

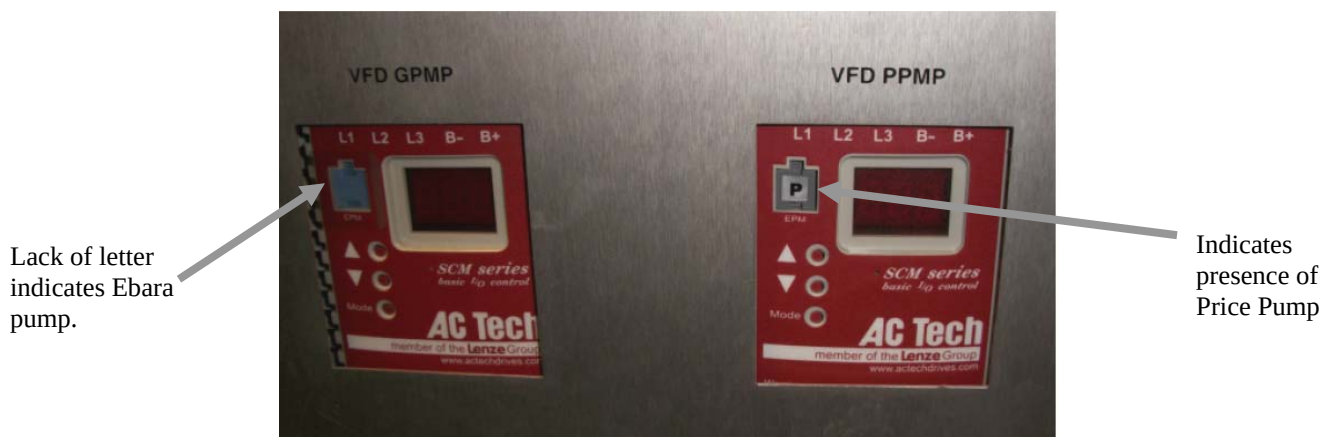


Photo 1 Detail of VFD EPM Chip in different Pump Style VFDs

Tools Required:

1. Pliers for VFD Chip (included with VFD Chip FRU Kit, Lytron PN 205-0212-27)

IMPORTANT: The HEC may have pumps from different manufacturers. (Price or Ebara). It is imperative that the pump type be exactly matched with the correct VFD chip. The Pump FRU kit will contain the correct VFD chip for the pump, as well as pliers to facilitate removal and replacement of the VFD chip.

An incorrectly-matched pump and VFD chip will result in either high pressure or low flows, risking equipment damage and/or systems errors.

Procedure:

1. Perform shutdown of system in accordance with Lytron manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.

Price pumps/chips and the Ebara pump/chips can be identified according to the following photos and descriptions:



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Price



Price Pump:

Irregular pump face shape.

Bolt at 9 o'clock position on pump face.

Price VFD Chip:

Identified with 'P'



Ebara



Ebara Pump:

Circular pump face shape.

Bolt at 12 o'clock on pump face.

Ebara VFD Chip:

Plain, colored chip.

NOTE: Suction hose is independent of pump type (Ebara cabinets may have white or blue hose type)

The gradient coil pump is located on the left-hand side of the HEC and corresponds to the VFD in the left most location in the Power Box. The power electronics pump is on the right-hand side of the HEC and corresponds to the VFD in the middle location in the Power Box. The right most VFD is for the chilled air blower and is not impacted by this replacement procedure.

If the style of chip matches the style of pump according to the above descriptions, the next section can be skipped and you may proceed with the next step in the installation procedure.

3. Remove the tape holding the chip in the front of the respective VFD in the Power Box.
4. Grab with the pliers and pull straight out.
5. Place removed chip in plastic bag to avoid mixing chips.
6. To install the new chip, align it according to its keyed shape and press in fully with your finger. Place a piece of tape (in the FRU kit) over the newly-installed chip.

(Note: If the VFD chip is missing, lost or broken from the Pump FRU, see the GE FRU Manual and order the correct VFD chip.)



Figure 6 Remove the existing VFD chip with replacing the Ebara pump with a Price Pump.



Figure 7 Complete installation. (This HEC has an Ebara gradient coil pump and a Price power electronics pump, as noted by the 'P' on the VFD chip in the power electronics VFD)

FINALIZATION

1. Clear the Lockout/Tagout in accordance with GE procedures.
2. Re-start HEC and turn pump ON. Add new DVMR coolant as necessary.
3. Check for any leakage. Repair if necessary.
4. Perform a TPS reset.
5. Ensure HEC is in a scan-ready state and not in the low power mode; this is done by prescribing a scan at least to the point where the scan protocol.
6. Return to the HEC and view the displayed speeds on the two pump VFDs.

Note if the VFD is displaying 'F1' the VFD chip is either not installed or not seated fully.

Review the charts below and circle 'YES' or 'NO' below, indicating you have confirmed the correct speed based on pump and system configuration:

GRADIENT COIL PUMP:		
<u>Pump Type</u> (Identify as described above):	EBARA	PRICE
<u>Displayed Value</u> <u>On left-most VFD</u> <u>('VFD GPMP')</u>	50 +/- 1	61 +/- 1

CORRECT GC Pump Speed? YES / NO

POWER ELECTRONICS PUMP:

Pump Type: (Identify as described above):	EBARA		PRICE	
RF-amplifier type in PGR cabinet (Liquid cooled is identified by presence of red and blue cooling hoses hooked up to the front of the RF amplifier. Hoses will not be present on air cooled RF Amplifiers)	Liquid	Air	Liquid	Air
Displayed value on middle VFD ('VFD PPMP')	50 +/- 1	43 +/- 1	61 +/- 1	53 +/- 1

Correct PE pump speed

YES / NO

7. If either of the pump speeds do not match the charts above, turn OFF the pumps and:

- A.Re-inspect that the VFD chips match the pump type.
- B.Ensure the HEC is in a scan-ready state and not in low-power mode.

If you do not have the proper VFD chips to match up with the pumps present, consult the FRU manual and order the required VFD chips.

End of Procedure



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820-0297 – Install Instructions Cryo Supply Header (P/N 205-0121-19) GE P/N 5376555

Cryo Supply Header Replacement

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	1 hour	15 mins

2 Overview

The Cryo Supply Header is a manifold located in the top portion of the Heat Exchanger Cabinet that routes facility-supplied coolant to the cryogen compressor and allows for filtering of the coolant.

- There are currently three configurations of this assembly in the product:
 - Configuration 1: Original style header contains a 150-micron plastic Y-filter that is removable with a hex head cap and can be seen in [Illustration 1](#).
 - Configuration 2: A version of the supply header simplified with a 150-micron basket filter and can be seen in [Illustration 2](#).
 - Configuration 3: The 150-micron screen is physically smaller than used in Configuration 2. The current version can be seen in [Illustration 3](#).
- This procedure outlines replacement of all 3 configuration headers.

Illustration 1: Configuration 1 Cryo Supply Header

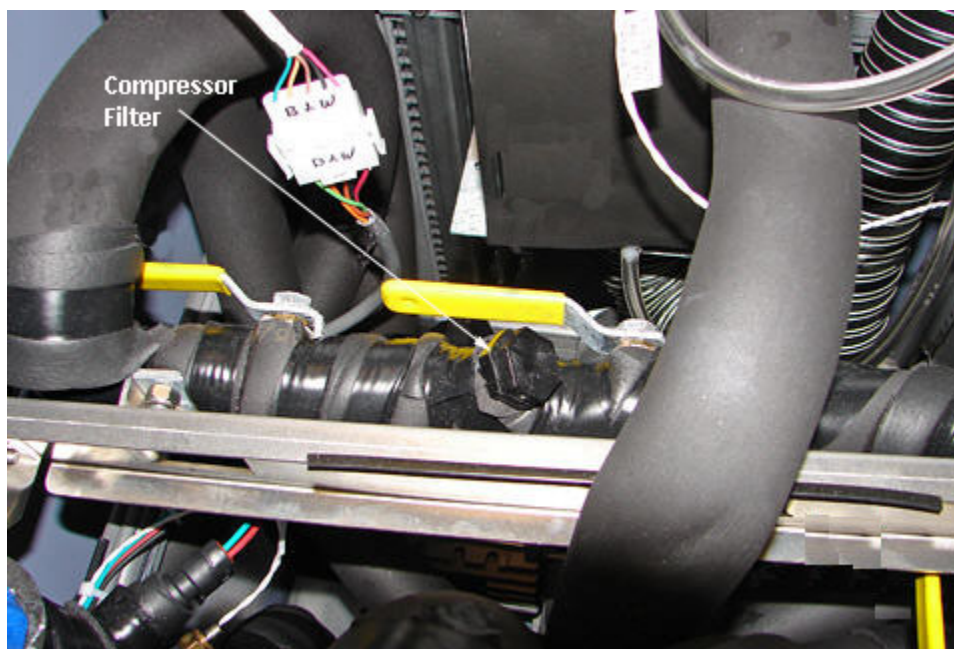


Illustration 2: Configuration 2 Cryo Supply Header

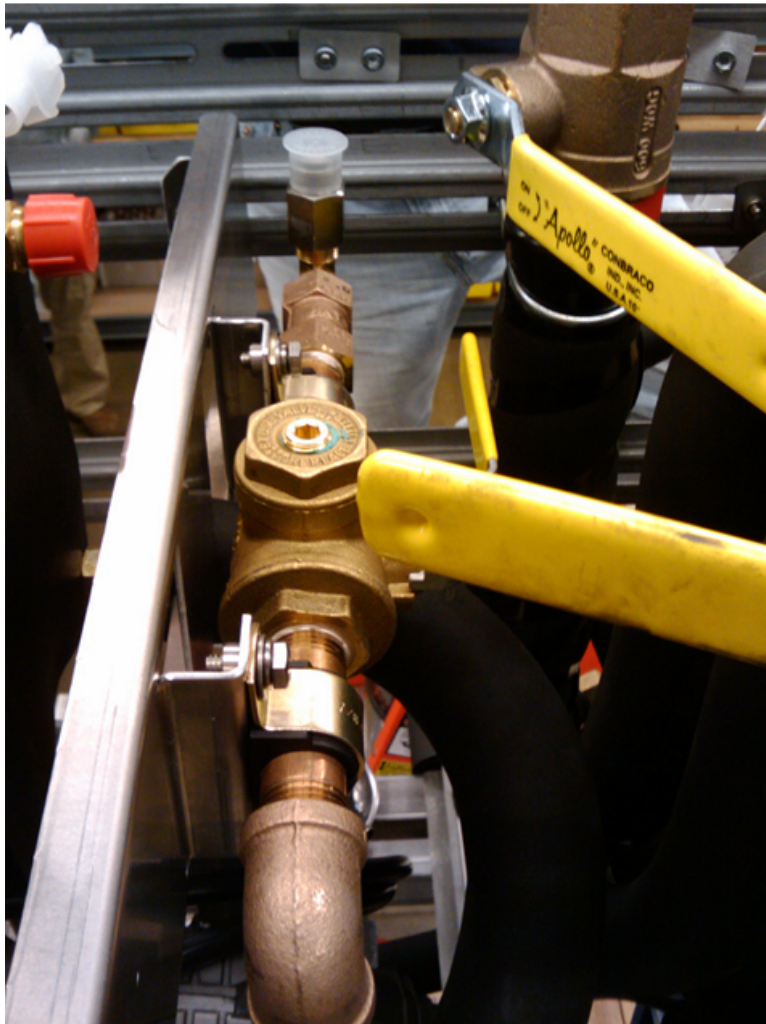
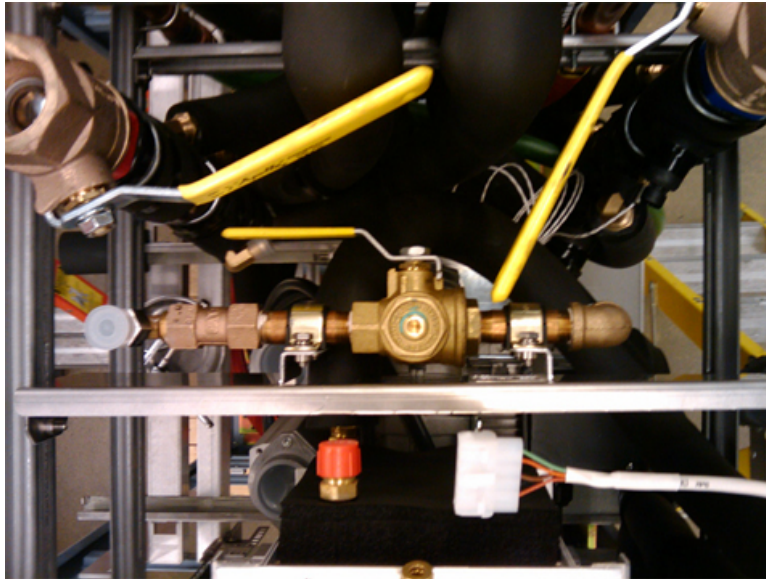
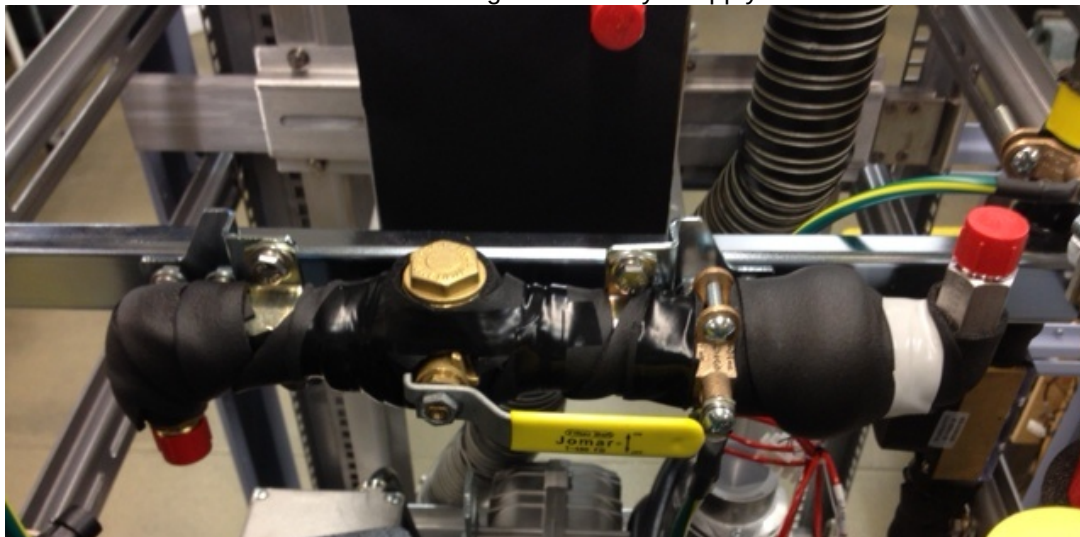


Illustration 3: Configuration 3 Cryo Supply Header



3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectively	Part#	Manufacturer
Adjustable Wrench	2	-	-	-
Service Platform	1	-	-	-
Coolant Draining Kit	1	-	-	-

3.2 Safety



PRESSURIZED LIQUID!

THERE IS PRESSURIZED LIQUID INSIDE THE HEC CABINET WHEN ENERGIZED.

Perform LOTO on the Heat exchanger cabinet. See the MR Service Safety Manual, PN 5452735.

4 Procedure

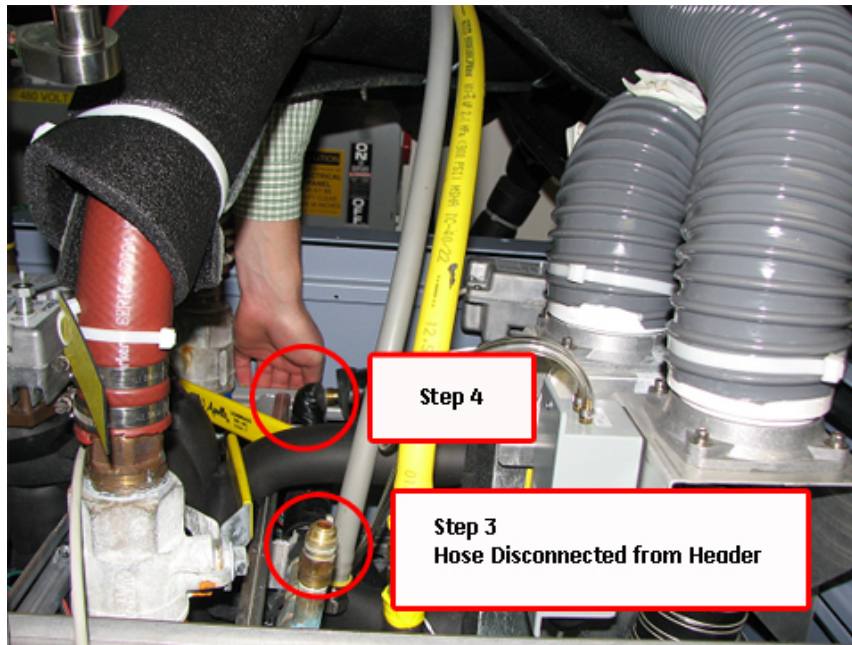
4.1 Cryo Supply Header Removal

NOTE: This section outlines removal for Configuration 1, but the process is the same all 3 configurations.

1. Perform [LOTO for the HEC](#). See MR Service Safety Manual, PN 5452735.
2. Drain the facility water from the cabinet per the [Coolant Draining](#) procedure following GE Procedures.

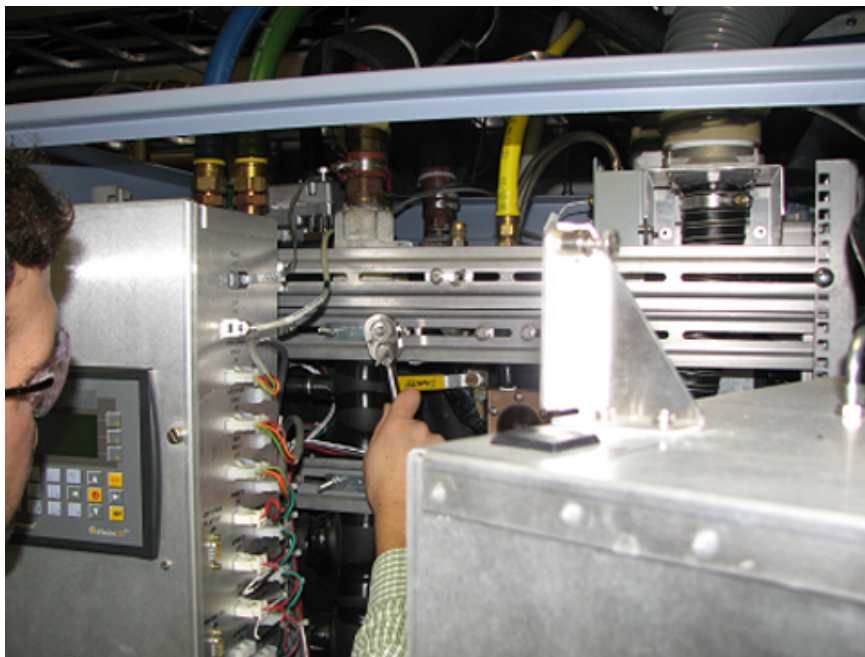
3. Remove the gray hose from the front side of the header by loosening the swivel joint. (See [Illustration 3](#).)

Illustration 3: Front Hose Removal



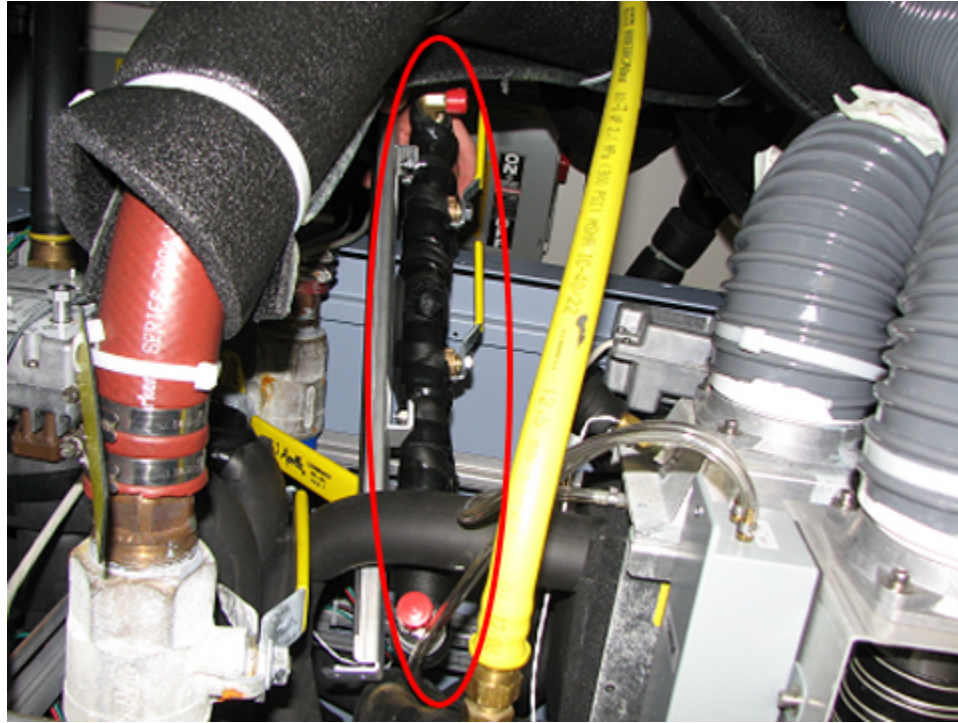
4. Remove the insulation and hose from the back side of the header by loosening the swivel joint.
5. Move the power box from the front of the cabinet out of the way by loosening the upper and lower screws that attach the box to the cabinet.
6. Loosen 4 bolts (2 in the front of the cabinet, 2 in the back) that secure the cryo supply header assembly to the cabinet rails. (See [Illustration 4](#).)

Illustration 4: Removing Cryo Supply Header Screws



7. Remove cryo supply header from top of cabinet. (See [Illustration 5](#).)

Illustration 5: Cryo Supply Header Removal



4.2 Cryo Supply Header Installation

1. Orient the cryo supply header with the hex cap facing up and attach rear hose to cryo header. **These should be finger tightened plus an additional quarter turn; with two wrenches, one for holding lower fitting in place and the other to tighten upper fitting.**
2. Fasten new cryo supply header by securing 4 previously removed screws to the cabinet rails.
3. Attach previously removed front hose and secure firmly to prevent leaking. **These should be finger tightened plus an additional quarter turn; with two wrenches, one for holding lower fitting in place and the other to tighten upper fitting.**
4. Return the power box to its original location by securing two screws originally loosened securing it to the cabinet brackets.
5. Ensure all valves are open to allow coolant flow through facility plumbing of the HEC.
6. Remove LOTO from HEC. See the MR Safety Manual, PN 5452735.

5 Finalization

1. Check the fittings for leaks and retighten if necessary.
2. Confirm coolant flow to the cryo compressor on the PLC box.
3. Confirm that cryogen compressor flow is greater than or equal to 7 L/min (pressure is less than 6 bar at the facility inlet).

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG.APPR'D	SRV APPRV'D
A	Initial Release	5/5/11	APT	HB
B	ECN #100952	11/08/13	NG	HB



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820-0300 – Installation Instructions for Facility Supply & Return Header AAVID PN: 205-0121-21 & 205-0121-22 (GEHC PN: 5392905 & 5392906)

Purpose: The purpose of this document is to demonstrate how to remove & replace the facility and supply headers for the HEC machine.

Scope: The document applies to HEC cabinet.

Resources Required: 2 hours (1 person)

Tools Required:

- AAVID Document #820-0233 (Electrical Box Installation Instructions)
- AAVID Document #820-0234 (Signal Box Installation Instructions)
- AAVID Document #820-0232 (Blower Installation Instructions)
- 1-1/4 in. Nonferrous Open-end or Adjustable Wrench
- 7/16 in. Nonferrous Deep-well Socket or Wrench
- Wet/Dry Vacuum or Bucket for Draining

Review entire procedure before beginning.

Removal Procedure (Facility Return Header):

1. Perform shutdown and apply Lockout/Tagout to the system in accordance with the GE *LOTO for Heat Exchanger Cabinet (HEC)* procedure.
2. Open the door to HEC. If necessary, the door can be removed from its hinges by lifting up on the door. Remove any grounding straps associated with the door.
3. Close the ball valve on the Facility Return Header.
4. Drain Facility side plumbing per GE draining procedures. Ensure any valve up-stream of the Facility Return Header ball valve is closed. Lockout/Tagout the effected ball valves.

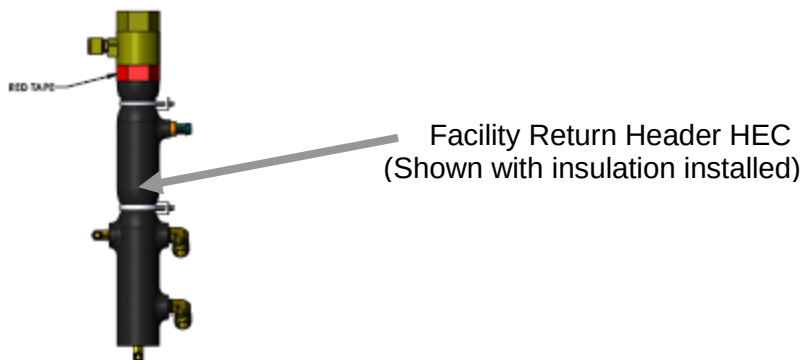


Illustration 1: Detail of Facility Return Header (shown without insulation for clarity.)

NOTE: When removing items weighing more than 35 lb., the hoist system must be utilized.

5. Moving/removing the Signal Box and Electrical Box to gain access to connection points on Facility Return header will be required. Refer to AAVID Procedures #820-0233 (Electrical Box) and #820-0234 (Signal Box) for details on removing these parts.
6. Remove the pressure transducer(s) from the header. Refer to Illustration #3 for header location.
7. Using the 1-1/4 in. wrench, remove the Facility Return Connection from the 3-way valves at the header. (Insulation may need to be removed at the header and the hose drained). Mark each 3-way valve hose to facilitate re-installation.
8. Remove the hose coming from the Cryo flow meter. Mark hose for identification purposes and to facilitate re-installation.
9. Remove the bottom bracket that secures the header to the HEC frame.
10. While supporting the header with one hand, loosen and remove the upper support bracket that secures the header to the HEC frame.
11. Remove Facility Return Header from the HEC.

Removal Procedure (Facility Supply Header):

1. Perform shutdown and apply Lockout/Tagout to the system in accordance with the GE *LOTO for Heat Exchanger Cabinet (HEC)* procedure.
2. Open the door to HEC. If necessary, the door can be removed from its hinges by lifting up on the door. Remove any grounding straps associated with the door.
3. Close the ball valve on the Facility Supply Header.
4. Drain Facility side plumbing per GE draining procedures. Ensure any valve up-stream of the Facility Supply Header ball valve is closed. Lockout/Tagout the effected ball valves.

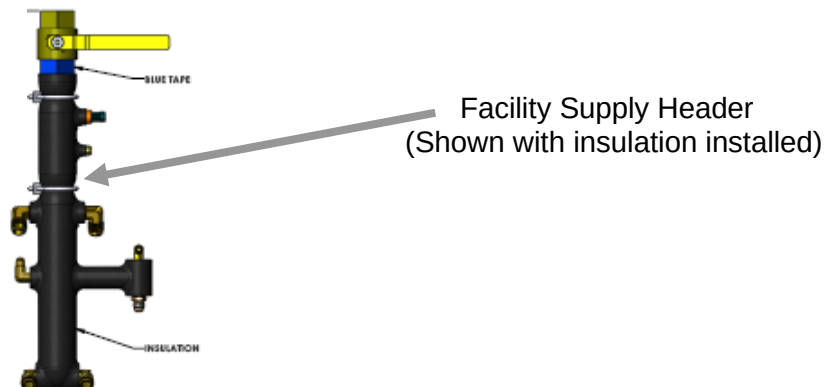


Illustration 2 Detail of Facility Supply Header

NOTE: When removing items weighing more than 35 lb., the hoist system must be utilized.

5. Moving/removing the Signal Box, Electrical Box, and Blower to gain access to connection points on Facility Supply header will be required. Refer to AAVID Procedures #820-0233 (Electrical Box), #820-0234 (Signal Box), and #820-0232 (Blower) for details on removing these parts.
6. Remove the pressure transducer(s) from the header. Refer to Illustration #3 for header location.
7. Using the 1-1/4 in. wrench, remove the Facility Supply Connections from the 3-way valves at the header. (Insulation may need to be removed at the header and the hoses drained). Mark each 3-way valve hose to facilitate re-installation.
8. Remove the rear cover to the HEC to gain access to the brackets that secure the Facility Supply Header to the HEC frame.
9. Remove the bottom bracket that secures the header to the HEC frame.
10. While supporting the header with one hand, loosen and remove the upper support bracket that secures the header to the HEC frame.
11. Remove Facility Return Header from the HEC.

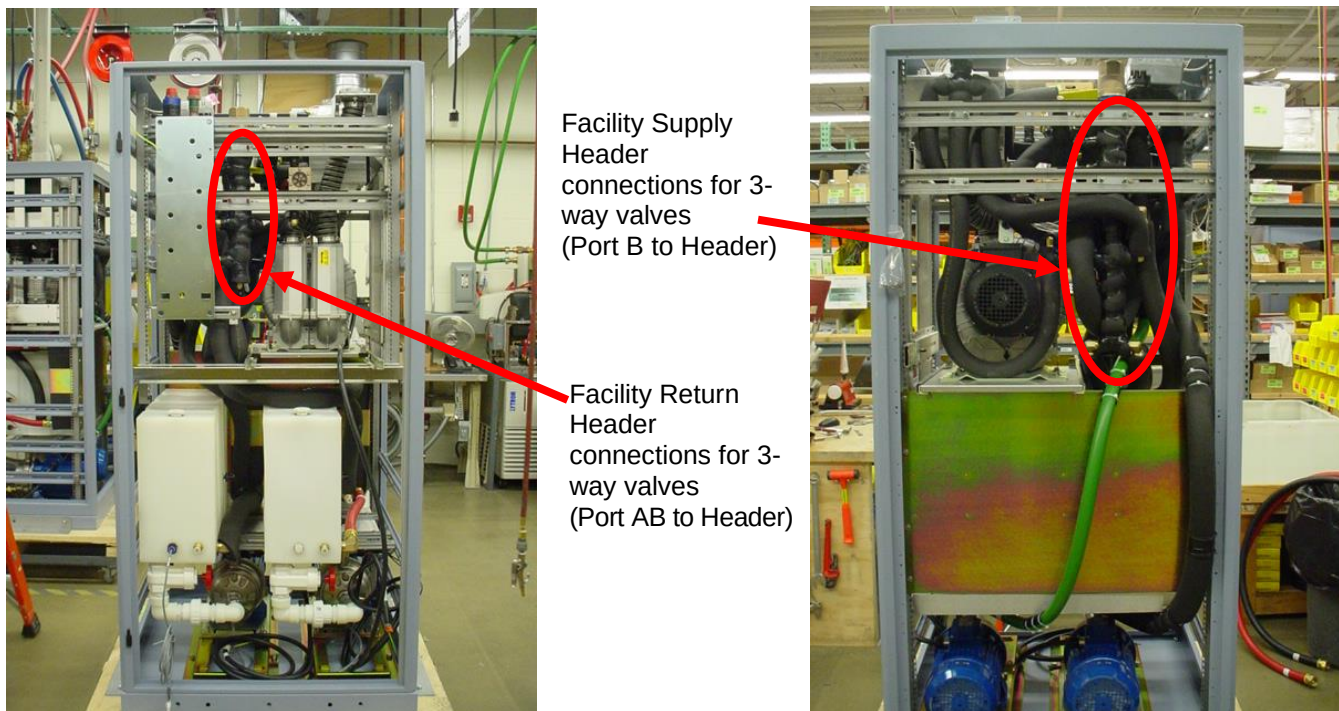


Illustration 3: Cabinet with Signal Box and Electrical Box Removed (Front and Rear Views)

Installation Procedure (Facility Return Header):

1. Secure the new header to the frame by using the 2 brackets saved from previous removal of old header. Tighten nuts until hand tight and then turn and additional $\frac{1}{4}$ turn.
2. Insert pressure transducer cable to the plug on the left side of the header.
3. Screw in the pressure transducer into the back of the header. Tighten until hand tight and then turn and additional $\frac{1}{4}$ turn.
4. Using a 1 $\frac{1}{4}$ " wrench, attach the previously marked 3-way valve connections. Tighten until hand tight and then turn and additional $\frac{1}{4}$ turn. (Another wrench may be needed behind the 1- $\frac{1}{4}$ in. wrench to protect the heat exchanger flange.)
5. Attach the Cryo flow meter connection to the bottom of the header. Tighten until hand tight and then turn and additional $\frac{1}{4}$ turn. (Another wrench may be needed behind the 1- $\frac{1}{4}$ in. wrench to protect the heat exchanger flange.)
6. Re-install the Signal Box (AAVID document #820-0234) and Electrical Box if removed (AAVID document #820-0233)
7. Remove any tubing utilized for draining in the previous steps.
8. Open the Facility Return Ball valve.
9. Remove Lockout/Tagout on the system in accordance with GE procedures.
10. Remove any Lockout/Tagout on any valve upstream of the Facility Return Header ball valve.
11. Restart the system.
12. Check for leaks around the manifolds of the mixing valve(s) that were replaced.
13. Replace door if removed. Connect any grounding straps that may have been removed.

Installation Procedure (Facility Supply Header):

1. Secure the new header to the frame by using the 2 brackets saved from previous removal of old header. Tighten nuts until hand tight and then turn and additional $\frac{1}{4}$ turn.
2. Insert pressure transducer cable to the plug on the left side of the header.
3. Screw in the pressure transducer into the back of the header. Tighten until hand tight and then turn and additional $\frac{1}{4}$ turn.
4. Using a 1 $\frac{1}{4}$ " wrench, attach the previously marked 3-way valve connections. Tighten until hand tight and then turn and additional $\frac{1}{4}$ turn. (Another wrench may be needed behind the 1- $\frac{1}{4}$ in. wrench to protect the heat exchanger flange.)
5. Re-install the Signal Box (AAVID document #820-0234), Electrical Box (AAVID document #820-0233), and Blower (AAVID document #820-0232) if removed.
6. Remove any tubing utilized for draining in the previous steps.
7. Remove Lockout/Tagout on the system in accordance with GE procedures.
8. Remove any Lockout/Tagout on any valve upstream of the Facility Supply Header ball valve.
9. Restart the system.
10. Check for leaks around the connections to the Facility Supply Header from all sources. Correct if found.
11. Re-install the rear cover to the HEC.
12. Replace front door if removed. Connect any grounding straps that may have been removed.

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial	5/2/19	HB	NZ
B	ECN# 202315 BOYD REBRANDING	4/13/20	JAH	HB
C	ECN#203207 REMOVE HEC VERSION NUMBER	5/11/2021	SM	MJG

820-0342 – Install Instructions for Chilled Air Heat Exchanger

Boyd PN 205-0121-26 GE PN 5858305

Boyd PN 205-0121-39 GE PN 5858306

Scope:

These instructions apply to the specific HEC serial numbers noted below.

205-0121-26 (GE PN 5858305) HEC Serial Number (S/N) below 897257-01

205-0121-39 (GE PN 5858306) HEC Serial Number (S/N) 897257-01 and above.

Tools Required:

- Philips and slotted screwdrivers
- Adjustable wrenches
- Allen wrench set

Time Required: Approximately 0.5 hour

NOTE: During this procedure the facility water may be shut off for approximately 1/2 hour. There will be no coolant flow to the Cryo-compressor during this time.

Procedure:

1. Perform shutdown of system in accordance with BOYD manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Shut facility water supply by closing the two ball valves located at the top of the HEC.
4. Drain coolant from facility water loop.

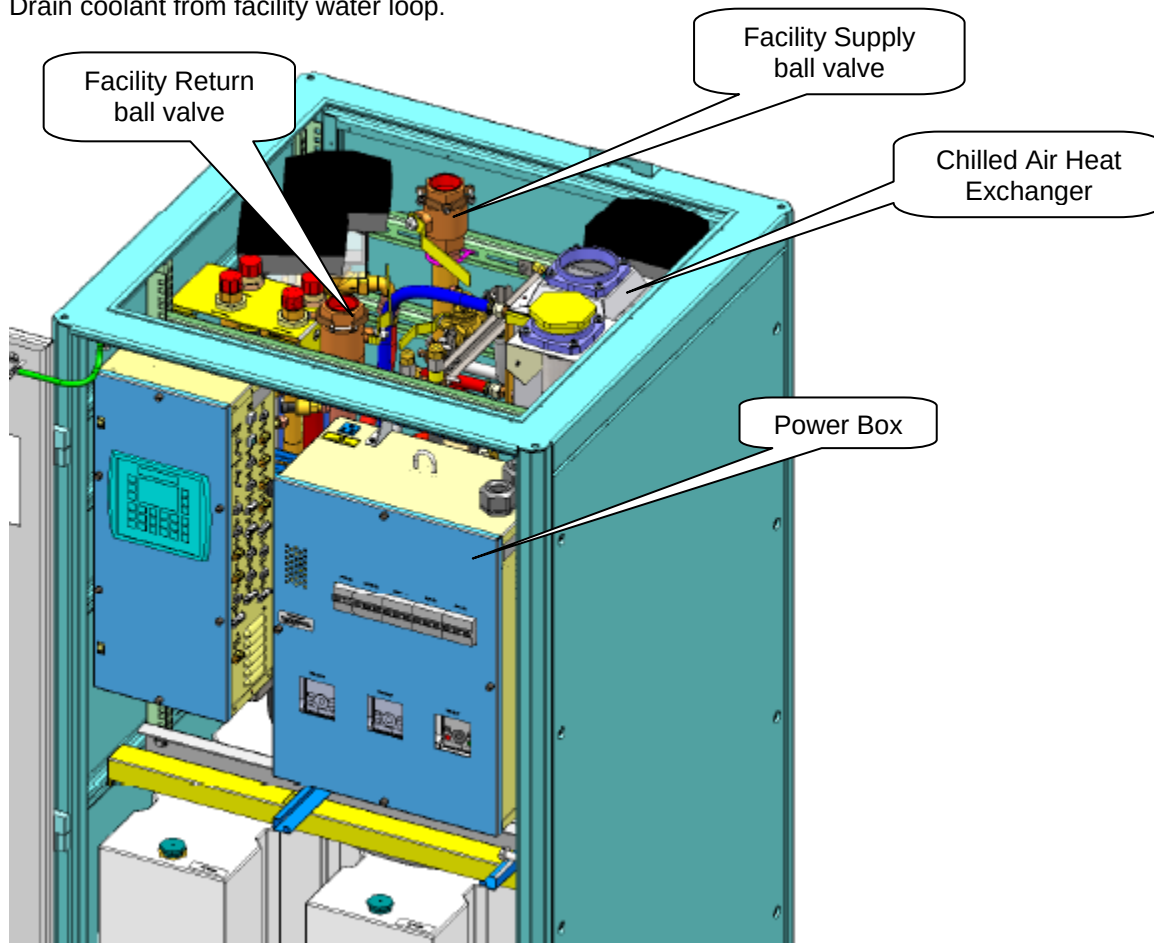


Figure 1

5. To access the heat exchanger and plenum from the front of the cabinet you may choose to slide the power box out of the way. Refer to instruction for FRU kit 5156319-100.
6. Remove the clamps securing plenum to the top and bottom of the Chilled Air Heat Exchanger.
7. Carefully loosen and remove the Swagelok connector securing the RTD to the top/side of the chilled air heat exchanger. Use two wrenches, one to stabilize, and one to remove.
CAUTION: Do not lose the Swagelok parts as they'll be used during installation process.
8. **NOTE:** Mark the water connections to the heat exchanger as they MUST go back onto the replacement in the correct order. Carefully remove the water connections to the Chilled Air heat exchanger. Use two wrenches when removing. One to stabilize the copper pipe from the heat exchanger and the 2nd to unscrew the flare fitting.

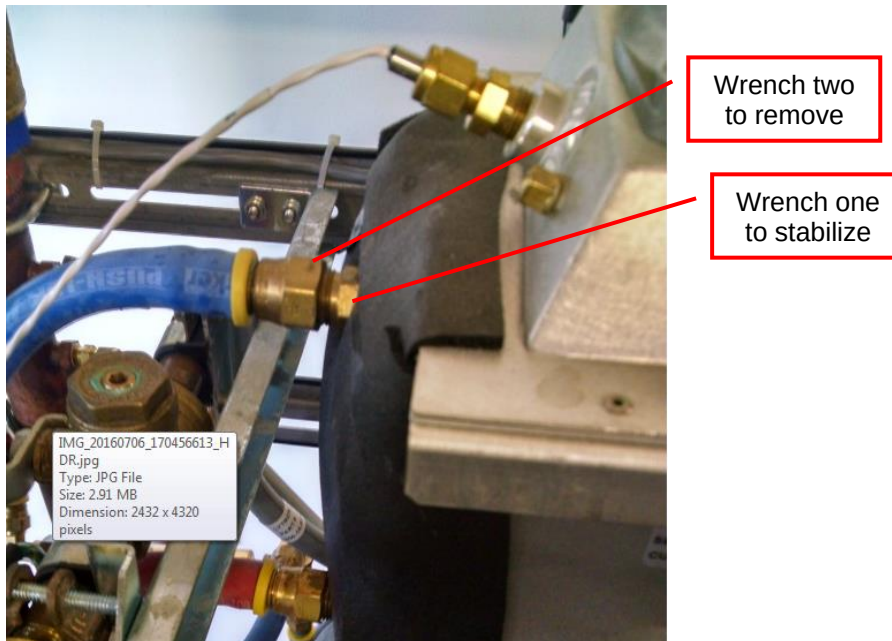


Figure 2

9. Remove the two socket head cap screws, washers and lockwashers securing the heat exchanger to the bracket as shown in figure 3. Save hardware as it will be used for installation.



Figure 3

10. Install replacement heat exchanger using hardware removed in step 9.

11. Connect the water hoses to the heat exchanger removed earlier. Ensure the location of the hose is the same as when removed in step 8. Be sure to use two wrenches. One to stabilize the connection of heat exchanger and the other to tighten the hose fitting. Tighten 'hand tight' plus ¼ turn.
12. Install the RTD with Swagelok removed in step 7.
13. Install the plenum to top and bottom of the heat exchanger assembly.
14. SLOWLY open 'Facility In' ball valve and check for leaks at the water connections.
15. Slowly open the 'Facility Out' ball valve and check for leaks at the water connections.
16. If power box was removed during repair process, place back into position and secure with captive hardware.
17. Power on cabinet and test per GE installation procedures.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial Release	7/7/16	JAH	HB
B	ECN# 202315 BOYD REBRANDING	4/13/20	JAH	HB
C	Revised per ECN# 203478	12/28/21	SM	SM



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820-60026-000 – Removal & Install Instructions GC/PE HX ASSY FRU KIT (BOYD PN 205-0121-28, 205-0121-40 & GE PN 5858303 & 5858304)

Scope:

These instructions apply to the specific HEC serial numbers noted below.

205-0121-28 (GE PN 5858303) HEC Serial Number (S/N) below 897257-01

205-0121-40 (GE PN 5858304) HEC Serial Number (S/N) above 897257-01 and above

Tools Required:

- M8 or M4 Wrench or equivalent (depending on HEC S/N – see detail above for applicable serial numbers)
- 5/16" nut driver (for hose clamps)
- 1 1/16" open end wrench
- 1 1/4" open end wrench
- #2 Philips screwdriver
- Bucket for draining
- Dollies for moving HEC cabinet (optional) GEHC PN 5307574

IMPORTANT: The HX (Heat Exchanger) replacement procedure is different for the two types of HEC cabinets that are currently in use. For HEC cabinet with part number **PN 205-0121-28 (units below SN 897257-01)** (old style), the HX will be replaced by accessing the cabinet from the front. For HEC cabinet part number **205-0121-40 (units with SN greater than 897257-01)**, the HX will be replaced from the rear.

IMPORTANT: If it is determined that the HX requires replacement from the rear, the HEC may need to be moved forward for proper replacement. Shipment dollies will be required for HEC cabinet relocation.

Removal Procedure for PN 205-0121-40 (for units with SN greater than 897257-01 and above, rear cabinet access is required)

1. Lockout/Tag out system from main disconnect in accordance with GE procedure 5452735.
2. Remove LCS7397GX door by opening door and removing door from its hinges.
3. Using shipping dollies move HEC forward to gain access to the rear panel. Cables on the top of the HEC may need to be removed.
4. Remove LCS7937GX rear panel by removing qty 4 Philips #2 screws (2 screws on left side, 2 screws on right side of panel). Refer to figure 1 below.
5. Remove the 2 ground connections on the inside of the back panel. Retain hardware for re-installation.
6. Lockout/Tagout and close Facility Supply and Return ball valves on top of system (Refer to diagram below) and drain heat exchanger to be removed (through ball valve on associated pump). Refer to BOYD Manual #820-0190 .

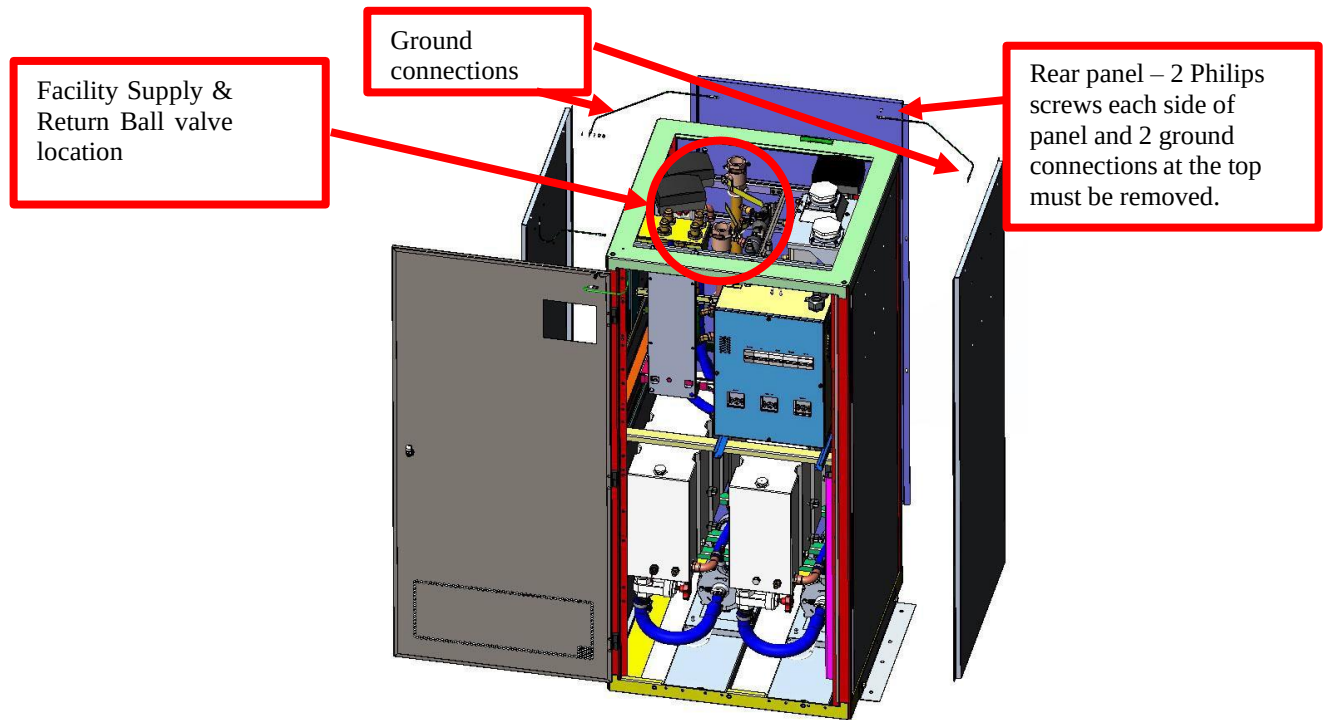
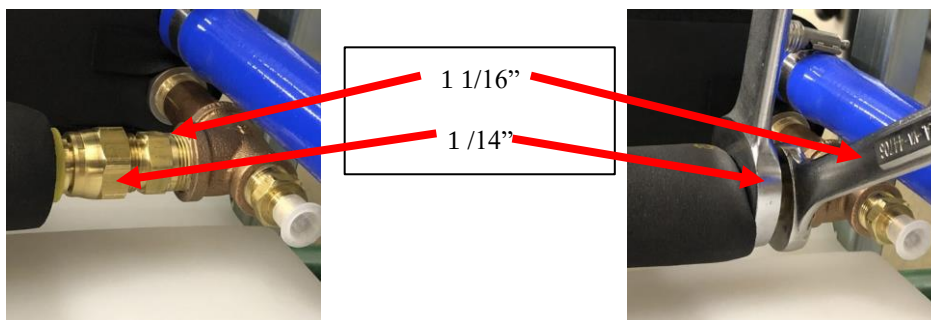


Figure 1 - Location of Facility Supply & Return Ball Valves

7. Drain coolant from tank & coolant loop of the associated heat exchanger to be replaced. Use a bucket to catch fluid from the pump drain hose and empty periodically so as not to overflow the bucket.
8. Place the bucket under each of the hoses to catch fluid remaining in the hose.
9. Pull back insulation on the hose connections to the heat exchanger.
10. Using the 1 1/16" open end wrench for support, remove hose going to Facility water manifold and mixing valve using the 1 1/4" open end wrench. See photos below.

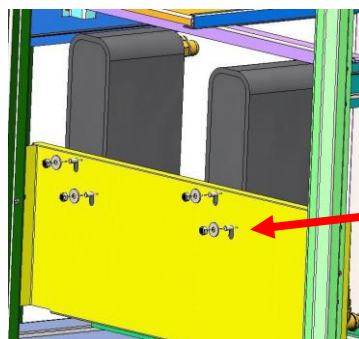


11. Using the 5/16" nut driver, remove remaining heat exchanger inlet and outlet hoses from the heat exchanger to be replaced. Note location of hoses for re-installation of new heat exchanger. ONLY remove the hoses that are attached to the heat exchanger as the replacement heat exchanger is supplied with fittings pre-installed.

NOTE

For Boyd PN 205-0121-28 (GE PN 5858303) M4 hardware is used.

12. Using the M8 wrench or equivalent, remove 2 nuts and flat washers from the rear of the cabinet frame to facilitate removal of the heat exchanger. See drawing below.



M8 nuts and flat washers (only remove for heat exchanger to be replaced).

Figure 2 - Heat Exchanger mounting nuts and washer detail

13. Remove heat exchanger from HEC.

End of Removal Procedure

Removal Procedure for PN 205-0121-28 (units below SN 894095-01 front cabinet access)

1. Lockout/Tag out system from main disconnect in accordance with GE procedure 5452735.
2. Remove LCS7397GX door by opening door and removing door from its hinges.
3. Lockout/Tagout and close Facility Supply and Return ball valves on top of system (Refer to diagram below) and drain heat exchanger to be removed (through ball valve on associated pump). Refer to BOYD Manual #820-0190 .

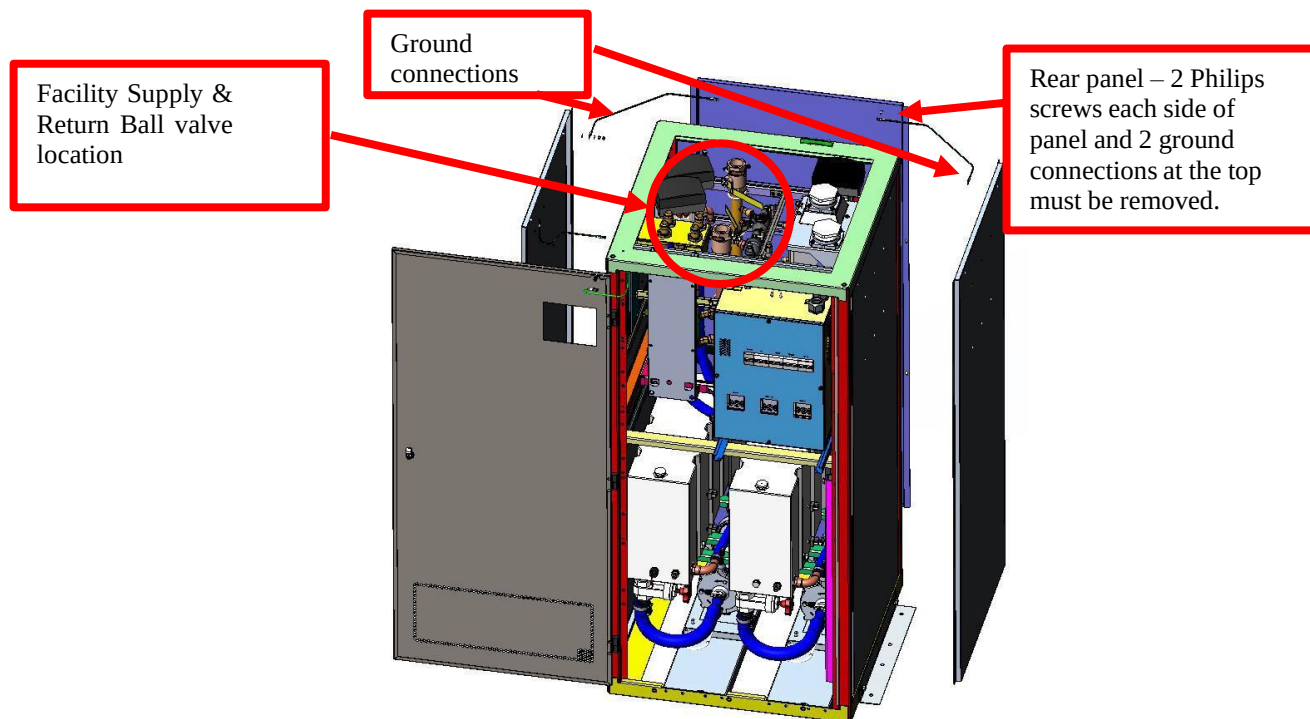
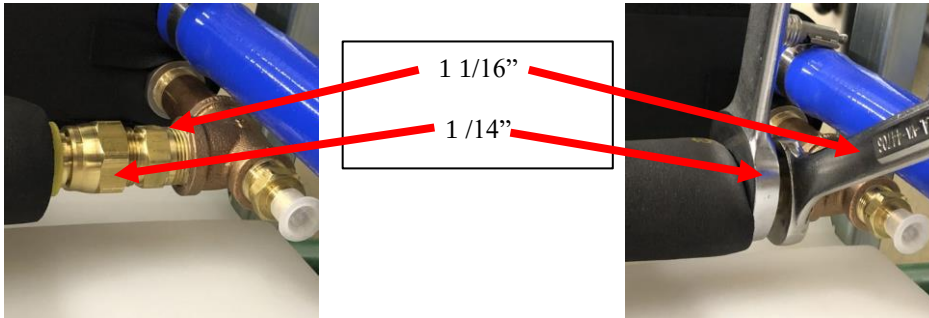


Figure 3 - Location of Facility Supply & Return Ball Valves

4. Drain coolant from tank & coolant loop of the associated heat exchanger to be replaced. Use a bucket to catch fluid from the pump drain hose and empty periodically so as not to overflow the bucket.
5. Place the bucket under each of the hoses to catch fluid remaining in the hose.
6. Pull back insulation on the hose connections to the heat exchanger.
7. Using the 1 1/16" open end wrench for support, remove hose going to Facility water manifold and mixing valve using the 1 1/4" open end wrench. See photos below.



Using the 5/16" nut driver, remove remaining heat exchanger inlet and outlet hoses from the heat exchanger to be replaced. Note location of hoses for re-installation of new heat exchanger. ONLY remove the hoses that are attached to the heat exchanger as the replacement heat exchanger is supplied with fittings pre-installed.

NOTE

For Boyd PN 205-0121-28 (GE PN 5858303) M4 hardware is used.

8. For GE PN 5858303, it is possible to remove the Tank Reservoir if more room is needed to remove the heat exchanger bracket and heat exchanger. See Procedure doc #820-60027-000 for removal instructions for the Tank before proceeding.
9. Using the M4 wrench or socket, remove 2 nuts on each side of the heat exchanger bracket. Support the heat exchanger while removing the nuts. See photo below.



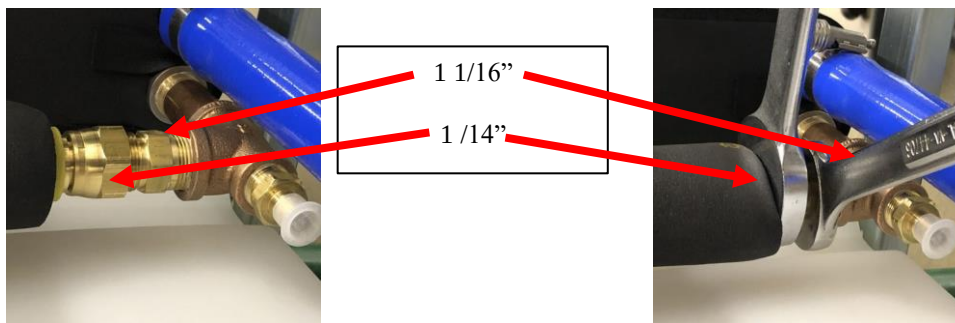
10. Remove heat exchanger bracket.
11. Remove heat exchanger.

Installation Procedure for PN 205-0121-28 (for units below SN 897257-01)

1. Install heat exchanger into the location where the old heat exchanger was removed (GC or PE side).
2. Support the heat exchanger while attaching the bracket. Secure heat exchanger with the bracket that was removed previously. Use an M4 wrench to secure the 2 nuts on either side of the bracket that is mounted to the screw studs protruding from the back plate.

NOTE For Boyd PN 205-0121-28 (GE PN 5858303) M4 hardware is used.

3. Using the 5/16" nut driver, re-attach the hoses removed earlier making sure to compare with the reverse order in which the hoses were removed.
4. Using the 1 1/16" open end wrench for support, tighten hoses going to Facility water manifold and mixing valve using the 1 1/4" open end wrench. See photos below.



5. If Tank Reservoir was removed to gain access to the heat exchanger, replace Tank Reservoir using installation document #820-60027-000
6. Close drain ball valve on associated pump to prevent further draining.
7. Open facility water ball valves (Supply & Return) on top of machine to allow system to fill.
8. Fill system in accordance with BOYD Manual #820-0190 .
9. Visually check for leaks on heat exchanger connections.
10. Re-install the ground connections on rear panel with retained hardware. Refer to Figure 4 below.
11. Re-install rear panel with retained hardware (4 places). Refer to Figure 3 below.

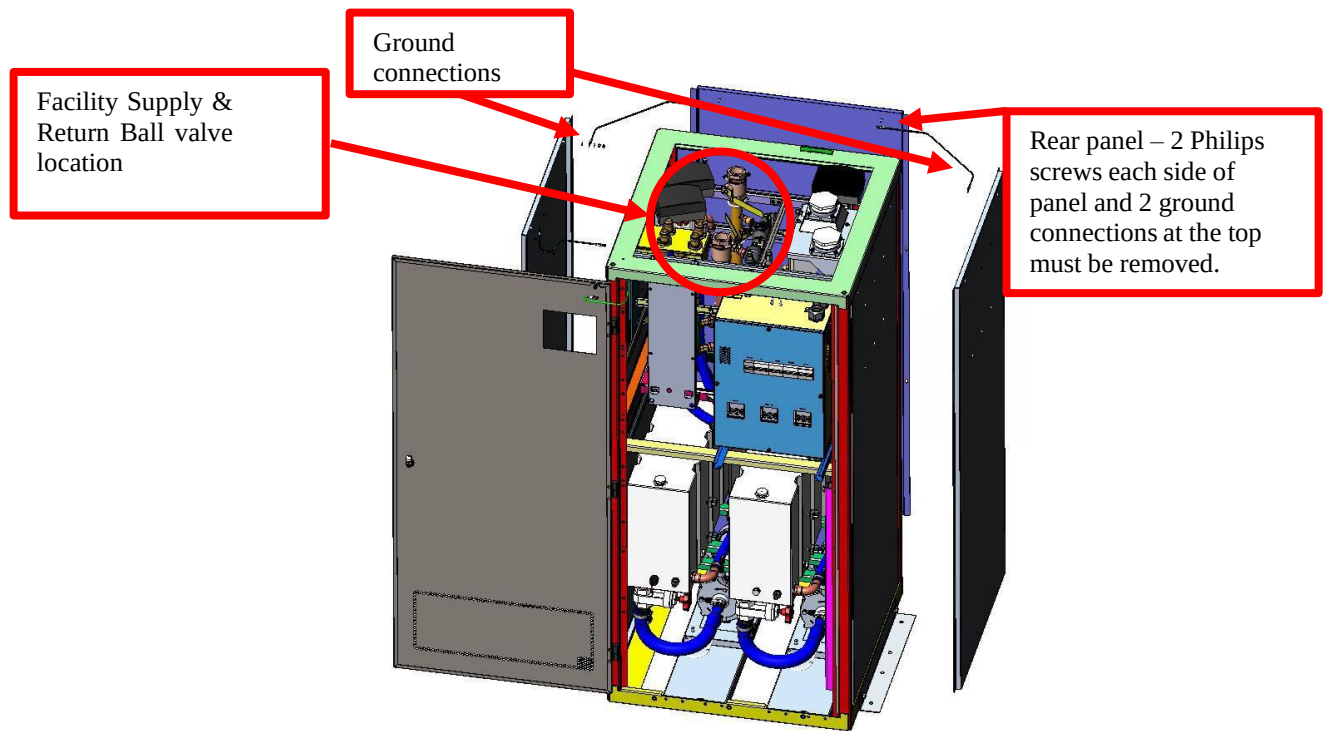


Figure 4

End of Installation Procedure**Installation Procedure PN 205-0121-40 (units with SN greater than 897257-01)**

1. Install heat exchanger into the location where the old heat exchanger was removed (GC or PE side).
2. Secure heat exchanger to location so studs on rear of heat exchanger line up with holes in mounting frame.

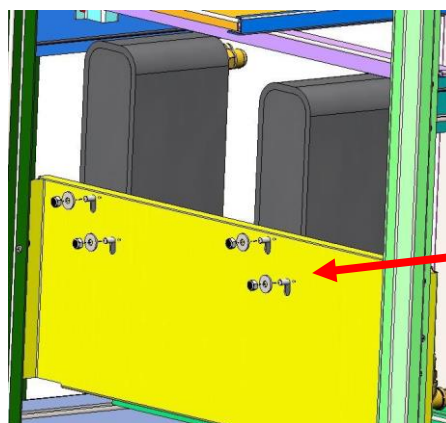


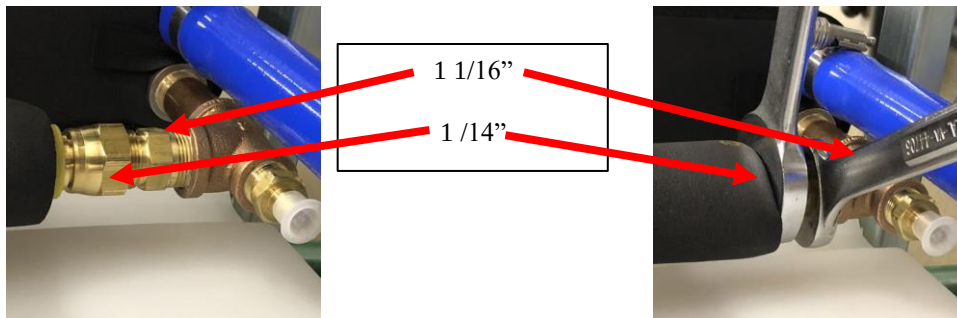
Figure 5

NOTE

For Boyd PN 205-0121-28 (GE PN 5858303) M4 hardware is used.

3. Attach the M8 flat washers and nuts to the studs protruding on the mounting frame. Screw to hand tight and then go $\frac{1}{4}$ turn past that point to secure.

4. Using the 5/16" nut driver, re-attach the hoses removed earlier making sure to compare with the reverse order in which the hoses were removed.
5. Using the 1 1/16" open end wrench for support, tighten hoses going to Facility water manifold and mixing valve using the 1 1/4" open end wrench. See photos below.



6. Re-install rear panel with retained hardware (4 places). Refer to Figure 6 below
7. Return HEC cabinet to original position.
6. Close drain ball valve on associated pump to prevent further draining.
7. Open facility water ball valves (Supply & Return) on top of machine to allow system to fill.
8. Fill system in accordance with BOYD Manual #820-0190
9. Visually check for leaks on heat exchanger connections.
10. Re-install the ground connections on rear panel with retained hardware. Refer to Figure 6 below.

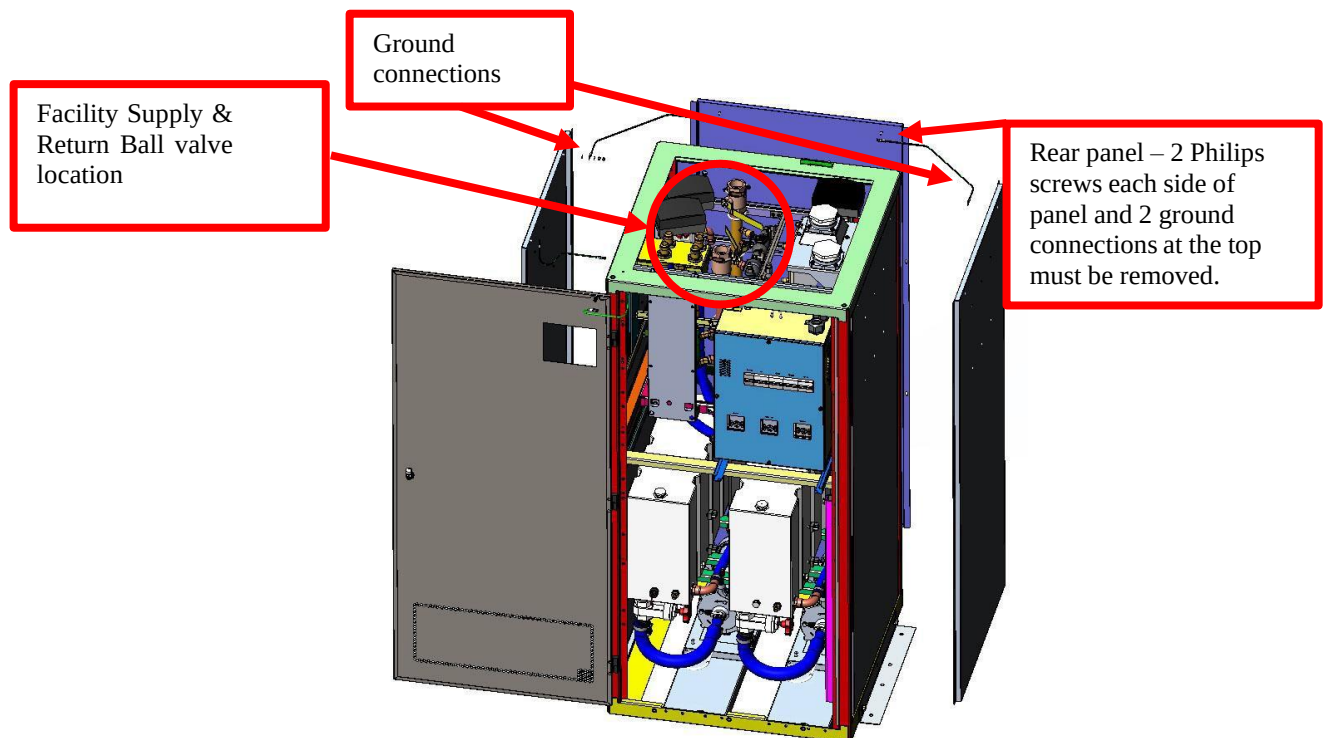


Figure 6

End of Removal Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial Release	3/19/2021	SM	MJG
B	GE EDITS & LCS7397G2 addition	8/4/2021	SM	MJG
C	REV. PER ECN 203478	12/28/21	SM	SM



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820-60027-000 – Install Instructions Reservoir Kit (w/level switches & valve) (PN 205-0121-29) GE PN 5861806

Tools Required:

- 5/16" nut driver (for hose clamps)
- 3/8" nut driver (for hose clamps)
- #2 Philips screwdriver
- Loctite® 565
- Bucket for draining

Removal Procedure:

1. Perform shutdown of system in accordance with AAVID Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735)
2. Lockout/Tag out system from main disconnect in accordance with GE procedures.
3. Remove LCS7397GX door by opening door and removing door from its hinges.
4. Lockout/Tagout and close Facility Supply and Return ball valves on top of system (Refer to Figure 1 below) and drain heat exchanger to be removed (through ball valve on associated pump). Refer to AAVID Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).

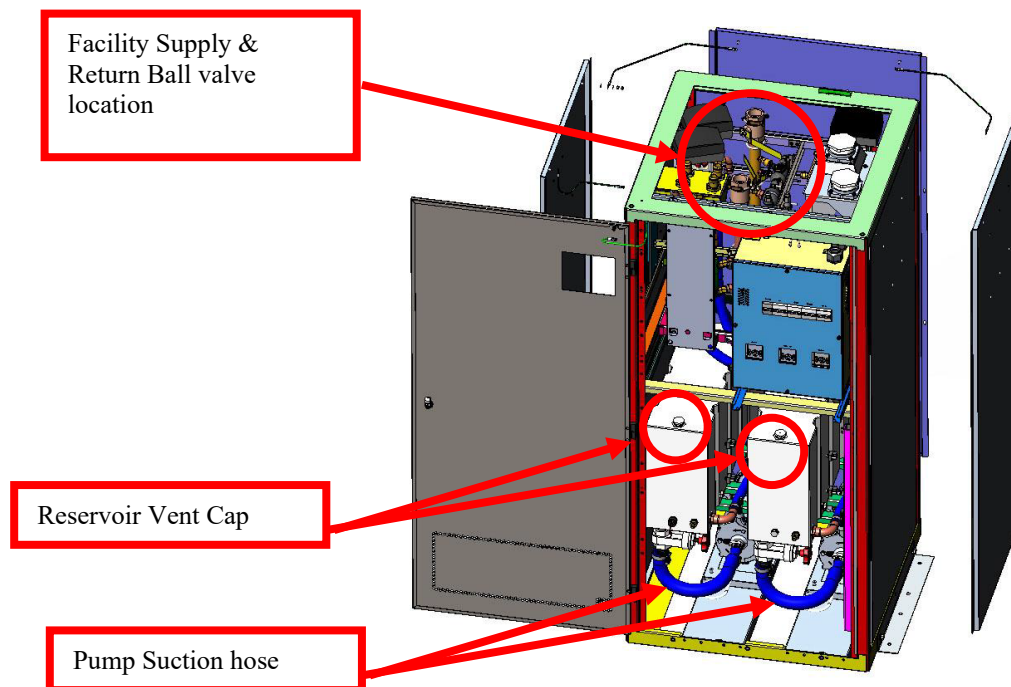


Figure 1 Location of Facility Supply & Return Ball Valves

1. Loosen vent cap by turning CCW 2 turns (loose enough to let air displace coolant).
2. Confirm the red colored gate valve under tank is fully open.
3. Drain coolant from tank & coolant loop (GC or PE side). Use a bucket to catch fluid from the pump drain hose and empty periodically so as not to overflow the bucket.
4. Open drain valve on pump associated with reservoir being removed. Let coolant drain into bucket. Close valve and empty bucket before it overflows. Continue draining reservoir until empty.

5. Using the 3/8" nut driver, remove pump suction hose clamp to facilitate removing suction hose from white gate valve with red handle (located underneath reservoir).
6. Remove pump suction hose from reservoir.
7. Using the 3/8" nut driver, loosen the clamp on the blue hose that connects the reservoir to the heat exchanger to facilitate removal side of reservoir. (do not remove the hose clamp but slide up the hose) Refer to photo 1 below.

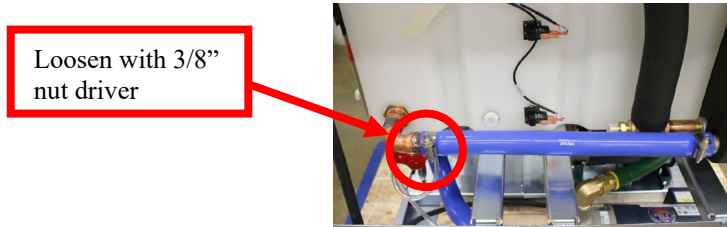


Photo 1 Detail of hose from heat exchanger

8. Remove blue hose from fitting on side of reservoir.
9. Remove spade lugs from tank level switches on side of reservoir. Note which leads goes to each switch before removal of leads.

NOTICE

10. Remove DI sensor on front of reservoir being careful to not damage the sensor. The sensor will be used on the new reservoir.
11. Before removing the reservoir, measure edge of tank to end of frame. The measurement will help place the new tank in the correct position. The measurement is different for each GC or PE tank (relative to the frame). Refer to photo 2 & 3 below. Alternately, one could mark the edge of the reservoir that is flush with the front and back support rails onto the rail with a marker. The new tank will be installed so that the reservoir edge is flush with the mark on the front and back support rails.

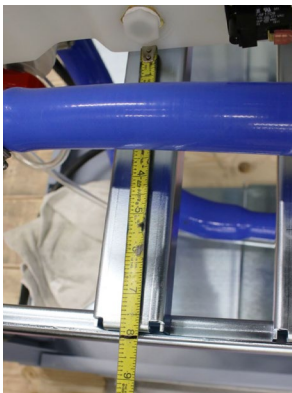


Photo 2 Detail of GC Tank = 8"



Photo 3 Detail of PE Tank = 1 3/8"

12. Using the #2 Philips screwdriver, remove the qty 4 #10-24 screws, lock washers and flat washers underneath the reservoir that is being removed. Refer to Figure 2 below. Retain hardware for re-installation.

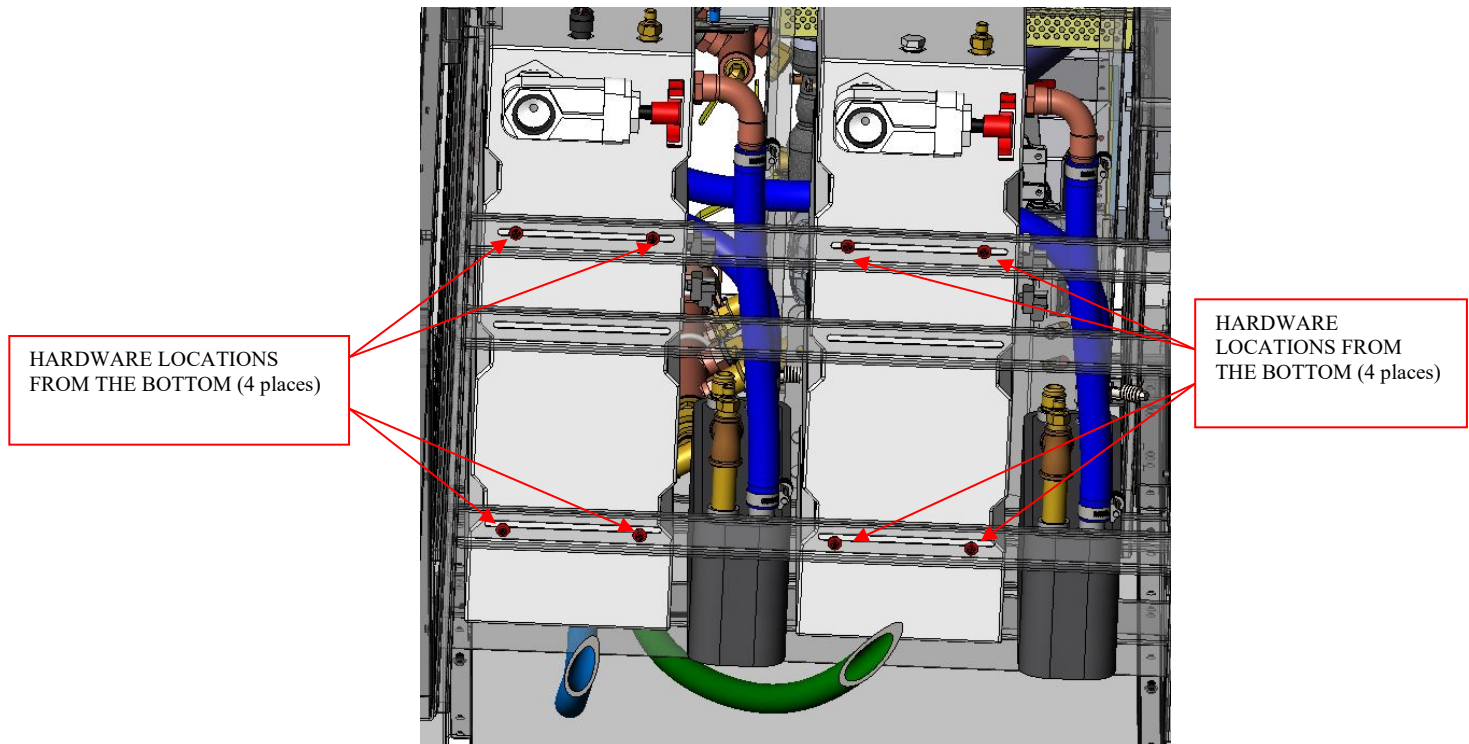


Figure 2 Detail of mounting screws for Reservoir (GC or PE side)

13. Remove the reservoir.

End of Procedure

Installation Procedure:

1. Place new reservoir onto the support rails. Position the reservoir so that it is (GC is 8", PE = 1 3/8") from the side of the cabinet as show in Step 11 of Removal Procedure above. Alternately, place flush with line as marked in Step 11 of Removal Procedure above.
2. Using #2 screw driver, secure reservoir with qty 4 #10-24 screws, lock washers and flat washers that were removed in Step 12 of Removal Procedure above. Refer to Figure 2 above for screw mounting locations.
3. Apply Loctite® 565 to DI sensor threads to prevent leaking and screw into the front of the reservoir.
4. Re-connect tank level switch leads making sure to adhere to switch positions that were noted before removal.
5. Re-attach blue hose from the heat exchanger to the fitting on side of reservoir. Slide the hose clamp back into place over the fitting. Using the 3/8" nut driver, tighten the clamp to hand tight and then go 1/4 turn further to make a secure connection.
6. Re-attach blue pump supply hose from gate valve to pump. Using the 3/8" nut driver, tighten the clamp hand tight and then go 1/4 turn further to make a secure connection.
7. Close drain valve on pump associated with reservoir being replaced.
8. Re-install vent cap by turning in a Clockwise (CW) direction.
9. Make sure red colored gate valve under tank is fully open.
10. Clear LockOut/Tagout tags and open ball valves for Facility Supply & Return hoses on the top of the HEC. Refer to AAVID Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).
11. Fill system in accordance with AAVID Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).
12. Check for leaks in the area of the reservoir just replaced. Repair if necessary.
13. Re-install front door on its' hinges.
14. Re-start the unit in accordance with AAVID Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial Release	3/24/2021	SJM	MJG



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820-60028-000 – Push Loc Hose Kit Install Instructions (PN 205-0121-44) GE PN 5857494

Tools Required:

- 1 1/16" Wrench
- 1 1/4" Wrench
- 1 1/2" Wrench
- 7/8" Wrench (qty 2)
- 3/4" Wrench
- Adjustable Wrench

Scope: This instruction will address the removal and replacement of several system hoses within the HEC. The draining and filling procedures included are common to hoses being replaced.

NOTE: As replacement procedure mirrors the removal process, only the general steps are listed as the hose locations remain the same.

Table 1: Kit Contents

Description	Boyd Part #	From	To	Insulation PN
Hose	200-1806-05	GC Pump discharge	GC Supply Header	530-0772-02
Hose	200-1806-06	PE Pump discharge	PE Supply Header	No Insulation
Hose	200-1806-07	Facility supply header	GC Heat Exchanger bottom	530-0772-01
Hose	200-1806-08	Facility supply header	PE Heat Exchanger bottom	530-0772-01
Hose	200-1806-09	GC heat exchanger, top	GC mixing valve-A	530-0772-01
Hose	200-1806-10	PE heat exchanger, top	PE mixing valve-A	530-0772-01
Hose	200-1806-11	Facility supply header	GC mixing valve-B	530-0772-01
Hose	200-1806-12	Facility supply header	PE mixing valve-B	530-0772-01
Hose	200-1806-13	GC mixing valve	Facility Return Header	530-0772-01
Hose	200-1806-14	PE mixing valve	Facility Return Header	530-0772-01
Hose	200-1806-15	Facility Supply Header	Cryo Supply Header	530-0772
Hose	200-1806-16	Cryo return header	Facility return header	530-0772
Hose	200-1806-18	Blower Mixing Valve Port A	Blower Heat Exchanger Return	530-0772
Hose	200-1806-19	Blower Mixing Valve Port B	Facility Supply Header	530-0772
Hose	200-1806-20	Blower mixing valve	Facility Return Header	530-0772
Hose	200-1806-21	Facility Supply Header	Blower Heat Exchanger Supply	530-0772
Insulation	530-0772-02	N/A	N/A	N/A
Insulation	530-0772-01	N/A	N/A	N/A
Insulation	530-0772	N/A	N/A	N/A

Table 2: Additional Parts needed for replacement

Description	GE Part Number
Electrical tape	46-258218P3
Teflon tape	46-252065P19
Insulation tape	5861808
Replacement coolant (4 gallons)	5174313-4
Tie Wraps	N/A
5 gallon bucket (qty = 2) to capture coolant	N/A

Removal Procedure:

1. Lockout/Tag out system from main disconnect in accordance with GE procedure 5452735.
2. Remove LCS7397GX door by opening door and removing door from its hinges.

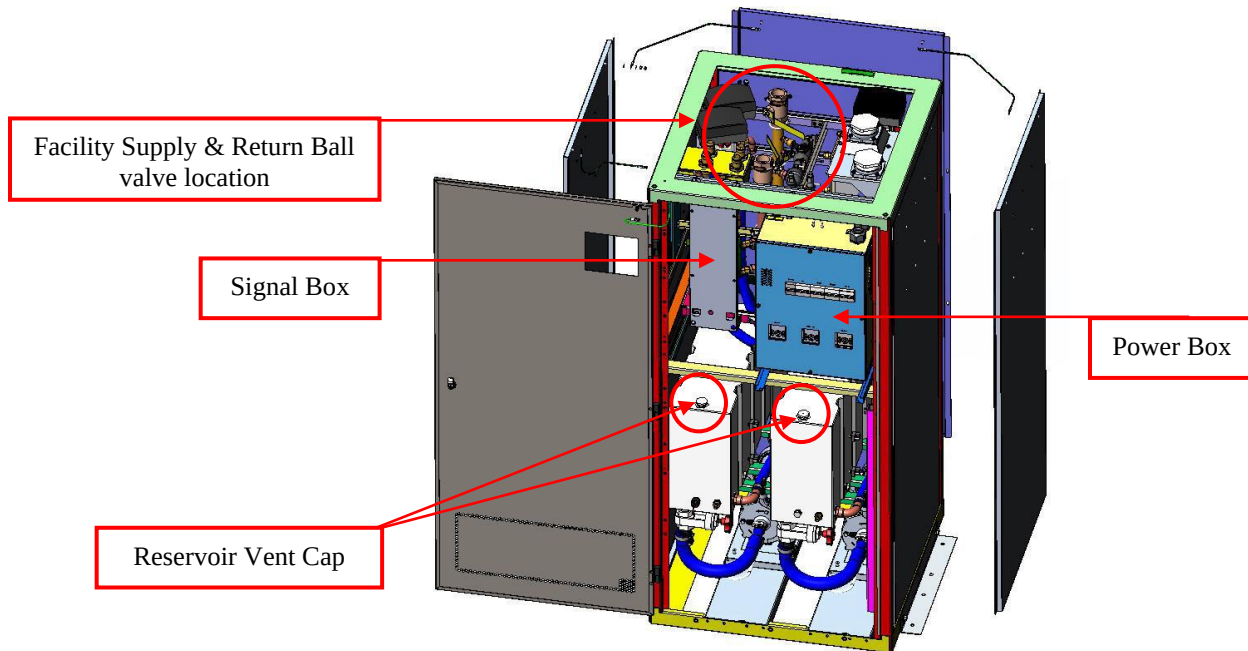


Figure 1 Location of Facility Supply & Return Ball Valves

3. Move Power Box out of the way by moving it on the guide rails. Depending on the hose being replaced, the Signal Box may also have to be removed to access the hose. Refer to Figure 1 for locations of Power and Signal Box.
4. Loosen vent cap on reservoir by turning CCW 2 turns (loose enough to let air displace coolant).
5. Verify red colored gate valve under tank is fully open.
6. Drain coolant from tank & coolant loop (GC or PE side). Use a bucket to catch fluid from the pump drain hose and empty periodically so as not to overflow the bucket. See Figure 2 below.

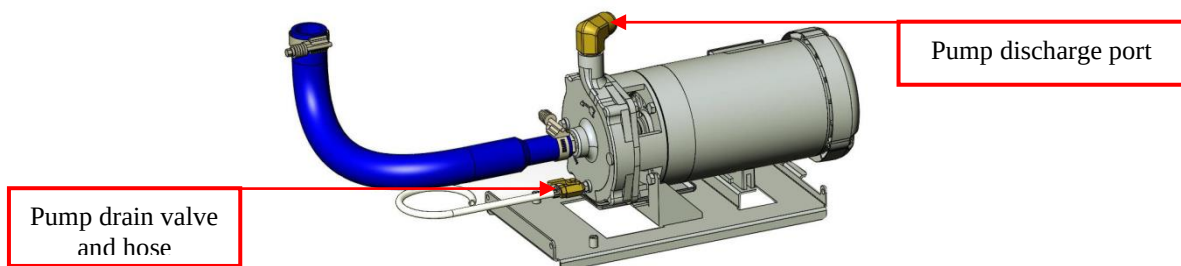


Figure 2 Pump drain & discharge detail

7. Remove thermal insulation from hose being replaced. Discard.

NOTE: Steps apply for both GC and PE hoses.

For all removal and tightening use supporting wrench to hold fitting (see Figure 3)

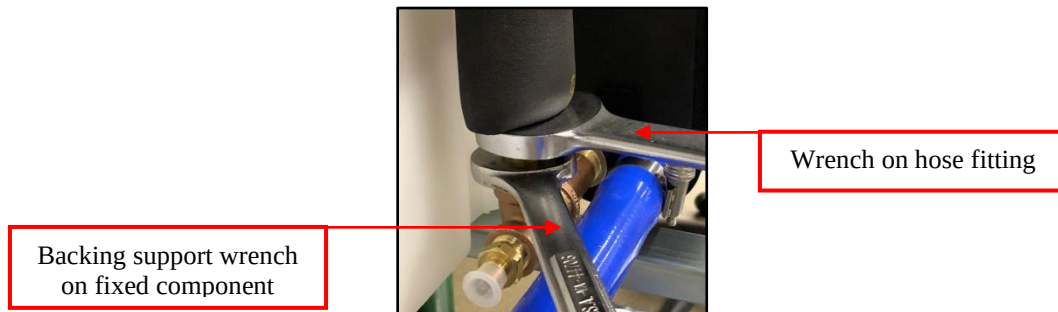


Figure 3 Wrench placement

I. Pump Discharge to Supply Header

PN 200-1806-05 GC (Blue hose) (Figure 4a)

PN 200-1806-06 PE (Green hose) (Figure 4b)



Figure 4a GC hose PN 200-1806-05



Figure 4b PE hose (shown attached to manifold PN 200-1806-06)

- a. Using the adjustable wrench and the 7/8" wrench, remove the pressure sensors and temperature probes from the hose & manifold to be removed. Coil up to prevent damage to the sensors and tie wrap to frame.
- b. Using the adjustable wrench as a backing support (figure 3), loosen the connector attached to the pump discharge port (GC or PE) with the 1/1/2" wrench. Refer to Figure 4 for discharge port hose identification.
- c. Using the 1 1/2" wrench, loosen flared fitting holding the GC (blue) (PE - Green) Supply hose to Supply manifold as shown in the Figure 5 below.
- d. Remove hose.

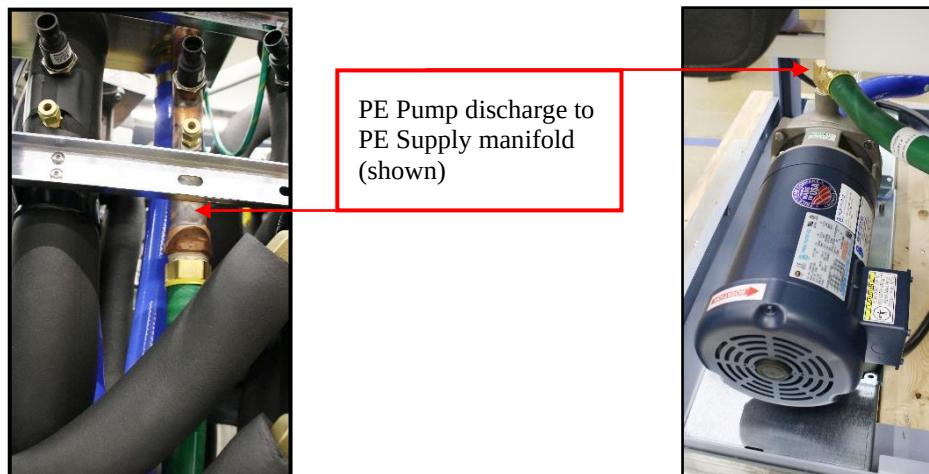


Figure 5 PE Pump Discharge to PE Supply manifold

II. Facility Supply Header to Heat Exchanger, bottom

PN 200-1806-07 GC - Blue

PN 200-1806-08 PE -Blue

- Push back insulation on hose to be removed to gain access to fitting.
- Using the 1 1/4" wrench and 1 1/16" wrench for support (Refer to Figure 3 above), loosen the fitting attached to the bottom heat exchanger port (GC or PE – blue hose).
- Using the 7/8" wrench and adjustable wrench as a support, remove GC or PE hose from Facility Supply Header. (Figure 6 below)
- Remove hose.

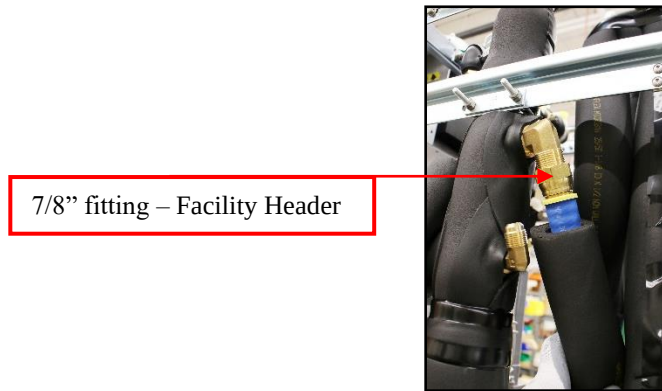


Figure 6 Facility Supply Hoses (PE & GC)

III. Facility Supply Header to Cryo Supply

PN 200-1806-15 - Gray

- Push insulation back on hose to be removed to gain access to the fitting.
- Using one 7/8" wrench for removal and another for support (Refer to Figure 3 above), remove hose from Facility Supply header.
- Using one 7/8" wrench for removal and another for support (Refer to Figure 3 above), remove hose from Cryo Supply header (Figure 7 below).
- Remove hose.

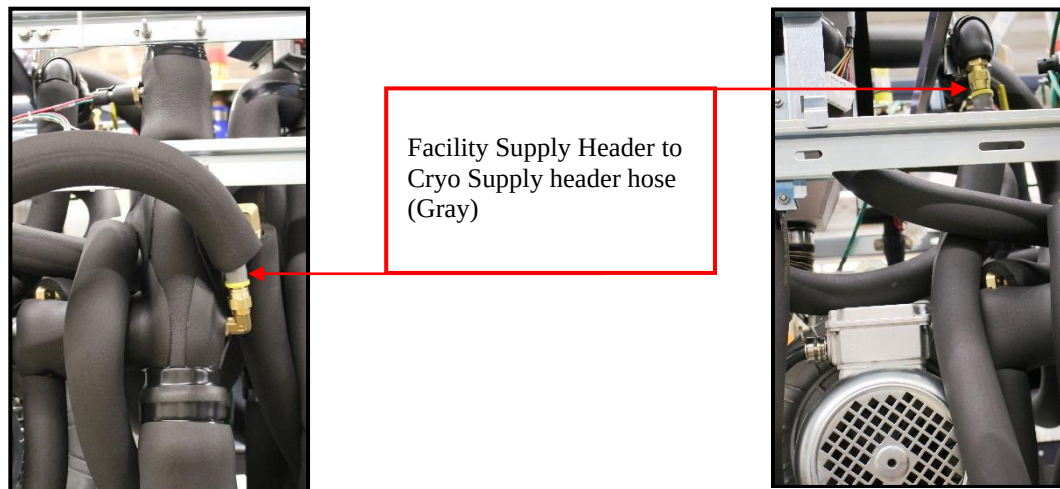


Figure 7 Facility Supply Header to Cryo Supply header detail

IV. Facility Return Header to Cryo Return Header

(PN 200-1806-16 – Yellow)

- Push insulation back on hose to be removed to gain access to the fitting.
- Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3 above), remove hose from Facility Supply header.
- Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3 above), remove hose from Cryo Supply header (Figure 8 below).
- Remove hose.

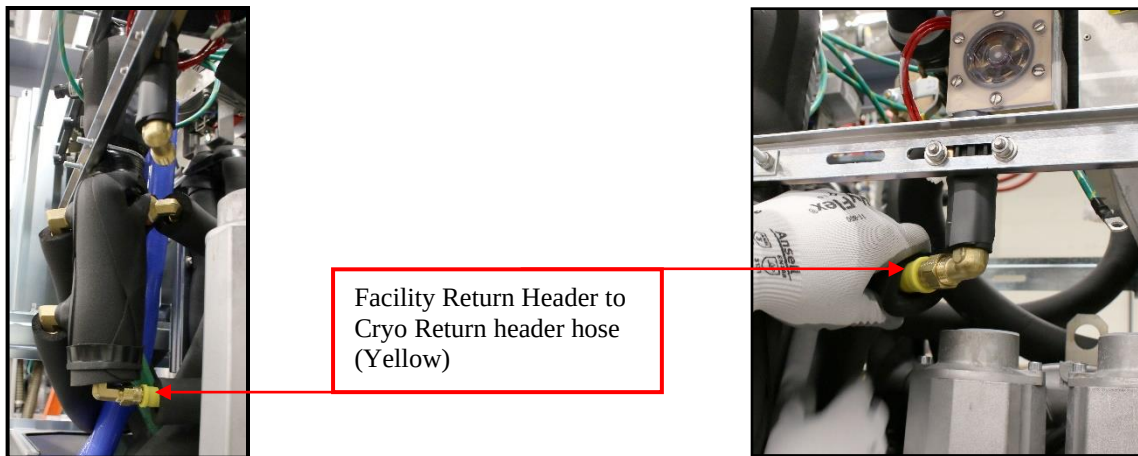


Figure 8 Facility Return Header to Cryo Return Header

V. Facility Supply Header to Blower Heat Exchanger Supply

PN 200-1806-21 - Blue

- Push insulation back on hose to be removed to gain access to the fitting.
- Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3 above), remove hose from Facility Supply header.
- Using a 7/8" wrench and 3/4" wrench for support (Refer to Figure 3 above), remove hose Blower heat exchanger (Figure 9 below).
- Remove hose.

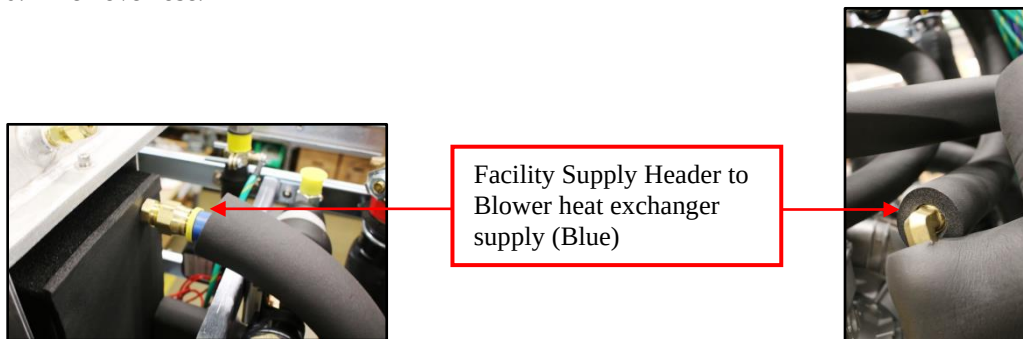


Figure 9 Facility Supply Header to Blower Heat Exchanger Supply

VI. GC/PE Mixing Valve to Heat Exchanger top (Port A)

(PN 200-1806-09 – RED GC) NOTE: GC hose Port A is shorter than corresponding hose for PE

(PN 200-1806-10 - RED PE)

- Push back insulation around the 'A' side of the mixing valve. Using a 3/4" wrench and an Adjustable for support loosen the 3/4" hose connector going to the 'A' port of the mixing valve. See Figure 10 below for port locations and destination reference (GC or PE side)

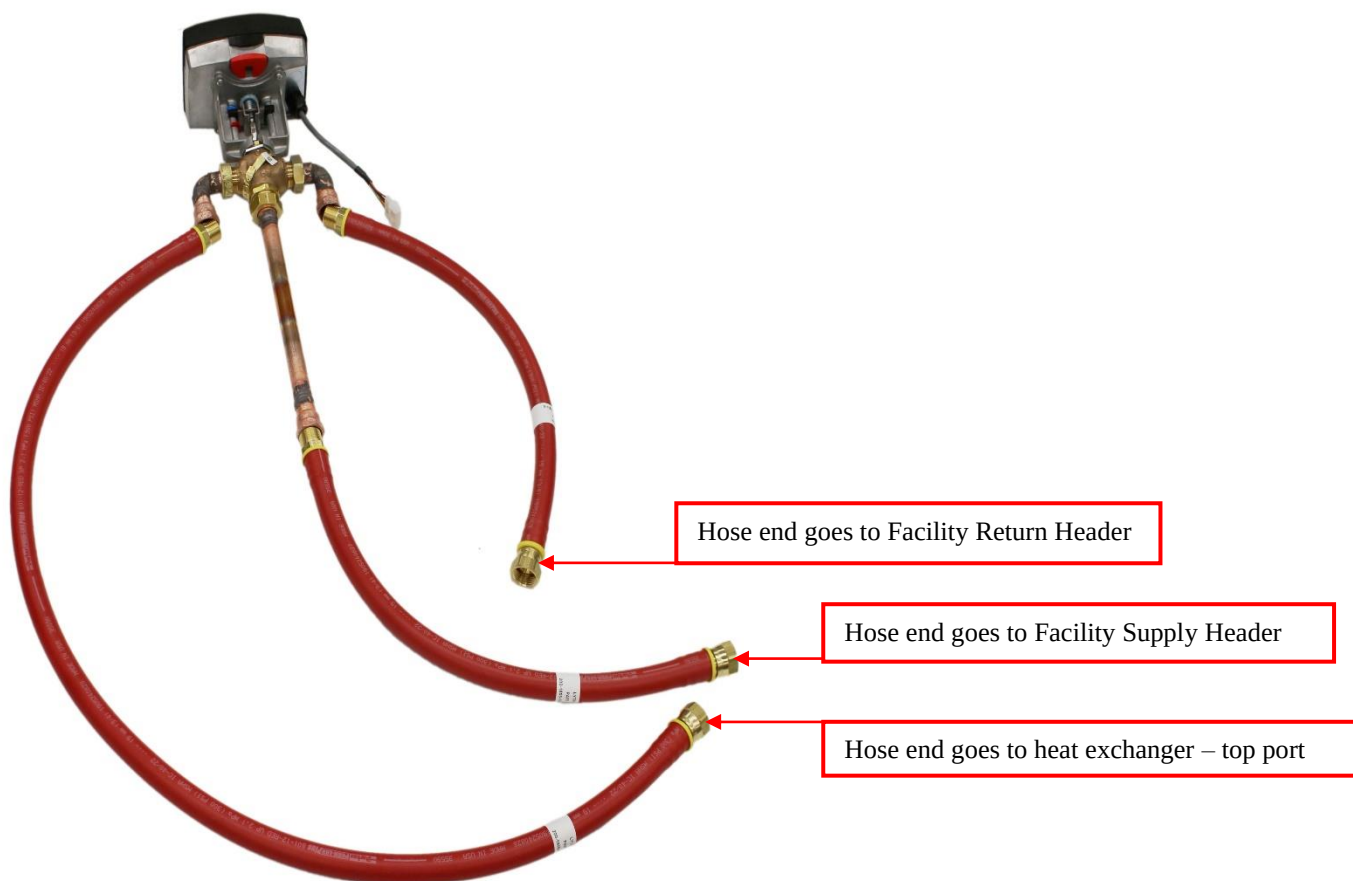


Figure 10 PE Mixing valve detail showing ports A, B, and AB for reference

- b. On opposite end of port 'A' hose, loosen and remove hose connection going to the top port of the heat exchanger (GE or PE heat exchanger). See Figure 11a & 11b

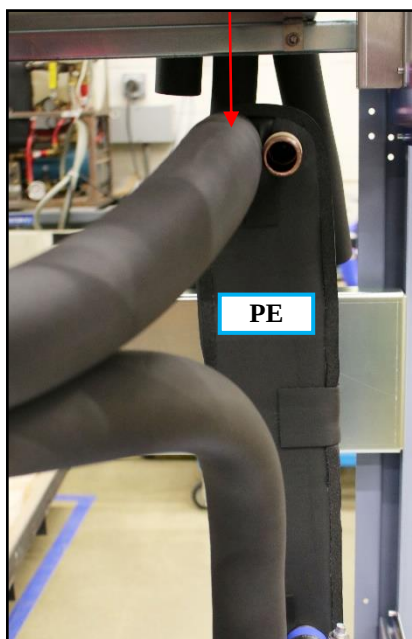


Figure 11a PE Port 'A' to PE HX top

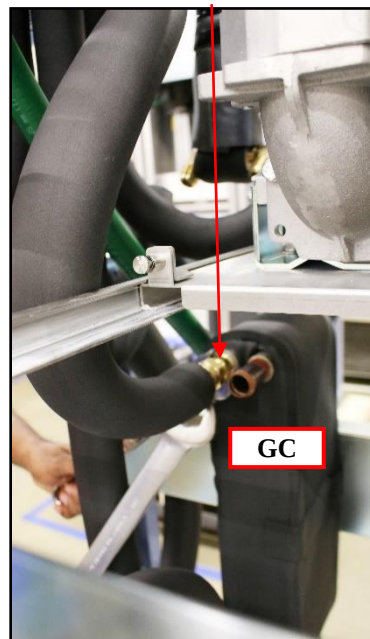


Figure 11b GC Port 'A' to GC HX top

VII. GC/PE Mixing Valve to Facility Supply Header (Port B)

(PN 200-1806-11 BLUE - GC) **Note:** Hose is BLUE only for GC Mixing Valve
(PN 200-1806-12 RED – PE)

- a. Push back insulation around the 'B' side of the mixing valve. Using a $\frac{3}{4}$ " wrench and an Adjustable wrench loosen the $\frac{3}{4}$ " hose connector going to the 'B' port of the mixing valve. See Figure 12 below for port locations and destination reference (GC or PE side)

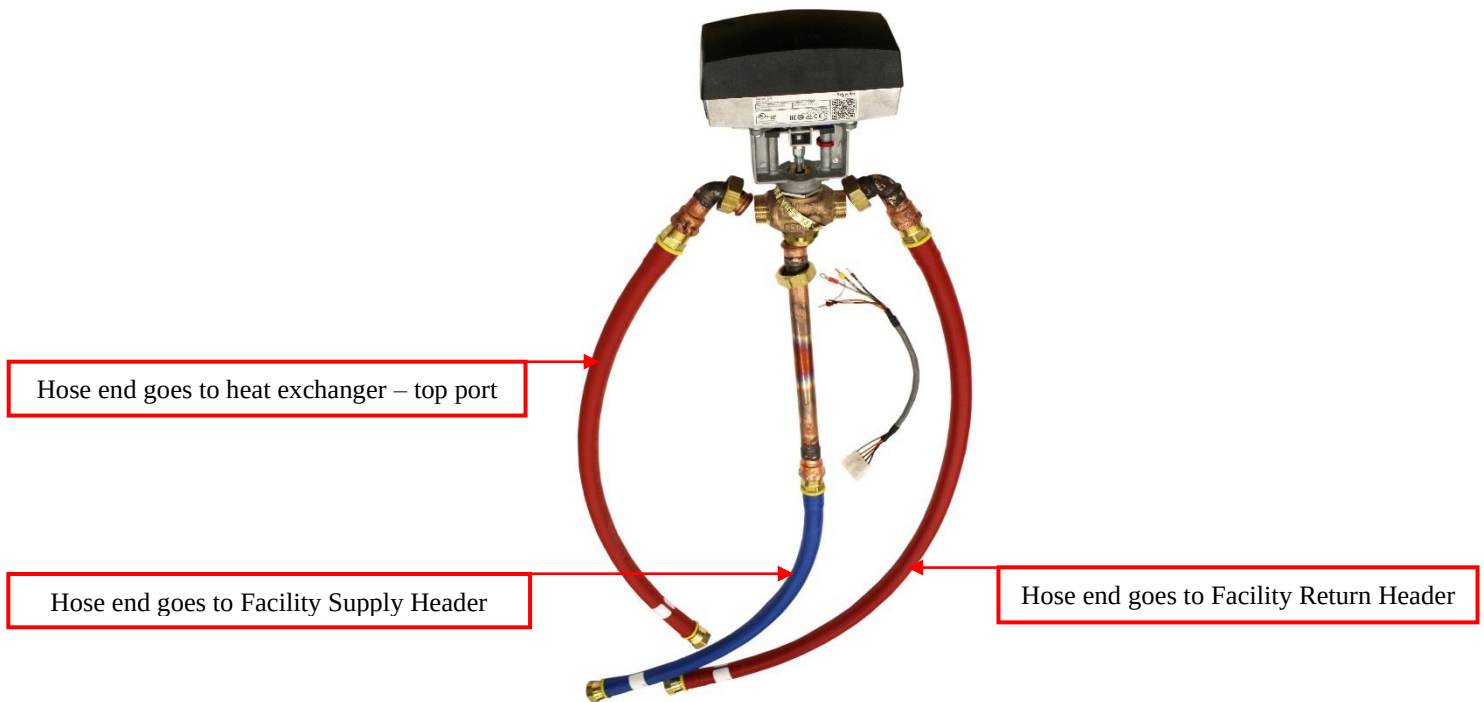


Figure 12 GC Mixing valve detail showing ports A, B, and AB for reference

- b. Using a $\frac{3}{4}$ " wrench and an adjustable wrench for support, loosen and remove hose connection going to the Facility Supply Header (for red PE hose & blue GC). See Figure 13a & 13b.
- c. Remove hose.



Figure 13a. PE Mixing Valve Port 'B' to Facility Supply Header

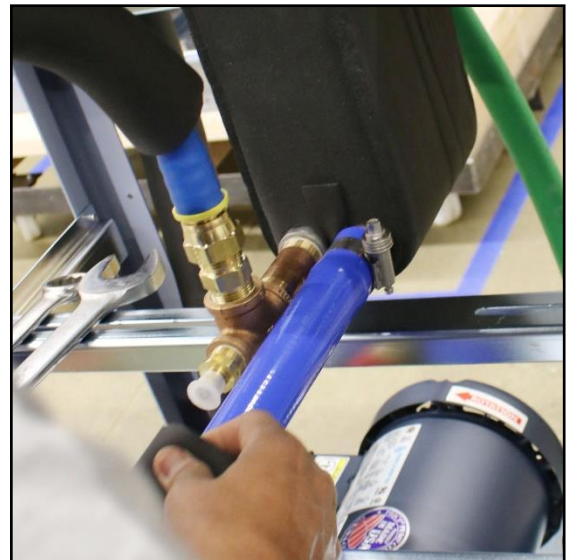


Figure 13b. GC Mixing Valve Port 'B' Heat Exchanger lower left port

VIII. GC/PE Mixing Valve (Port AB) to Facility Return Header

(PN 200-1806-13 – RED GC)

(PN 200-1806-14 – RED PE)

- a. Push back insulation around the 'AB' side of the mixing valve. Using a 3/4" wrench and an adjustable wrench for support, loosen the 3/4" hose connector going to the 'AB' port of the mixing valve. Refer to Figure 10 & 12 for port locations and destination reference (GC or PE side)
- b. Using a 3/4" wrench and an adjustable wrench for support, loosen and remove hose connection going to the Facility Return Header (for red PE hose & red GC). See Figure 14 below.
- c. Remove hose.

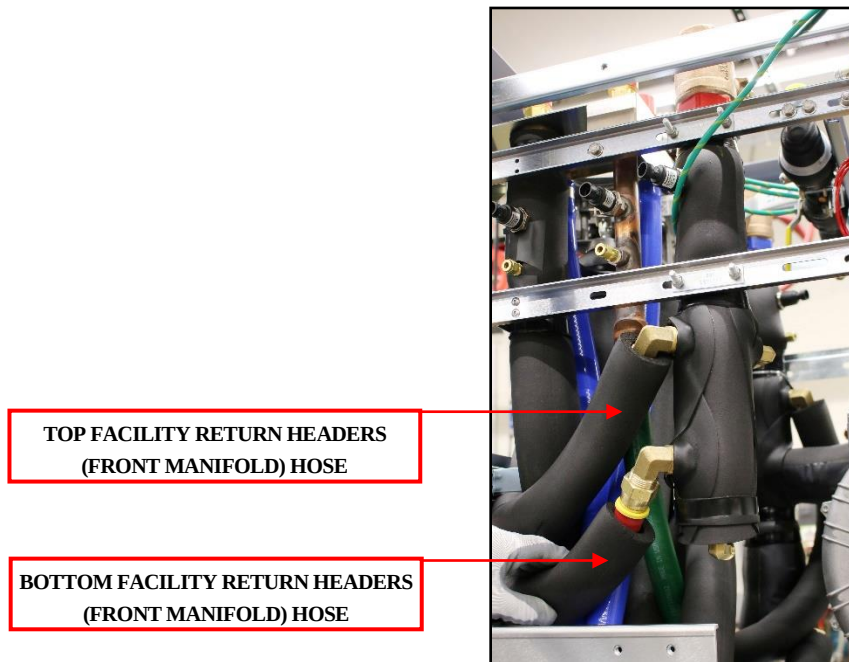


Figure 14. GC & PE Mixing Valve Port 'AB' to Facility Return Manifold

IX. Blower Mixing Valve (Port 'A') to Blower Heat Exchanger Return

(PN 200-1806-18 RED)

- a. Refer to Figure 15 below for port locations for the mixing valve.
- b. Push insulation back on hose to be removed to gain access to the fitting.
- c. Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3), loosen and remove hose from Blower heat exchanger return. (Figure 16 below).
- d. Remove hose.

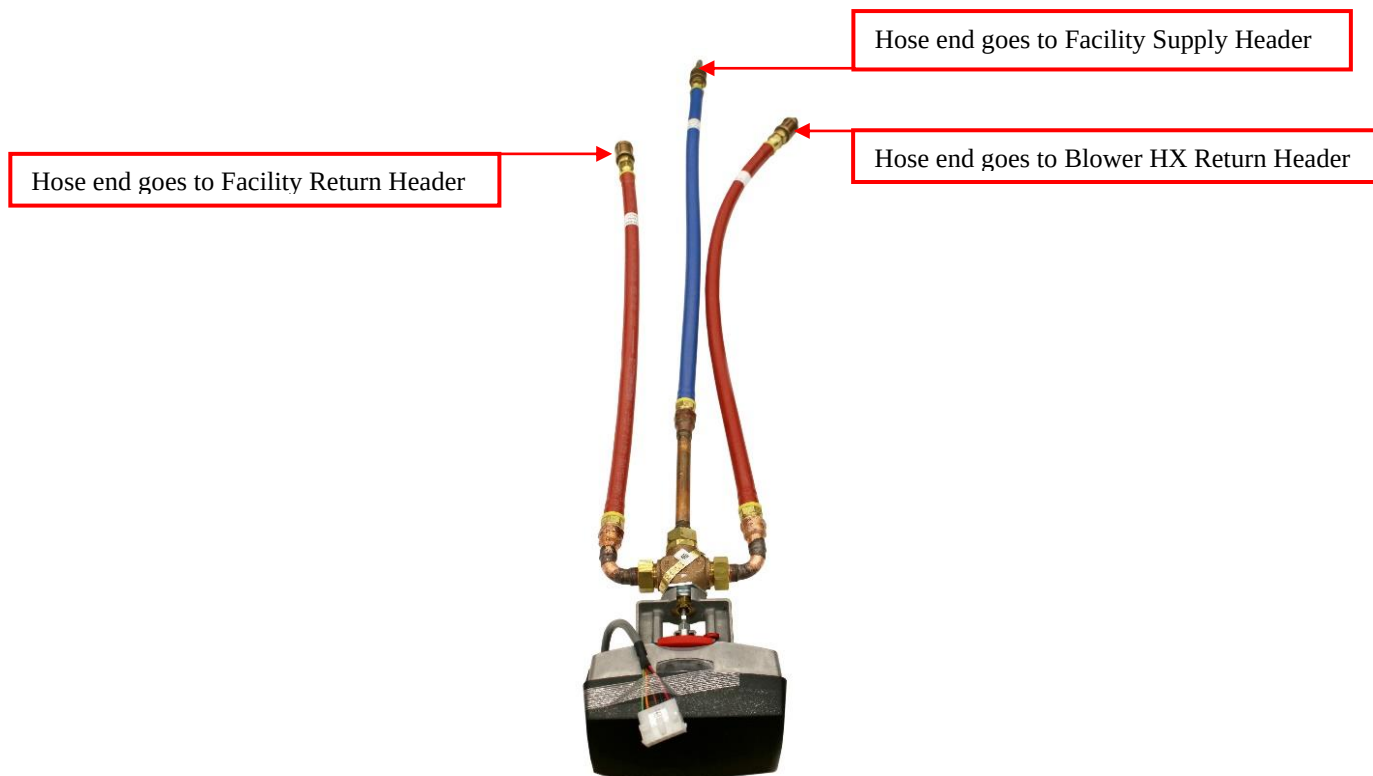


Figure 15 Blower Mixing valve detail showing ports A, B, and AB for reference

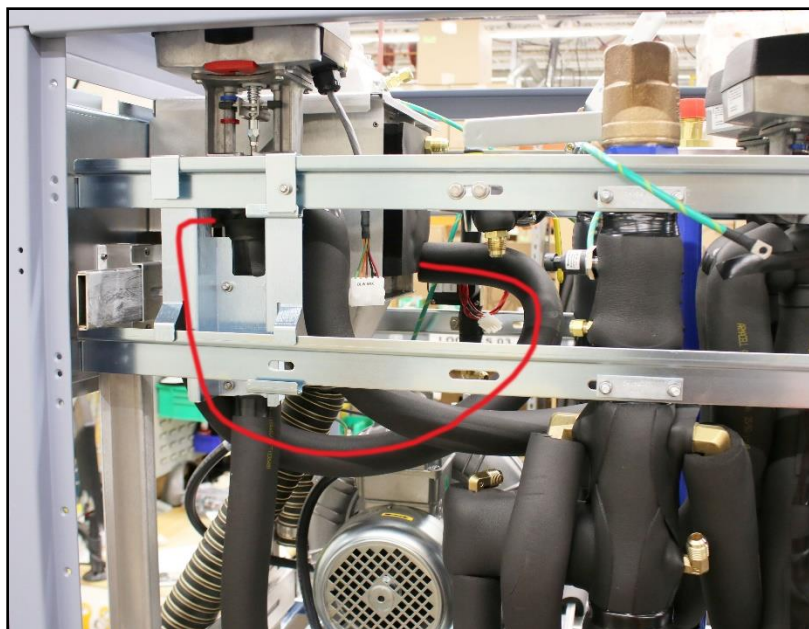
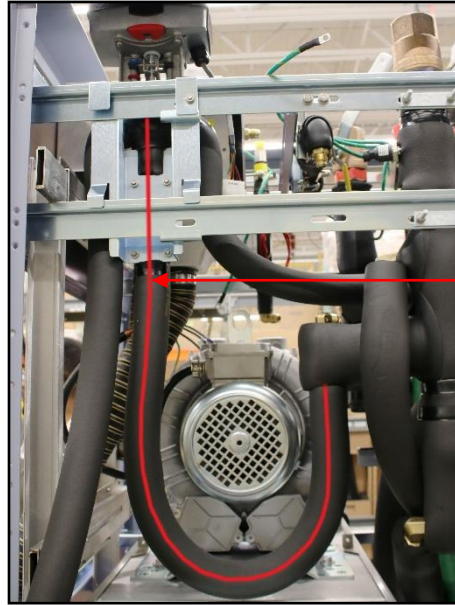


Figure 16 Blower Mixing Valve (Port 'A') to Blower Heat Exchanger Return

X. Blower Mixing Valve (Port 'B') to Facility Supply Header

(PN 200-1806-19 BLUE)

- a. Refer to Figure 15 for port locations for the mixing valve.
- b. Push insulation back on hose to be removed to gain access to the fitting.
- c. Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3 for reference), loosen and remove hose from Facility Supply Header. (Figure 17 below).
- d. Remove hose.



Blower Middle Hose end goes to Facility Supply Header Return Header

Figure 17 Blower (Port 'B') to Facility Supply Header

XI. Blower Mixing Valve (Port 'AB') to Facility Return Header

(PN 200-1806-20 RED)

- a. Refer to Figure 15 for port locations for the mixing valve.
- b. Push insulation back on hose to be removed to gain access to the fitting.
- c. Using a 7/8" wrench and an adjustable wrench for support (Refer to Figure 3 for reference), loosen and remove hose from return header (Figure 18 below).
- d. Remove hose

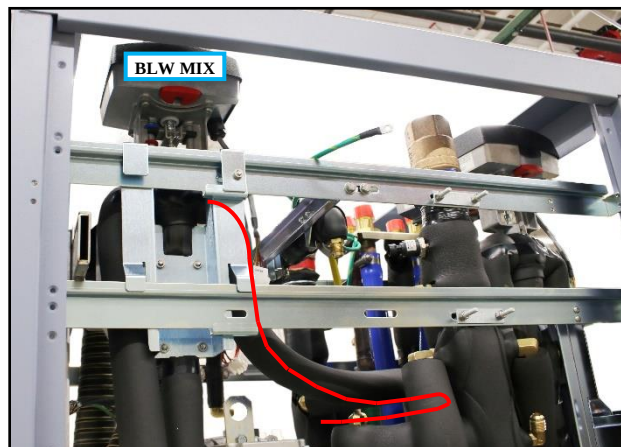


Figure 18. Blower Mixing Valve (Port 'AB') to Facility Return Header

End of Procedure

Installation (General Procedure):

- Using the Table 1 below, determine the correct size of insulation to insert hose being replaced.

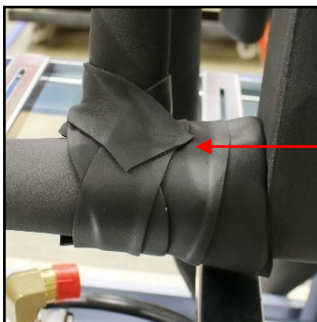
Table 1

Hose p/n	Qty	Hose Dia	Hose Length	Hose color	Insulation dia.	Insulation PN	2" Insulation tape
200-1806-05	1	1"	62 ¾"	Blue	1 3/8"	530-0772-02	530-2327
200-1806-06	1	1"	66 ¾"	Green	No Insulation	No Insulation	No Insulation
200-1806-07	1	¾"	39"	Blue	1 1/8"	530-0772-01	530-2327
200-1806-08	1	¾"	45 ¾"	Blue	1 1/8"	530-0772-01	530-2327
200-1806-09	1	¾"	38 1/4"	Red	1 1/8"	530-0772-01	530-2327
200-1806-10	1	¾"	55 1/2"	Red	1 1/8"	530-0772-01	530-2327
200-1806-11	1	¾"	23"	Blue	1 1/8"	530-0772-01	530-2327
200-1806-12	1	¾"	23"	Red	1 1/8"	530-0772-01	530-2327
200-1806-13	1	¾"	27 3/4"	Red	1 1/8"	530-0772-01	530-2327
200-1806-14	1	¾"	2201/2"	Red	1 1/8"	530-0772-01	530-2327
200-1806-15	1	1/2"	32 ½"	Gray	7/8"	530-0772	530-2327
200-1806-16	1	1/2"	17 ½"	Yellow	7/8"	530-0772	530-2327
200-1806-18	1	1/2"	29"	Red	7/8"	530-0772	530-2327
200-1806-19	1	1/2"	27 1/2"	Blue	7/8"	530-0772	530-2327
200-1806-20	1	1/2"	25"	Red	7/8"	530-0772	530-2327
200-1806-21	1	1/2"	24 ½"	Blue	7/8"	530-0772	530-2327

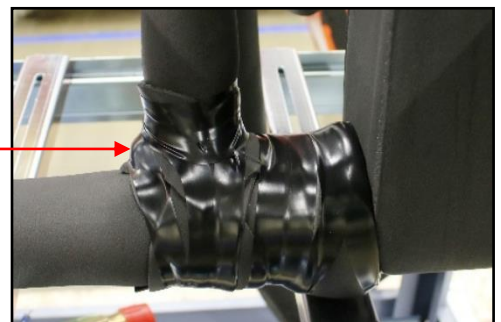
- Slide hose into correct insulation sleeve (see Table 1 above). The length of insulation should include an extra 3" on each end to provide a good friction fit at the ends.
- Apply several wraps of Teflon tape to threads of end connections.
- Secure hose to each connection point from where it was removed originally. Use back up wrench as was used originally to support connection during removal. **Tighten each connect to hand tight and then turn another 1/4 turn to secure.**
- Wrap 2" Wide insulation tape around fitting joints, end connection and where separate pieces of insulation meet.
- Wrap electrical tape over the 2" tape to completely cover the 2" insulation tape. Refer to photos below for examples.



Detail showing insulation, 2" Insulation tape and electrical tape to cover.



Detail showing insulation, 2" Insulation tape and electrical tape to cover.



7. Re-Install the Power box or the Signal box that was removed previously to help gain access to the hose being replaced.
8. Close drain valve on pump.
9. Remove Lockout/Tagout and open Facility Supply and Return ball valves on top of system (Refer to Figure 1 in Removal procedure above) Refer to Boyd Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).
10. Re-fill the system in accordance to Boyd Manual #820-0190 (Refer to GE MR Service Safety Manual 5452735).
11. Tighten vent caps on reservoir tanks.
12. Check system for leaks, repair if found.
13. Re-attach door panel by re-hanging onto hinges. Re-attach any side panel that was removed previously.
14. Return system to normal operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	INITIAL RELEASE	6/18/2021	SJM	MJG
B	PER ECN203709	2/11/2022	SJM	MJG
C	PER ECN 203833	6/17/2022	SJM	MJG

55 Dragon Court, Woburn, MA 01801, Tel: (781)933-7300, www.boydcorp.com

820-60035-000 – Install Instructions HEC Air Hose Kit (PN 205-0121-45) GE PN 5860300

Tools Required:

- Philips #2 screwdriver
- 1/4" nut driver

Procedure:

Removal of the HEC Blower air hoses:

1. Perform shutdown of system in accordance with Boyd manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedure 5452735
3. Open the front door to the HEC.
4. Pull Electrical Box forward by loosening 2 thumb screws to the left of the cabinet (top and bottom) securing it to the frame.

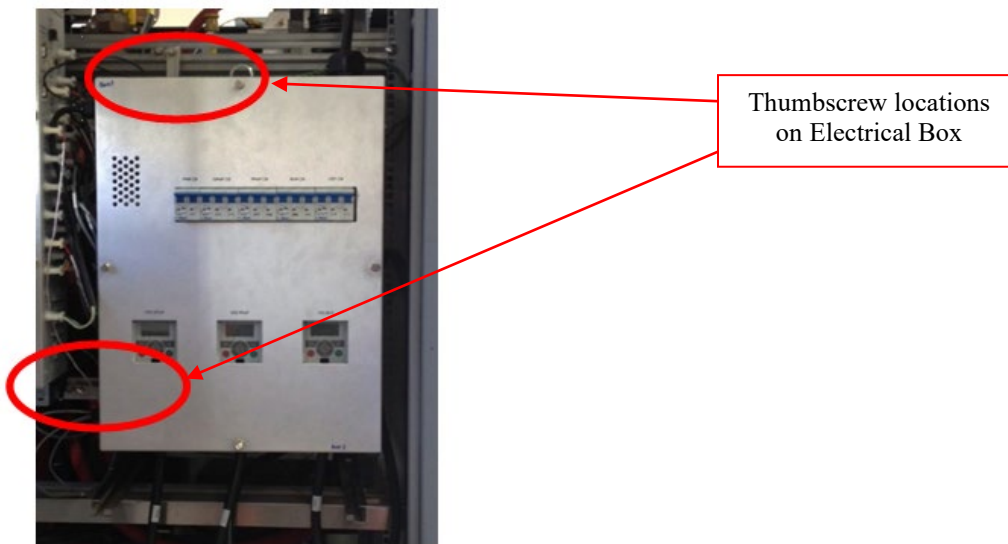


Figure 1 Electrical Box Thumbscrew location detail

5. Pull Electrical Box forward and using hinges, swing out of the way to the right to gain access to the blower hoses
6. Using a 1/4" nut driver, remove hoses from blower (both suction and discharge, noting which one runs to the Air Heat Exchanger and which runs to the Air Heat Exchanger mounting frame (18" hose). Refer to Figure 2 on next page.
7. Repeat loosening hose clamps on the bottom of the Air Heat Exchanger (top of 15 3/8" hose) and the hose that attaches to the Air Heat Exchanger mounting frame that runs to the suction side of the blower (18" hose)

End of Procedure

Installation of HEC Blower air hose kit:

1. Perform shutdown of system in accordance with Boyd manual #820-0190.
2. Lockout/Tag out system in accordance with GE procedures.
3. Open the front door to the HEC.
4. Pull Electrical Box forward by loosening 2 thumb screws to the left of the cabinet (top and bottom) securing it to the frame. Refer to Figure 1 on previous page.
5. To identify the correct hose length, stand in front of the blower from the front of the HEC. The right hand hose is 15 3/8" whereas the left hand hose is 18". Refer to Figure 2 below.

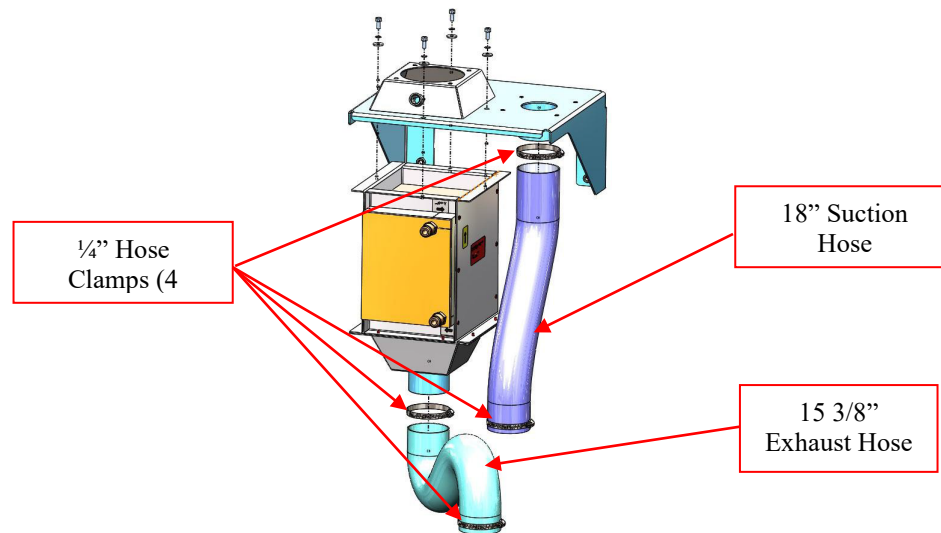


Figure 2 Blower Hose Detail

7. Install hoses using a 1/4" nut driver to tighten hose clamps on the bottom of the Air Heat Exchanger (top of 15" hose) and the hose that attaches to the Air Heat Exchanger mounting frame that runs to the suction side of the blower (18" hose)
8. Using a 1/4" nut driver, install hoses to blower (both suction and discharge, noting which one runs to the Air Heat Exchanger and which runs to the Air Heat Exchanger mounting frame (18" hose). Refer to Figure 2.
9. Rotate Electrical Box back and secure thumbscrews to attach to frame using the thumbscrews. Refer to figure 1.
10. Close the front door to the HEC.
11. Clear Lockout/Tag out system in accordance with GE procedures.
12. Perform startup of system in accordance with Boyd manual #820-0190.
13. Return HEC to online operation.

End of Procedure

Revision History

REV	ECN# / REASON FOR CHANGE	DATE	ENG APPRV'D	SERV APPRV'D
A	Initial Release	4/27/2022	SJM	MJG

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

TECHNICAL BULLETIN

Technical Bulletin Unique Identifier	DOC2534765
Title	Replacement Instructions for Variable Frequency Drive (VFD) in Lytron(Boyd) HEC
Subject	Replacement Instructions for Variable Frequency Drive (VFD)
Applicability (e.g., affected product, process)	PETMR, MR450, MR450W, MR450W GEM, Artist, MR750, MR750W, Architect
Date (DD-MMM-YYYY)	05/24/2021
Originating Business (select one)	☐ GEHC Digital (Imaging and Care Area Solutions, Value Based Care) ☐ Clinical Care Solutions (Ultrasound, Lifecare Solutions) ☒ Imaging (Magnetic Resonance, Molecular Imaging and Computed Tomography, Detection Guidance Systems, Global Services) ☐ Other: _____
Originating Modality	MR
Details	Replacement Instructions for Variable Frequency Drive (VFD)
Tracking Number	SVC-2020-0255
Affected Service Publications	NA
Resolution	See below
Technical Bulletin POC	100035593 Bora Guven

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Technical Bulletin DOC2534765

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Replacement Instructions for Variable Frequency Drive (VFD)

Required FRUs:

GEHC PN: 5862361 (GC & PE Pump VFD)

GEHC PN 5862362 (Blower VFD)

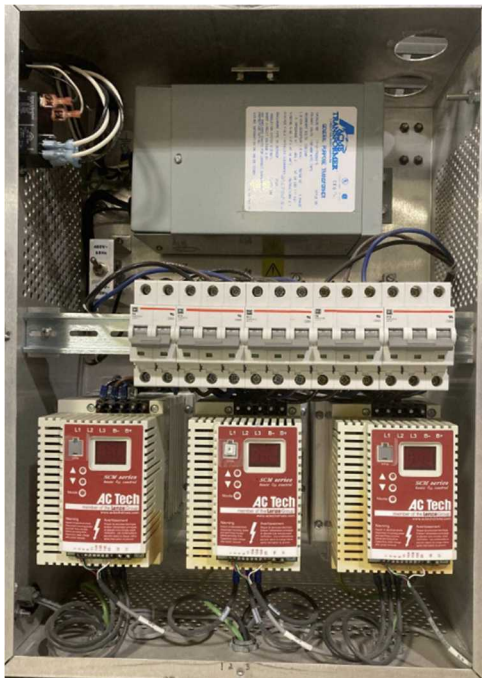
Tools Required:

- #2 Philips screwdriver
- 3/32" Flat blade screwdriver
- 1/4" Flat blade screwdriver

WARNING: HIGH VOLTAGE PRESENT, LOTO FROM MAIN DISCONNECTS.



This replacement procedure is for only for Power Box containing Eaton VFDs. Do not attempt to install Eaton VFD into Power Box containing AC Tech VFD. If AC Tech VFDs need to be replaced, the entire Power Box will need to be replaced GEHC PN 5156319.



AC TECH



EATON

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Technical Bulletin DOC2534765

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Removal Procedure (GC & PE Pump VFD, Blower VFD):

1. Perform shutdown of system, shut down power by switching the 5 circuit breakers in the electrical box to the off position
2. Lockout/Tag out system at the Main Distribution Panel to remove power from the HEC.
3. Remove HEC front door by opening door and removing door from its hinges.
4. Remove the Power Box exterior cover panel by loosening 4 screws (refer to Photo 2 below) with 1/4" flat blade screwdriver.

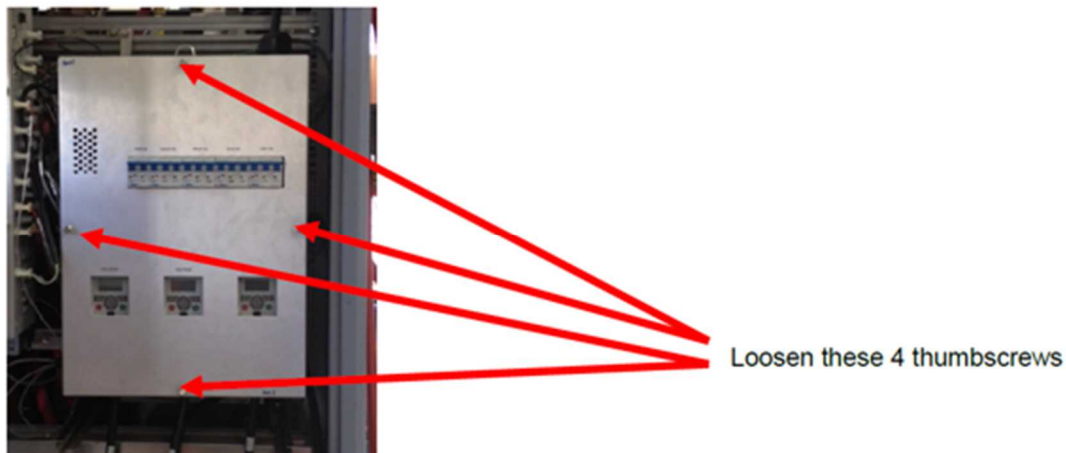


Photo 2

5. Lift the hinged cover to the VFD terminal section and snap cover into the up position as shown in the photo below, use a small flat head screwdriver. This will provide access to the control wires, power input cables and the DIP switch bank. Refer to Photo 3 below.

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Technical Bulletin DOC2534765

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Photo 3 Terminal Cover in Closed and Open positions

6. Remove power leads for device the VFD is controlling. These leads are labeled U-1, V-2, and W-3 as well as GND connection from the Gradient Coil (GC) VFD, or Power Electronics (PE) 5HP VFD or Blower 3HP VFD. These leads are on the right-hand side of the black terminal board as shown in Photo 3a below. **Note each location for re-installation of the new electrical Variable Frequency Drive (VFD).** – Refer to Photo 4a & 4b as well as Table 1 below.

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Technical Bulletin DOC2534765

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Photo 4a

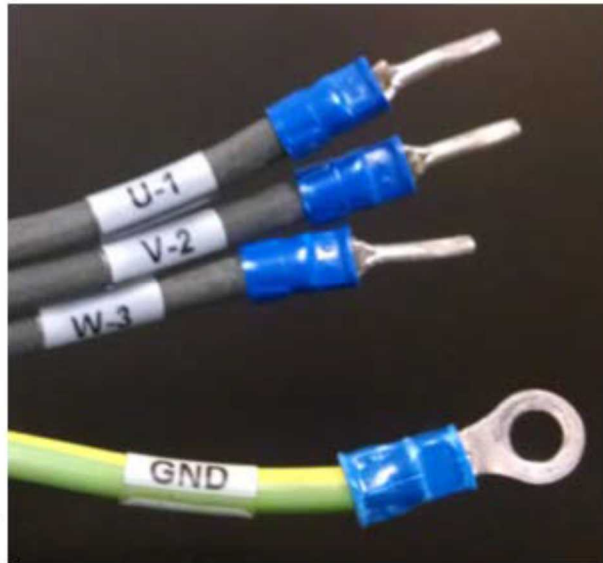


Photo 4b

Power Wire Detail for VFD

Pump/Blower Power Wires		
	VFD Terminal	Function
Lead callout	Eaton VFD	Device Supply
U-1	U/T1	L1 Power
V-1	V/T2	L2 Power
W-1	W/T3	L3 Power

Table 1

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

7. Remove VFD control wires with a 3/32 screwdriver or equivalent. These leads are on the left hand side of the terminal board at the bottom of the VFD as shown in Photo 5 below. **Note each color and location for re-installation of the new electrical Variable Frequency Drive (VFD). The schematic at the end of this document also calls out the correct color for L1, L2 and L3.**

VFD Incoming
Power leads for
L1(Black),
L2(Brown)
&L3(Blue) from
breaker

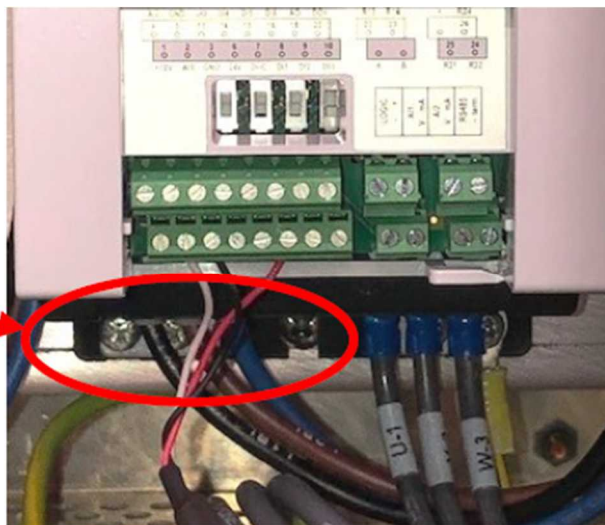


Photo 5 VFD Power Lead Detail

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Technical Bulletin DOC2534765

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8. Using the 3/32" screwdriver, remove the control wires for the VFD as described in the table below

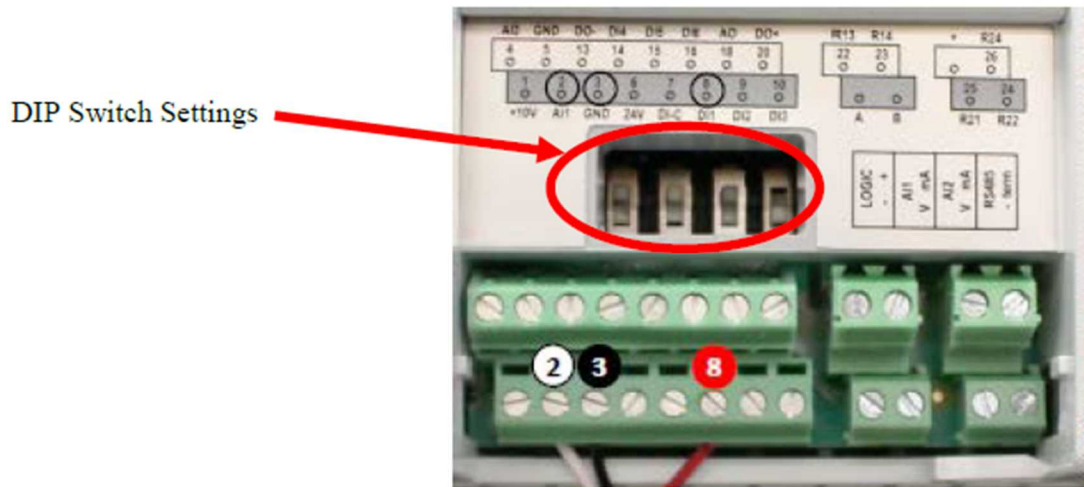


Photo 6 Control Wire Detail

Control Wire locations (reference only)		
	VFD Terminal	Function
Lead Color	Eaton	
Red	8	Motor Start
Black	3	Ground
White	2	4-20 mA analog input

Table 2

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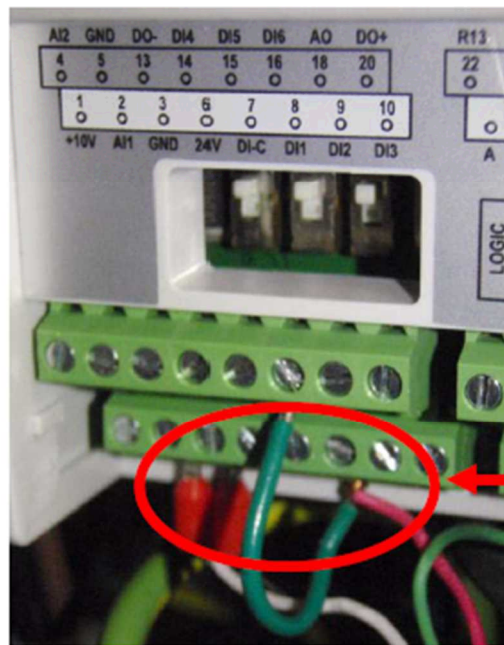
Technical Bulletin DOC2534765

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If either the GC or PE pump is of type Ebara, also remove the green jumper wire specifically for that pump. Save for re-installation in the new VFD. Refer to photo of pumps on page 16 below.



Green Jumper Wire
(required for an Ebara
pump)

Photo 7 Ebara Pump Jumper Wire Detail

9. Remove jumper for Ebara pump (if needed) between terminals #8 & #16 as shown in Photo 6 above. Save for re-installation on new VFD.
10. Using the Philips screwdriver, loosen the two screws (CCW 1/4 turn) above and two below which secure the VFD to back panel and remove from the Power Box. The screws are not fully removable and stay attached to the back panel.

DO NOT TRY TO REMOVE SCREWS FROM THE BACK PANEL (This will result in having to install a new POWER BOX).

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

11. **NOTE:** The Blower VFD has two screws at the top and only 1 screw at the bottom. Retain grip on blower VFD while loosening the top screws first followed by the bottom screw and then tilting the VFD forward to facilitate removal. Refer to Photo 7 below.

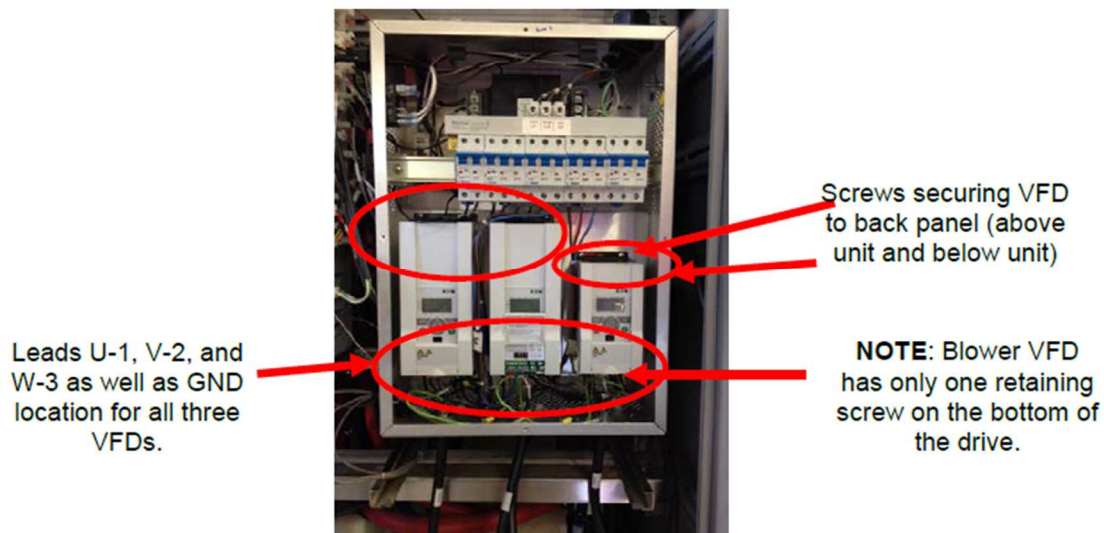


Photo 7 Interior of Power Box detail

12. This completes the removal of the VFD from the Power Box.

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Technical Bulletin DOC2534765

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Applicable to Device QMS Fraction

Installation Procedure (GC & PE Pump VFD, Blower VFD):

Installation of new VFD :

1. Using the Philips screwdriver, install the new Variable Frequency Drive (VFD) in the appropriate space of the Power Box (GE or PE VFD pump location) by screwing two screws above and two below to secure it to the back panel. Refer to Photo 2 below.

2. **NOTE:** The Blower VFD has two screws at the top and only 1 screw at the bottom. Retain grip on blower VFD while lowering the VFD onto the bottom screw to rest before installing the 2 upper screws. Tighten all screws to complete placement of the VFD. Refer to Photo 2 below

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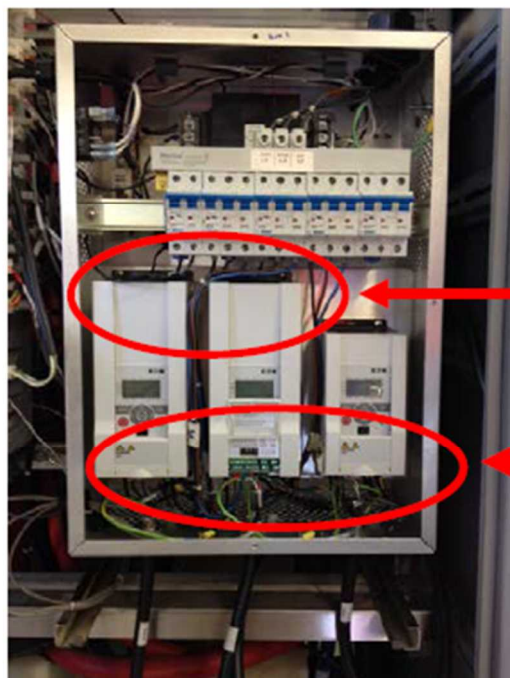
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Technical Bulletin DOC2534765

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Screws securing VFD to back panel (above unit and below unit)

Leads U-1, V-2, and W-3 as well as GND from all three VFDs.

Photo 2 Installation Detail

3. Wire the power going to pump or blower to the appropriate Variable Frequency Drive (VFD). Wires are phase sensitive and need to be installed in correct location (see labels on wires). For Eaton VFD, U-1 connects to U/T1 on the drive, V-2 connects to V/T2 on the drive, & W-3 connects to W/T3 on the drive. All wire connections are summarized in Table 1 below

It is critical that the motor phasing is correct. Failure to install the power wires in the correct sequence will cause the pumps and blower to run backward. This will likely cause damage to VFD and require replacement.

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Technical Bulletin DOC2534765

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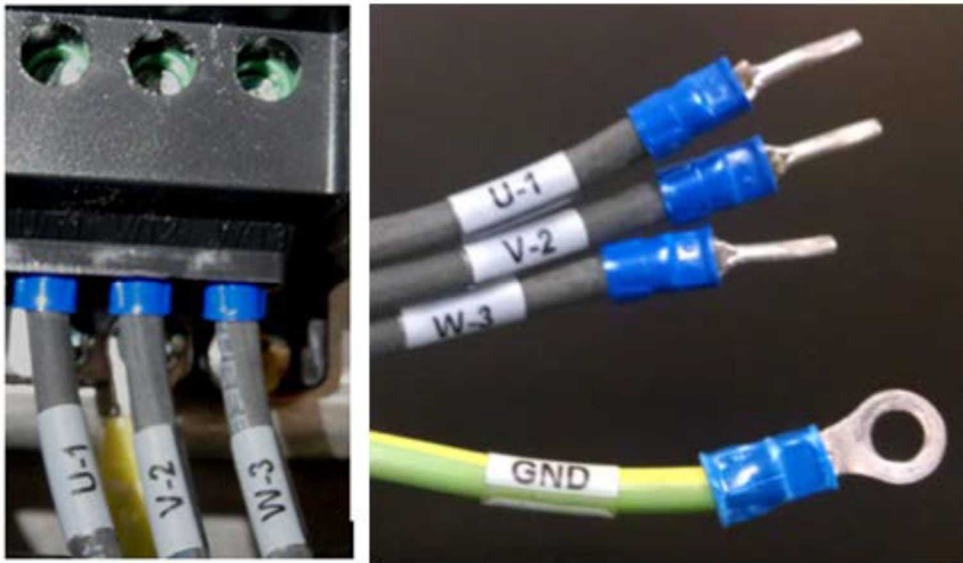


Photo 3a & 3b Pump or Blower power leads detail.

Pump/Blower Power Wires		
	VFD Terminal	Function
Lead callout	Eaton	Line Voltage
U-1	U/T1	L1 Power
V-2	V/T2	L2 Power
W-3	W/T3	L3 Power

Table 1 Power Wire Detail

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Technical Bulletin DOC2534765

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4. Wire the pump or blower control wires to the appropriate Variable Frequency Drive (VFD). Wires are installed on terminals 2, 3, and 8 as shown in the Photo 4 and Table 1.

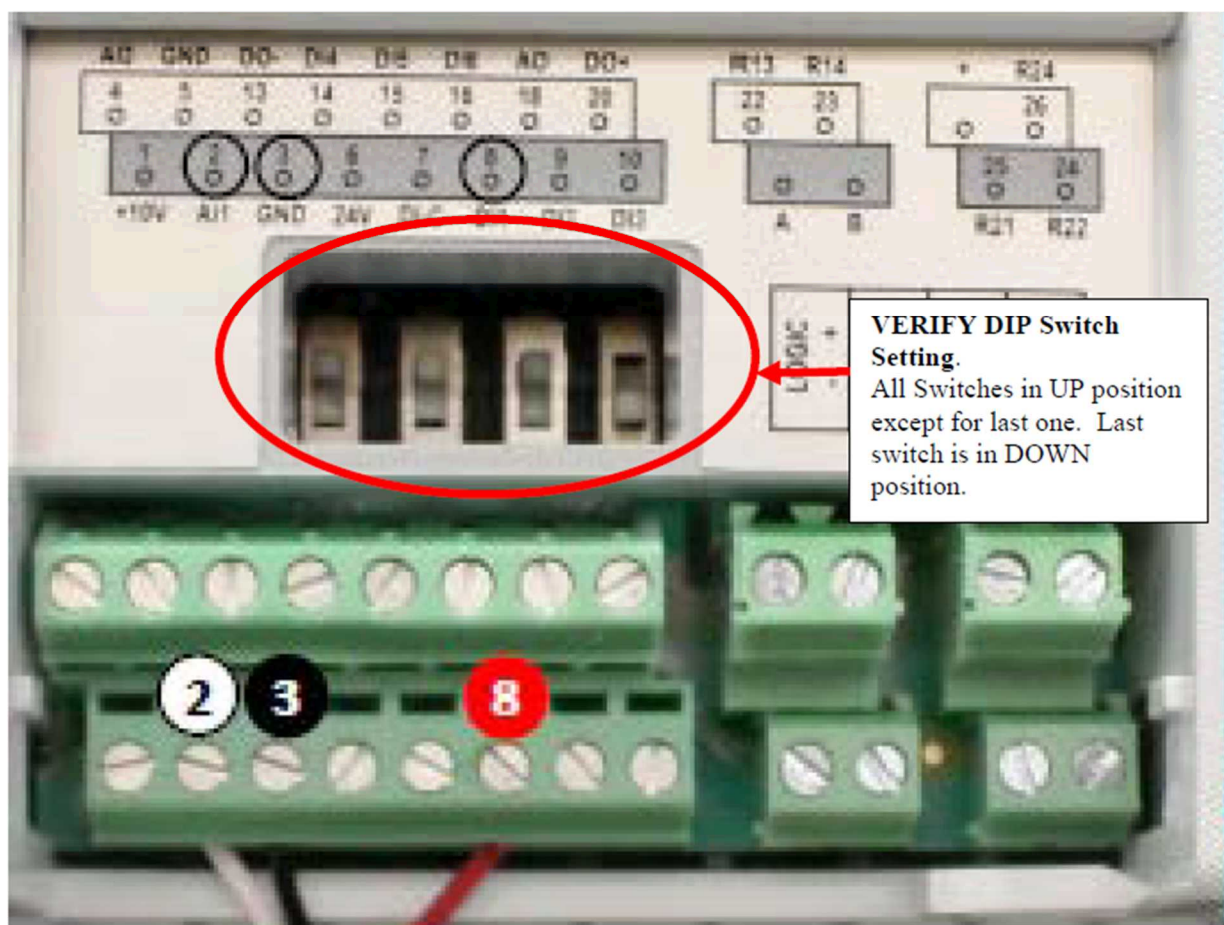


Photo 4 – Control wire terminal locations and DIP Switch positions

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Technical Bulletin DOC2534765

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Control Wire locations (reference only)		
	VFD Terminal	Function
Lead callout	Eaton	
red	8	Motor Start
black	3	Ground
white	2	4-2- mA analog input

5. **Verify DIP switch positions before continuing.** Only the right most DIP switch should be in the DOWN position (for both pump and blower VFD). All other DIP switches should be in the UP position. Refer to Photo 4 above.
6. Install power cable which supplies the VFD with power from the breaker with a 3/32 screwdriver or equivalent. These leads are secured to the left hand side of the terminal board at the bottom of the VFD as shown in Photo 5 below. **Verify proper color and position of the new electrical Variable Frequency Drive (VFD) power leads matches the order of removal from Step 7 of Removal Procedure above and as noted on the schematic at the end of this procedure.**

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

VFD Incoming
Power leads for
L1(Black),
L2(Brown)
&L3(Blue)
from breaker

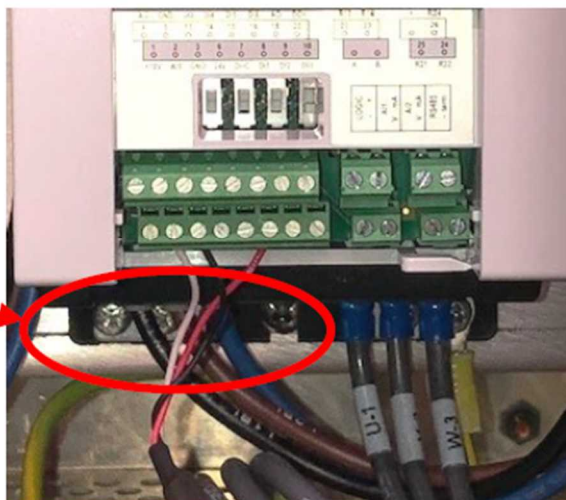


Photo 5 VFD Power Lead detail

If either the GC or PE pump is of type Ebara, also install the green jumper wire specifically for that pump. Refer to photos below as well as verifying pump type (Price or Ebara)

CAUTION: BEFORE YOU BEGIN, VERIFY THE PUMP TYPE AS EITHER A PRICE PUMP OR AND EBARA PUMP. IDENTIFICATION IS NECESSARY FOR COMPLETION OF NEXT SECTION.

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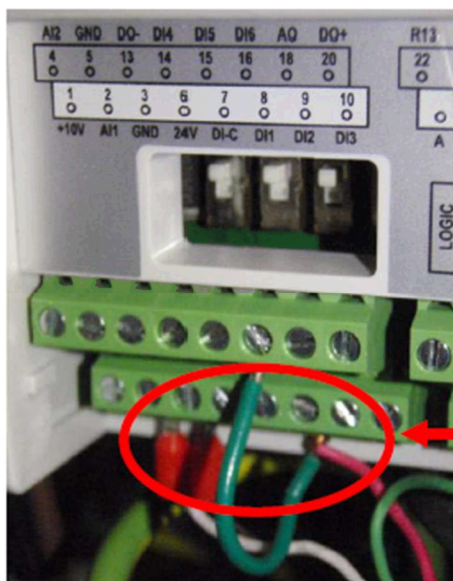
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Install Green Jumper Wire (Only required for an Ebara pump)

Photo 6 Ebara Pump Jumper Wire Detail



Price



Ebara

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

CAUTION: OBSERVE PROPER SAFETY PROCEDURES WHEN INSIDE THE ELECTRICAL CABINET. HIGH VOLTAGE IS PRESENT WHEN COVER IS REMOVED.

12. Remove LOTO from HEC at the Main Distribution Panel.
13. Turn on power by switching the 5 circuit breakers in the electrical box to the on position.
14. To check proper rotation of pumps & blowers (including cryo compressor), turn on breaker momentarily to observe the direction of the individual motor or blower. If rotation is reversed, verify wiring in Step 10 above. To correct a reverse rotation, switch two of the three phases for the appropriate breaker and re-check rotation is correct.
15. After verifying correct rotation of pump or blower, pull terminal cover down and snap into place over the terminal board.
16. Replace the cover to the Power Box by securing the 4 thumbscrews using a flat bladed screwdriver.

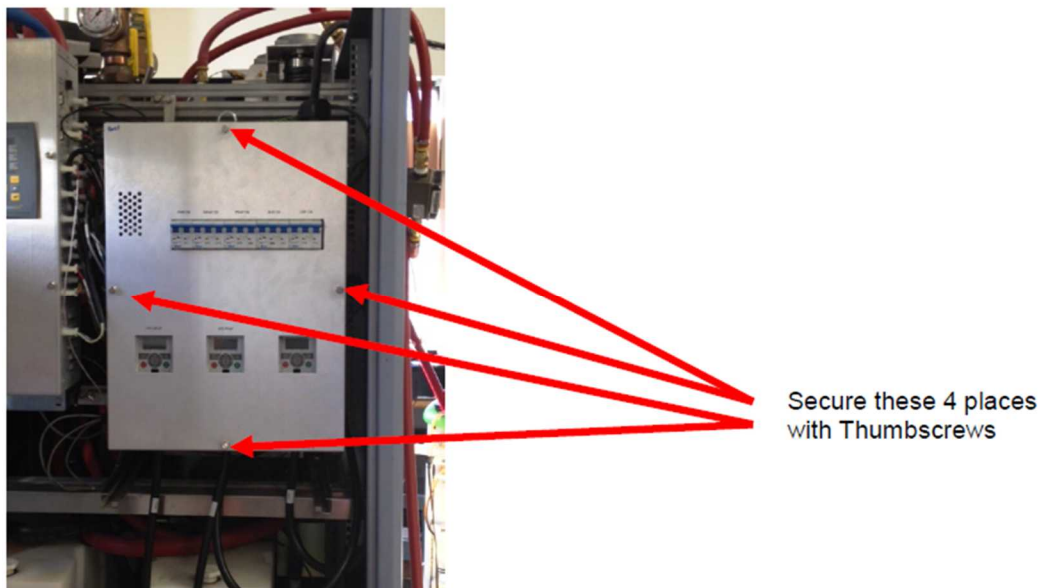


Figure 7 Front Cover in place (Thumbscrew locations).

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Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

16. Start system in accordance with startup procedures

FINALIZATION

1. Re-start HEC and turn pump/blower ON. Add new DVMR coolant as necessary.
2. Check for any leakage. Repair if necessary.
3. Perform a TPS reset.
4. Ensure HEC is in a scan-ready state and not in the low power mode; this is done by prescribing a scan at least to the point where the scan protocol has been downloaded.
5. Return to the HEC and view the displayed speeds on the two pump VFDs and the blower VFD.

GRADIENT COIL (GC) PUMP:

Pump Type (Identify as described above):	EBARA	PRICE
Displayed Value On left-most VFD (“VFD GPMP”)	50 +/- 1	61 +/- 1

POWER ELECTRONICS (PE) PUMP:

Pump Type: (Identify as described above):	EBARA		PRICE	
RF-amplifier type in PGR cabinet (identify by presence of red and blue cooling hoses hooked up to front of RF-amplifier. Hoses will not be present on air cooled RF amplifiers.	Liquid	Air	Liquid	Air
Displayed value on middle VFD (“VFD PPMP”)	50 +/- 1	43 +/- 1	61 +/- 1	53 +/- 1

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GEHC_GQP_12.08.002_T001



Technical Bulletin DOC2534765

Applicable to Device QMS Fraction

BLOWER:

BLOWER (Right most VFD)	60 +/- 1 (WB) or 70 +/- (NB)
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6. If either of the pump speeds does not match the charts above, turn OFF the pumps and:
 - A. If the pump is an Ebara pump, check corresponding VFD for that jumper wire is installed and wired to the correct terminals.
 - B. Ensure the HEC is in a scan-ready state and not in low-power mode.
7. Further Troubleshooting information can be found in the GE Service Methods document "HEC Troubleshooting".

SERVICE NOTE APPROVALS

Service Note route approvals per GEHC_GQP_02.01_R002 by Service, QA, and Engineering representatives.

Rev	Page	Description of Content Changed	Reason for Change or Change Control Number	Author Name / SSO	Date
1	All	Initial Entry	Initial Entry	B. Guven 100035593	05/24/2021